

Premetric Bosonic Response Structure in the Standard Model

Intrinsic Gauge Laws, Higgs Incidence, and Response Networks

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Abstract

The identity of a Standard Model carrier, the responses it supports, and the coordinates used to represent those responses are different objects. We separate them at a fixed premetric comparison target, using the AASC gauge, Higgs, and Yang–Mills theorem interfaces with their source hypotheses and conditional realization domains preserved.

A complete typed source grammar presents the intrinsic admissible response class. Its contextual signature records every admitted test after every finite typed continuation, including definedness and failure. Equality of these signatures gives the canonical response quotient; licensed constructors descend on their exact domains, and all semantic uses factor uniquely through the quotient. Neither particle labels nor a desired response count enters this construction.

The full Standard Model carrier chain determines three gauge-response roots—charge, color, and pre-vacuum weak—and one Higgs bridge/fixation identity. The Higgs supports distinct order, deformation, and fixation mechanisms. Restricting one common Standard Model source tuple constructs an actual mixed Higgs–electroweak readiness witness, structural residual charge transport, and a simultaneous compatible gauge assembly. Residual transport uses nonmetric support and an admitted response reduction; it does not assume a photon or a numerical vacuum value. On that source-realized fibre, every assembly has the same carrier-root occupation, with separately typed cardinalities $(3;1)$.

The common source comparison also fixes the complete structured gauge-law reduct: transport and loop-response assignments, representation and charge structures, anomaly probes, and residual interfaces with their full partial domains. Fixed-sibling covariance is proved for intrinsic mixed instances, retaining their actual attachments and reflecting their domains in both directions. Every compatible assignment has a finite source-operation network representation that preserves its actual inputs, relative incidences, and observations. Thus uniqueness of the law record is separated from equality of responses at different admitted inputs. The latter has an exact complete-signature criterion and is not assumed. Every actual common ledger also admits injective set-codescent on its declared packet overlaps.

The resulting premetric atlas preserves the positive Yang–Mills extended-support layer and the ambient gravity necessity while excluding both from an undifferentiated boson count. Metric, physical-vacuum, polarization, resonance, and confinement realizations remain separately licensed constructions over this structural domain.

Keywords: admissibility; premetric structure; Standard Model; carrier identity; Higgs bridge; contextual quotient; compatible realization.

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1 The fixed comparison fibre and source interfaces

1.1 The premetric comparison target

We classify the response architecture of the source-certified gauge and Higgs roots before vacuum-fixed particle, pole, polarization, or confinement projection. The classification distinguishes four response sorts:

$$b \in \mathfrak{B}_B := \{Q, C, \text{EW}^{\text{pre}}, H\}.$$

Here Q denotes the Abelian charge-response locus, prior to physical residual electromagnetic realization; C denotes color transport; EW^{pre} denotes the weak-chiral network before vacuum descent; and H denotes the Higgs order–deformation–fixation locus. The four-element index is a sort declaration, not a four-carrier or four-particle claim.

Global-form-sensitive conclusions are evaluated fibrewise. Let $\mathfrak{G}_{\text{src}}$ be the family of global-form data for which the import certificates below are instantiated, and fix $\Gamma \in \mathfrak{G}_{\text{src}}$. No proof combines witnesses from different values of Γ .

Definition 1.1 (Core BCRS-I target). *The core target is*

$$\tau_B(\Gamma) = (D, K_D, \text{RPI}_D, G_{\text{AASC}}, \partial D, \mathfrak{B}_B, \Gamma, \mathfrak{M}_B, \text{Succ}_B).$$

D is the common same-domain object; K_D is the AASC kernel; RPI_D is the structural realized-interior certificate described below; G_{AASC} is the ambient premetric gravity role; and ∂D is the boundary/localization interface. The mechanism signature $\mathfrak{M}_B$ permits transition, transport, loop residue, representation/charge response, anomaly, product compatibility, residual/vacuum readiness, color self-response, extended-support, Higgs order/deformation/fixation, boundary, continuation, readiness, and failure operations. It specifies the comparison language, not its inhabitants. Succ_B declares how changing that language or its anchors produces a successor target.

The target precedes the source registry, candidate set, semantic-use algebra, presentation quotient, role assignment, live residue, and cardinality analysis. None of these is a target parameter. Nor does τ_B contain a photon, gluon, $W^\pm$, Z^0 , physical Higgs, generator count, polarization count, mass, width, propagator, numerical coupling, vacuum coordinate, detector record, or confinement result.

Theorem 1.2 (Target and granularity finality). *The family $\{\tau_B(\Gamma)\}_{\Gamma \in \mathfrak{G}_{\text{src}}}$ fixes the four response questions independently of every later response object and count. Altering a target anchor, kind sort, admitted mechanism type, global-form fibre, boundary test, continuation operation, or successor rule after inspecting a live residue is a target update, not an inference in the original fibre.*

Proof. The fixed-domain comparison class preserves boundary typing, the evaluated fragment, the standing partition, the lawful quotient interface, continuation, and role typing [4, Def. 0.1]. Definition 1.1 specializes precisely those data to the BCRS-I question. The structural exhaustion theorem then places any standing-relevant change in boundary typing, act classification, quotient, or continuation; it supplies no fifth same-scope locus [4, Thms. 7.1–7.4]. A change of $\mathfrak{B}_B$, $\mathfrak{M}_B$, or Γ changes one of those coordinates. Since no later display or count occurs in τ_B , it cannot select the target without reversing this dependence. $\square$

1.2 The structural physical-interior antecedent

The gauge theorems use a fixed realized physical interior, understood as the following structural role organization. Write

$$\mathbf{L}_D = (\text{NonVacRec}_D, [B_D]_{\equiv B}, (\text{Adj}_D, N_D, \text{LCD}_D), \text{Cmp}_D, \text{BoundTrans}_D, \Phi_D, \Pi_D, \text{GaugeRole}_D, \text{MassResp}_D, \Omega_D, (\text{Time}_D, \text{Caus}_D))$$

for the R1–R12 role ladder, and let

$$\text{RPICert}_D := (\mathbf{L}_D, \text{Joint}_D, \text{Coher}_D, \text{Closure}_D, \text{Minimal}_D).$$

Joint_D witnesses joint instantiation and cross-compatibility of the ladder. R12, Theorem 6.1, maps that witness to the final three fields [49]. In that source, *interior* means the integrated standing-bearing role organization of D , not a spatial container, manifold interior, metric tensor, field equation, or matter inventory.

Proposition 1.3 (Exact realized-interior antecedent). *For every BCRS-I source instance, RPI_D means the image of an inhabited RPICert_D under the R12 synthesis theorem. Hence the fixed-realized-physical-interior hypothesis appearing in Gauge Realization, SM-2, and SR7 is retained and discharged by a named antecedent. No metric, smooth chart, QFT presentation, or physical boson projection is thereby inserted into τ_B .*

Proof. R12 defines the realized physical interior as a role package and proves its coherence, closure, and operational minimality from the jointly instantiated R1–R12 ladder [49, Defs. 3.1–3.5 and Thm. 6.1]. The same source states that metric tensors and spacetime manifolds are later specializations of comparison, localization, transport, and temporal roles. The full antecedent record therefore supports the role-level conclusion without importing a metric realization. $\square$

The gravity necessity theorem supplies G_{AASC} at this same premetric level. It is needed because a nondegenerate physical comparison arena cannot be supported by a null or ungoverned background, but it is not a member of a gauge or Higgs response fibre.

Proposition 1.4 (Ambient-gravity firewall). *Deleting G_{AASC} removes a target-enabling necessity condition for the claimed physical comparison class. Nevertheless no source- or type-preserving map sends G_{AASC} to a BCRS response occupant. Counting it as a bosonic carrier or as a fourth gauge response is ill-typed.*

Proof. The gravity necessity theorem makes the quotient/constraint role necessary for reusable physical identity, admissible redescription, and continuation [50, Thms. 7.10, 8.2 and Cor. 8.3]. A BCRS response, however, must have a root in the imported gauge/Higgs carrier interface and an output sort in $\mathfrak{M}_B$. The ambient gravity coordinate has neither. It remains an anchor of τ_B , not an element of the response algebra. $\square$

1.3 The inherited fermionic result

The fermionic classification supplies the cross-arc interface

$$\mathbb{H}_{III \rightarrow BC}^{2,0} = (\mathbb{F}_{\text{FINAL}}^{2,0}, \text{ProofClass}_{U01:U17}, \text{StatusTarget}_{III}, \\ \text{SourceFingerprint}_{III}, \text{ClaimCeiling}_{III}^0, \text{SuccessorRule}_{III})$$

from [1, Sec. 10]. In BCRS this interface fixes cross-arc identity, especially the equality of the one Higgs carrier, and preserves the scope of the fermionic result.

Lemma 1.5 (No bosonic generation from U18). *The U18 record licenses the shared identity $C_H = \mathcal{H}_{\text{br/fix}}$ and the inherited boundary data. It proves no BCRS source completeness, response quotient, gauge response cardinality, product-network disposition, confinement, unstable pole, or physical common apex.*

Proof. Those statements are outside the U18 claim ceiling. Its successor rule requires a newly tagged independent proof for every stronger assertion. All BCRS semantic, quotient, incidence, and compatibility objects below are new and are not images of a fermionic count. $\square$

1.4 Conservative theorem interfaces

Each source theorem enters through an interface retaining its full antecedent witness.

Definition 1.6 (Instantiated theorem interface). *For a source theorem $T_i : H_i \rightarrow O_i$, an interface into $\tau_B(\Gamma)$ is a tuple*

$$\mathcal{E}_i = (j_i, w_i, \rho_i, \text{NoProm}_i),$$

where j_i identifies the source target with a declared face of the fixed BCRS fibre, $w_i \in H_i$ is its full antecedent witness, $\rho_i : O_i(w_i) \rightarrow \overline{O}_i$ is a typed structural reduct, and NoProm_i records the fields that the reduct may not promote. The expression $\rho_i(O_i(w_i))$ is defined only after w_i is supplied.

Thus a reduct may forget a chart, basis, smooth representative, numerical value, pole label, or downstream projection while retaining the proved structural role. It may not turn a conditional conclusion into an unconditional one. The admissible import object $\mathbb{I}_B(\Gamma)$ is the fibre product of the following instantiated interfaces over their common $D, K_D, \text{RPI}_D, \partial D, \Gamma$, and C_H coordinates.

Interface	Retained object	Exact boundary
R1–R12 and gravity	$\text{RPI}_D, G_{\text{AASC}}$	The full role-ladder and gravity antecedents remain attached; no metric projection or graviton is imported [49, 50].
Gauge-root vocabulary	$C_Q, C_{\text{col}}, C_{\text{wk}}^{\text{pre}}$ as typed target roots	The root names are a conservative structural vocabulary. No SM-2 closure, primitive-status deletion, SR7 assembly, photon/gluon/ W/Z particle, pole, generator dimension, or numerical value is imported [13].
Gauge A1–A6, PSM1–PSM6, SR1–SR6	Transition, transport, loop, representation/charge, anomaly, product, and residual/vacuum structural coordinates	Each witness is instantiated directly from its own source theorem; smooth or realized hypotheses stay in that component. The interfaces share only the fixed target coordinates at this stage. No common-ledger or SR7 conclusion is inferred. Only their role-level outputs are projected into the BCRS algebra [20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37].
Yang–Mills endpoint	Fixed local theorem object and its distinct extended-support sector layer	The endpoint’s support, deformation, global-form, and completion-filter hypotheses remain attached. Neither confinement nor a free asymptotic gluon is imported [47].
Higgs H17–H22 and U18	One $C_H = \mathcal{H}_{\text{br/fix}}$, with order, deformation, and fixation interfaces	Identity, scale, vacuum, extension, and projection boundaries are retained; no second Higgs, vev selector, numerical potential, or scalar pole is imported [39, 40, 41, 42, 43, 44, 1].
Role-Occupancy Closure	The abstract candidate–quotient–eligibility–compatibility–orbit theorem schema	No BCRS candidate, live residue, orbit, or count is imported [8].

SR7 and the SM carrier-exhaustion sequence are independent source theorems. Their antecedents contain no BCRS response quotient, compatibility predicate, or assignment count. The direct interfaces above suffice for response formation; the independently established SR7–SM-2 chain may also supply a

common source realization, as used in Definition 10.18. Its coarse carrier structure does not replace the finer BCRS response analysis. The later comparison is a sound forgetful map with an explicit gauge-assignment extension fibre, not an assumed equivalence of the two moduli objects. The SR7 antecedent record comprises a fixed realized physical interior, K_D , the complete Realized Physics ladder, Gauge Realization A1–A6, PSM1–PSM6, SR1–SR6, and the fixed nondegenerate pre-mass comparison regime [38, Prop. 3.1 and Thm. 19.1]. The integrated SYM and standalone Standard Model presentation quotients are not identified with the BCRS response quotient. In particular, a source carrier count cannot decide a BCRS response-signature equivalence.

Theorem 1.7 (Conservative import finality). *For every $\Gamma \in \mathfrak{G}_{\text{src}}$, the preferred source handoffs determine an inhabited $\mathbb{I}_B(\Gamma)$. Every retained conclusion is an image $\rho_i(O_i(w_i))$ with its full antecedent witness attached. Consequently:*

- (i) *no metric or realized hypothesis is erased;*
- (ii) *no conclusion-equivalent BCRS quotient, compatibility object, live residue, assignment orbit, or response count is imported;*
- (iii) *the three gauge roots are fixed target identities, while their BCRS response fibres remain to be formed;*
- (iv) *C_H is one and the same Higgs bridge/fixation identity in the Higgs and U18 records; and*
- (v) *the Yang–Mills extended-support result remains positive structural information with its completion boundary intact.*

Proof. The R12 witness gives the common realized-interior face by Proposition 1.3. The A1–A6, PSM1–PSM6, and SR1–SR6 interfaces are instantiated directly from their respective source theorems; their common $(D, K_D, \text{RPI}_D, \partial D, \Gamma)$ coordinates are fixed target identifications rather than a joint-survivor conclusion. SM-2 is used only to align the conservative gauge-root vocabulary. In particular, neither its Proposition 4.1 nor its Theorems 16.2, 18.1, or 22.4 is used here, since those results include SR7 in their proof packages. The Higgs endpoint and U18 agree on $C_H = \mathcal{H}_{\text{br/fix}}$. The Yang–Mills main theorem supplies the fixed local object and the distinct theorem-scope extended-support layer over the color root. Equalities used in the fibre product are therefore source-theorem equalities or fixed target identifications, not similarities of labels. Definition 1.6 keeps every w_i in the dependent domain of ρ_i , so reduct formation cannot discharge a missing premise. Finally, none of the retained codomains is a BCRS source algebra, quotient, live residue, or assignment orbit; those objects first appear in Sections 2–14. $\square$

1.5 Logical dependencies

The proof order is

$$\begin{aligned}
 \tau_B &\prec \mathbb{I}_B \prec \mathcal{E}_B \prec \mathcal{G}_B \prec \mathcal{S}_B \prec \text{Act}(\mathcal{S}_B) \\
 &\prec \mathcal{Q}_B \prec \text{Packets}_B \prec (E_B, J_B, \text{Live}_B) \\
 &\prec \text{Assign}_B / \text{Aut}_B \prec \text{ROC} \prec \text{Count}_B \prec \text{Projection}_{\text{phys}}.
 \end{aligned} \tag{1.1}$$

$\mathcal{E}_B$ is the registry-independent semantic comparison algebra and $\mathcal{G}_B$ is the source-generator family. Defining the semantic class without source tags makes its subsequent source coverage a theorem, rather than a condition of membership.

Principle 1.8 (Fixed-authority no backflow). *At a fixed $\tau_B(\Gamma)$ and import instance, later compatibility, cardinality, metric, vacuum, pole, confinement, or detector information cannot create, delete, identify, or recount an earlier source or standing class. A load-bearing change is either a successor target or a dependency inversion.*

Counterexample criterion 1.9 (Scope and import failure conditions). *Theorem 1.2 or Theorem 1.7 fails if the target contains its own source registry or role count; if the four sorts are regrouped after seeing a residue; if global-form fibres are spliced; if an imported theorem loses a hypothesis; if the structural interior is confused with a metric background; if U18 or SYM-1 generates a bosonic count; if the Yang–Mills nonlocal sector is discarded or promoted to confinement; if gravity is counted as a response; or if a second same-scope Higgs identity is introduced.*

2 The independent response semantics

2.1 Imported roots and response sorts

Write $\mathfrak{G}_{\text{adm}} := \mathfrak{G}_{\text{src}}$. The conservative import theorem supplies four root objects

$$\text{Root}_B(\Gamma) = \{C_Q, C_{\text{col}}, C_{\text{wk}}^{\text{pre}}, C_H\},$$

typed respectively over $Q, C, \text{EW}^{\text{pre}}, H$. These source-certified carrier identities support the response analysis: which standing-bearing presentations they admit, which presentations are semantically identical, and which can coexist in a compatible live assignment.

Definition 2.1 (Gauge role index). *The gauge-root typing map is*

$$\text{rt}_g : \{C_Q, C_{\text{col}}, C_{\text{wk}}^{\text{pre}}\} \longrightarrow \mathbf{R}_g^0 := \{r_Q, r_{\text{col}}, r_{\text{wk}}\},$$

with the evident sort-preserving values. $\mathbf{R}_g^0$ is an independently fixed role index; it asserts neither that a response occupant exists nor that an occupant is unique or jointly compatible.

The role index fixes the classification questions; the cardinalities of their compatible live fibres are a separate question for the response analysis.

2.2 Structural witnesses and semantic candidates

Definition 2.2 (Raw submissions and standing-semantic witnesses). *A raw response submission is a finite typed compositional diagram over the fixed roots and mechanism profiles of $\tau_B(\Gamma)$, together with proposed effect, boundary, continuation, readiness, and failure values. Well-typedness alone does not give those proposed values standing.*

A standing-semantic atom of profile μ is instead a pair (a, ℓ) , where a consists of actual structural data and ℓ witnesses the following intrinsic laws on their declared domains:

- (i) *All participating data have the fixed domain, root identities, boundary, and global form. A many-input atom has one common anchor witness; separately anchored marginal data do not suffice.*
- (ii) *Each asserted positive output is an admitted consequence of those structural data. Its effect and boundary coordinates are its actual target-test values, not independently assignable labels. Equal standing presentations give equal consequence values.*
- (iii) *Every admitted restriction, composition, or continuation preserves the declared root and the relevant field, particle, quotient, transport, and boundary witnesses. Domain changes are recorded as target updates and never as same-target operations.*
- (iv) *A response distinction has standing only through an admitted distinguishing witness. A quotient-identifiable difference is skin; deleting a witnessed distinction is not an admissible quotient.*

- (v) *The mechanism-specific preservation laws below hold, including the full intrinsic refinement predicates of any specialized operation actually performed. A predicate is supplied by its complete witness tuple, not by a passing flag.*
- (vi) *A negative output includes a witness of the failed preservation or neutrality law. An unexecuted conditional output retains its exact input type and implication without asserting an inhabitant of that type. An unavailable antecedent is not a witness of failure and is not assigned a passing value.*

These laws refer to structural objects, admitted tests, and preservation relations. They do not refer to theorem names, source tags, registry membership, a desired cardinality, or existence of a source-algebra term.

The mechanism laws in clause (v) specify the following structural requirements, independently of their source certificates.

- (L1) *Transition and transport.* Local presentation arrows preserve the fixed invariant-role family; identities, inverses, restriction, and composition are used only on their admitted interfaces. Transport assigns boundary-fixed comparison classes to admitted continuation morphisms and preserves bounded transport, field mediation, particle identity, quotient comparison, and endpoint compatibility, without selecting a local representative.
- (L2) *Loop and representation response.* A loop includes its admitted endpoint-closing datum. Its residue is the actual composite comparison class; a nontrivial residue must be boundary-detectable, quotient-stable, transport-witnessed, endpoint-covariant, role-preserving, and same-domain. Representation actions preserve the transition, transport, and loop response witnesses. Charge identity is equality under all admitted charge-response probes.
- (L3) *Witnessed specializations.* Smooth or infinitesimal notation is subordinate to admitted smoothable continuation interfaces. It preserves local transition projection, transport, identity-neighborhood comparison, first-order quotient descent and regularity, commutator closure, and operational-loop persistence under admitted refinement. Algebraic actions, subcarriers, sums, tensors, conjugates, handed splits, dimensions, and multiplicities are retained only when their realization maps preserve the corresponding response witnesses, locally refinable and finitely witness-presentable operations, and all transition, transport, loop, field, particle, quotient, and boundary data. A representation label, basis, scalar coordinate, or internal index has no independent standing. Abelian coordinates preserve the charge-witness ledger and quotient-commutative response; normalization is skin unless witnessed. Ratios, neutrality, sign, conjugation, discrete classes, periodic response, and central-quotient claims each retain their own witnesses. Discrete or lattice operations must be locally finitely witness-presentable and response-composition-compatible; anomaly and vacuum remain separate guards.
- (L4) *Products and coupled sectors.* Sector labels have standing, co-scope, and quotient-distinctness witnesses. Independent-product status requires coherent transitions, assembled transport, and quotient-trivial mixed transition, transport, and loop residue. Boundary-detectable, quotient-stable, role-preserving mixed residue instead has coupled-sector status. Coupling is not itself failure: it must remain locally refinable, quotient-stable, transport-coherent, and finitely representable along admitted chains. Field and particle roles, anomaly and vacuum guards, and no-selector conditions remain attached to the common ledger.
- (L5) *Obstruction and residual response.* An anomaly value is the obstruction class of an actual admitted local-to-global assembly probe. A nonneutral value obstructs same-domain gauge standing. Algebraic sums require aggregation readiness; traces must be basis-independent and witness-preserving; nonlinear charge forms are probe-induced; mixed forms use jointly stable prior ledgers; signed

contributions and multiplicities are response-witnessed; global or index coordinates preserve endpoint comparison. Measure, phase, counterterm, and inflow claims must retain their witness and exact repair-domain status. Algebraic neutrality is equivalent to neutrality of the actual obstruction ledger. Vacuum compatibility requires a non-null support profile and its full preservation ledger; an executed residual response additionally requires a witnessed vacuum-mediated admissible reduction. The residual ledger is induced by that reduction and inherits or lawfully reduces transition, transport, loop, representation, charge, anomaly, product/coupled, and Lie-type witnesses. Its support and deformation baseline, local finite witness-presentability, and quotient minimality are retained: no coarser quotient erases witnesses and no finer unwitnessed distinction is assigned standing. Neither an equation template nor an arbitrary zero certifies neutrality.

- (L6) *Extended color support.* A theorem-scope extended-support datum is attached to the fixed local color object and Γ , respects the admitted support/deformation relation, and retains its actual extended-to-local incidence. Only this structural reduct is retained; vacuum vectors, Hamiltonians, spectral data, and mass-gap values are not premetric response fields.
- (L7) *Higgs and mixed incidence.* Order, deformation, and fixation uses lie over one fixed C_H , preserving the Higgs-domain target, invariant bundle, gauge inheritance, residual normal form, vacuum support, deformation mediation, identity continuity, and redescription stability. A representative or downstream route cannot select primitive standing. A mixed Higgs–charge–weak incidence retains one joint structural witness with those exact input restrictions. A conditional electroweak, metric, confinement, polarization, or excitation port is not an executed physical output. The existence of the joint incidence witness is not implied by the existence of its marginals.

Throughout, actual structural values are intrinsic pointed records under the independently licensed witness-preserving presentation comparisons. All relative incidences, attachment classes, guards, and conditional domains are retained. Exact input matching is equality of these intrinsic records with their shared-interface witness, not equality of raw presentation names. Section 11 constructs this source descent and proves that fixed-input domains and all standing observations are preserved. This prior source skin relation is not the later complete BCRS signature quotient.

Let $\mathcal{E}_B^{\text{raw}}(\Gamma)$ denote raw submissions and let $\mathcal{E}_B(\Gamma)$ denote finite typed compositional diagrams of standing-semantic atoms, with every grafting accompanied by the common interface and preservation witnesses required by its profile. For a diagram ξ , the effect, boundary, continuation, and failure records are obtained by evaluating its actual witnesses under all admitted target tests. These records are not extra free parameters.

Admissibility, rather than source-registry membership, determines the semantic class. A raw submission has a diagnostic evaluation when its evaluation domain is formed. Witnessed positive laws give a positive response atom; a proved obstruction gives a diagnostic atom; an unresolved obligation retains its precise unexecuted type. These distinctions specify standing conditions, not a decision procedure for arbitrary submissions.

The full port-sort family is

$$\mathfrak{S}_B = \mathfrak{B}_B \sqcup \mathfrak{A}_B,$$

where $\mathfrak{B}_B$ is the distinguished family of four peer root kinds and $\mathfrak{A}_B$ contains the auxiliary output types already declared by the mechanism signature, including product ledgers and $\text{Inc}_{H,EW}^{\text{pre}}$. These are structural interface sorts, not additional carrier roots. A mixed diagram can have an auxiliary output while retaining its incidence to the original input roots.

Definition 2.3 (Typed many-input operations). *For $n \geq 0$ and a profile $(b_1, \dots, b_n; b)$ in the full port-sort family $\mathfrak{S}_B$, let*

$$\mathcal{O}_B(b_1, \dots, b_n; b)$$

be the ways of grafting the outputs of n semantic presentations to a mechanism vertex of output sort b , carrying the common-interface and preservation witnesses of Definition 2.2. Unary operations include transition, transport, loop, representation/charge, boundary, continuation, and failure attachment. The product constructor is genuinely many-input. The single Higgs–electroweak constructor has profile

$$(H, Q, \text{EW}^{\text{pre}}; \text{Inc}_{H, \text{EW}}^{\text{pre}}).$$

Its charge and weak restrictions are two typed input legs of one mixed incidence; they are not separate generators and the constructor is not an arrow inside one kind fibre.

Grafting finite typed diagrams is associative, the one-edge diagrams are units, and input permutation acts only where the target preservation witness declares a lawful symmetry. Thus $\mathcal{E}_B(\Gamma)$, with the families $\mathcal{O}_B$, is a typed finite-diagram system with partial operations on their witnessed common-interface domains. Composition is associative wherever those domains match; arbitrary well-profiled tuples are not thereby executable. The many-input structure retains the joint domains of product and Higgs incidence, which cannot be expressed by four disjoint categories with only unary arrows.

Lemma 2.4 (Well-typed semantic composition). *Every licensed grafting of members of $\mathcal{E}_B(\Gamma)$ is again a member of $\mathcal{E}_B(\Gamma)$. An attempted graft with mismatched kind, boundary, global form, continuation, or failure ports is undefined rather than coerced.*

Proof. Finite compositional grafting preserves the diagram type. Its retained common-interface witness identifies all shared inputs, including boundary, global form, and the single Higgs identity where applicable. Its preservation witness gives the restriction and continuation laws and states that the output tests are the composites of the actual input tests. Consequently all clauses of Definition 2.2 remain satisfied. Port matching without this witness is insufficient and does not define a licensed operation. $\square$

2.3 Support, response, and guard codomains

The direct gauge interfaces separate common support, response roots, and obstruction or vacuum guards. This typing can be compared with the later SM-2 eight-field notation

$$C_{\text{gauge}} = \{C_{\text{gauge-order}}, C_{\text{transport}}, C_{\text{curv/hol}}, C_{\text{charge-response}}, \\ C_{\text{color-transport}}, C_{\text{weak-chiral}}, C_{\text{anomaly}}, C_{\text{gauge-vacuum}}\}.$$

This is heterogeneous interface notation, not an eight-element response base. Its use here does not import SM-2’s closed certificate or the SR7 premises of that certificate. Its structural reduct has the typed decomposition

$$\begin{aligned} \text{Support}_g &= \{C_{\text{gauge-order}}, C_{\text{transport}}, C_{\text{curv/hol}}\}, \\ \text{RootResp}_g &= \{C_{\text{charge-response}}, C_{\text{color-transport}}, C_{\text{weak-chiral}}\}, \\ \text{Guard}_g &= \{C_{\text{anomaly}}, C_{\text{gauge-vacuum}}\}. \end{aligned}$$

The first line supplies common order/transport/obstruction support; the second supplies the three imported response roots; the third supplies a failure guard and a residual/vacuum interface. An element in one line cannot be reassigned to another while preserving its source codomain and certificate predicate.

Proposition 2.5 (Typed gauge-interface stratification). *The displayed decomposition is forced by the output types of the direct gauge interfaces. In particular, neither common support nor a truth-valued guard is a peer occupant of $\mathbf{R}_g^0$, and the eight-field package cannot be counted as eight gauge responses.*

Proof. A1–A3 supply the common transition, transport, and loop-residue spine; A4 and the witness-preserving product/representation specializations supply typed response data over the already fixed sector roots; A5 and A6 supply obstruction and vacuum-compatibility interfaces [20, 21, 22, 23, 27, 28, 24, 25]. These fields have distinct dependent codomains. A type-preserving projection to $\mathbf{R}_g^0$ is defined exactly on the middle row. Reassigning a support or guard would discard its certificate codomain and hence would not be a morphism of the imported record. $\square$

The Higgs package has a different shape. Its one root C_H supports the three mechanism ports

$$\text{Ord}_H, \quad \text{Def}_H, \quad \text{Fix}_H.$$

They are fields over one identity, not three roots and not gauge responses. Excitation and pole data remain downstream ports.

2.4 The semantic response class

Definition 2.6 (Broad bosonic response class). *The semantic response class $\mathcal{R}_B^{\text{sem}}(\Gamma)$ is the full subalgebra of $\mathcal{E}_B(\Gamma)$ whose rooted diagrams have at least one response, obstruction, readiness, continuation, or failure output admitted by $\mathfrak{M}_B$. It includes discrete, continuous, functional, composite, and higher-arity diagrams. Only intrinsic witness admission is required; no requirement of activity, source generation, presentation quotient, compatibility, or terminal realization is imposed at this stage.*

Theorem 2.7 (Semantic class formation). *For every fixed $\tau_B(\Gamma)$, the algebra $\mathcal{R}_B^{\text{sem}}(\Gamma)$ is defined independently of the source registry and of every familiar boson display. Its four root sorts, many-input operation profiles, standing tests, continuation semantics, and failure codomains are fixed before source coverage, activity reduction, quotienting, or counting.*

Proof. The data are exactly the fields of Definitions 1.1, 2.2, 2.3, and 2.6. None mentions a source identifier or a later cardinality. Lemma 2.4 supplies closure under every licensed constructor. The imported roots identify the questions being asked, but they do not supply a presentation, live occupant, or response count. $\square$

Counterexample criterion 2.8 (Semantic-setting failure conditions). *Theorem 2.7 fails if a candidate is admitted only because its theorem tag occurs in a registry; if product or Higgs incidence is treated as an untyped unary arrow; if a shared support or guard field is counted as a peer response; if the three Higgs mechanism ports multiply C_H ; if the residual electromagnetic output is declared occupied before vacuum descent; or if a particle, generator, polarization, mass, pole, or confinement display enters the semantic definition.*

3 Source coverage and the complete semantic-use algebra

3.1 Instantiated source generators

Source certification distinguishes a theorem family from each of its lawful instantiations.

Definition 3.1 (All-instance source certification). *For a mechanism profile μ , a certified source generator retains a standing-semantic atom a , a full instantiation h of the relevant source hypotheses, and the source conclusion certifying that atom or its witness-preserving structural reduct. Write it as*

$$g_{\mu,a,h,c}, \quad c : \text{Cert}_{i,\mu}(a; h), \quad \text{ev}_B(g_{\mu,a,h,c}) = a.$$

The generator family ranges over all lawful instances, not one chosen solution of an existence theorem. If a source conclusion is stated in a richer representative language, its certified reduct preserves every retained target witness. Theorem identity, proof term, and file provenance are certificate data and not semantic coordinates.

Thus g_{tr} abbreviates the indexed family $\{g_{\text{tr},a,h,c}\}_{a,h,c}$; it does not identify its different admitted instances. The same convention applies throughout.

Definition 3.2 (BCRS source-generator family). *The family $\mathcal{G}_B(\Gamma)$ contains exactly the certified schemas with the following output roles:*

$$\begin{aligned} \mathcal{G}_{\text{spine}} : & \quad g_{\text{ord}}, g_{\text{tr}}, g_{\text{curv}}, g_{\partial}, g_{\text{cont}}, \\ \mathcal{G}_Q : & \quad g_{\text{rep}}, g_Q, g_{Q\text{res}}, g_{\text{pol}}, \\ \mathcal{G}_{\text{col}} : & \quad g_{\text{col}}, g_{\text{adj}}, g_{\text{self}}, g_{\text{YM}}^{\text{loc}}, g_{\text{YM}}^{\text{ext}}, g_{\text{conf}}, \\ \mathcal{G}_{\text{wk}} : & \quad g_{\text{wk}}, g_{\text{chiral}}, g_{\text{cc}}, g_{EW\,H}^{\text{pre}}, \\ \mathcal{G}_H : & \quad g_H, g_{H\text{ord}}, g_{H\text{def}}, g_{H\text{fix}}, g_{\text{exc}}, \\ \mathcal{G}_{\text{guard}} : & \quad g_{\text{anom}}, g_{\text{prod}}, g_{\text{vac}}, g_{\text{fail}}. \end{aligned}$$

The fixed global-form coordinate Γ and the ambient gravity role are target anchors, not response generators.

The source-use register and this generator family have different indexing roles. A register row names a use of a certified package; it is not an independence claim about a syntactic primitive. In particular, the transition and order uses are restrictions of the same g_{ord} package from A1. The charge generator g_Q carries the witnessed Abelian charge structure, whereas $g_{Q\text{res}}$ carries its actual structural residual use only on the admitted A6–PSM6 reduction domain. The fixed Γ and gravity entries index the target and have no generator-deletion interpretation. These distinctions follow from the displayed family and the source output types; they introduce no new carrier or response role.

The source dependencies are independent of the response count:

Semantic atom	Exact source coverage	Nonpromotion boundary
order, transition, transport, loop	R8 and Gauge A1–A3 [20, 21, 22]	no gauge choice, potential, field-strength component, or loop display is a new response root
representation, charge, Abelian refinement	Gauge A4, PSM3–4, and SR3–4 [23, 28, 29, 34, 35]	no basis, dimension, normalization, or photon label is promoted
anomaly, product, residual/vacuum	Gauge A5–A6, PSM2/5/6, and SR1/2/5/6 [24, 25, 27, 30, 31, 32, 33, 36, 37]	guards and compatibility predicates are not response occupants
color, adjoint/self-response, local and extended support	Gauge A1–A3, PSM1–PSM3, SR2–SR3, and Yang–Mills Main Theorem II.1 [20, 21, 22, 26, 27, 28, 33, 34, 47]	no generator dimension, color index, confinement, jet, or free-gluon state
weak-chiral, charged-current, and Higgs–electroweak incidence readiness	Gauge A4/A6, PSM2/3/6, SR2/3/6, and the Higgs bridge interface [23, 25, 27, 28, 31, 33, 34, 37, 39]	one mixed incidence-readiness generator; no W/Z , mass, mixing coordinate, Goldstone, polarization, pole, or duplicate Higgs count
Higgs root, order, deformation, fixation	H17, H18, H22, and U18 [39, 40, 44, 1]	one C_H ; no vev selector, second identity, numerical potential, or pole
boundary, continuation, readiness, failure	FE3, FE4, Bivalence, fixed-domain closure, and the exact source endpoint boundaries [9, 10, 6, 4, 11]	readiness does not execute projection; failure is retained rather than repaired

Definition 3.3 (Source term algebra). $\mathcal{S}_B(\Gamma)$ is the finite many-sorted source-term system on $\mathcal{G}_B(\Gamma)$, with the licensed partial operation domains and precisely the operation profiles of Definition 2.3. A term is therefore a finite well-typed compositional tree; product and Higgs incidence may have several inputs. The semantic syntax uses the same compositional convention. Physical continuation loops are certified loop atoms, not cyclic term syntax. The evaluation homomorphism

$$\text{ev}_B : \mathcal{S}_B(\Gamma) \longrightarrow \mathcal{R}_B^{\text{sem}}(\Gamma)$$

replaces each generator by its certified semantic atom and each formal constructor by the corresponding semantic grafting.

Cross-kind product and Higgs incidence enter the term algebra only through their declared many-input profiles. Ill-typed composites are excluded.

3.2 Coverage of the independent semantic class

Source completeness is evaluated against $\mathcal{R}_B^{\text{sem}}(\Gamma)$, which was defined before and without $\mathcal{G}_B$.

Lemma 3.4 (Pre-SR7 atomic source coverage). *Every standing-semantic atom of a declared target mechanism profile has a source certificate with the same retained structural and diagnostic values. The certificate is positive, obstruction-bearing, or conditional according to the atom’s actual status. This statement does not turn an unresolved raw submission into a classified standing witness.*

Proof. Fix an atom before choosing any source tag. Its intrinsic witness tuple satisfies the appropriate row of the following classification, whose hypotheses contain neither the source registry nor a BCRS response count.

- (i) For transition and transport, the retained continuation site, local arrows, and preservation witnesses instantiate A1’s transition predicate and A2’s comparison-role clauses. A1, Theorem 7.1, and A2, Theorems 7.1 and 10.1, certify the same arrows and transport comparison classes [20, 21].
- (ii) For loop, representation, and charge response, the actual loop composite and witness evaluations instantiate A3, Theorems 11.1 and 13.1, and A4, Theorems 11.1 and 13.1. A4’s equality test is equality of all admitted response outputs, not equality of labels [22, 23].
- (iii) For smooth, product, algebraic, Abelian, anomaly-coordinate, and residual specializations, collect the atom’s already witnessed refinement fields. The exact classification and exhaustion pairs are PSM1, Theorems 10.4–10.5 and 11.1; PSM2, Theorems 10.2, 11.2, and 12.1; PSM3, Theorems 13.1 and 14.1; PSM4, Theorems 14.1 and 15.1; PSM5, Theorems 14.1 and 15.1; and PSM6, Theorems 15.1 and 16.1 [26, 27, 28, 29, 30, 31]. Each positive clause preserves the very witness ledger being certified. Each negative branch retains its actually failed clause. PSM2’s coupled branch remains a lawful coupled ledger, not an independent-product or failure certificate. A source’s downstream-incomplete branch retains an unexecuted guard and does not certify failure of a later theorem.
- (iv) The obstruction and vacuum witnesses instantiate A5, Theorem 10.1, and A6, Theorems 10.1 and 12.1 [24, 25]. Executed SR1–SR6 tests use their complete common-ledger antecedents and evaluate the same witness tuple; their simultaneous-survival results are never replaced by marginal compatibility [32, 33, 34, 35, 36, 37].
- (v) The fixed color-root and extended-support witness instantiate the declared structural reduct of the Yang–Mills endpoint, Main Theorem II.1(A)–(E) and Theorems V.12, VI.12, VII.8, and VIII.2 [47]. This certifies theorem-scope support and incidence without importing its metric or spectral output coordinates.
- (vi) Higgs atoms instantiate the one-carrier bridge and identity interfaces: H17, Theorems 7.4 and 7.6, Lemma 8.3, and Theorem 11.4; H18, Theorems 5.6 and 8.1; and H22, Theorems 5.2, 8.3, and 9.1 [39, 40, 44]. For a mixed incidence the structural input includes the joint bridge witness package and its actual charge and weak restrictions. The bridge normal-form classifier certifies its standing location; witness-preserving restriction supplies the same retained incidence. This is a classification of a given full witness, not a construction of one from marginal roots or from a necessary survivability condition. Theorem 10.16 separately asks whether this exact joint domain is inhabited.
- (vii) Boundary, continuation, and diagnostic atoms are evaluations of the same instantiated source witnesses under admitted fixed-domain continuations. FE3/FE4 and fixed-domain factorization preserve their declared consequences [9, 10, 4]. A new test or an unlicensed continuation changes the target rather than supplying a counterexample inside the fixed semantic class.

Each row supplies a source instance for the existing lawful atom, not a replacement atom with a newly selected value. These rows exhaust the declared mechanism profiles. No step invokes the closed SM2 certificate, SR7, role occupancy, or cardinality. The profile list is finite, but the certificate families range over all admissible witness values, including continuous and functional ones. $\square$

Theorem 3.5 (Source coverage). *Evaluation covers the registry-independent standing-semantic response class:*

$$\text{ev}_B : \mathcal{S}_B(\Gamma) \rightarrow \mathcal{R}_B^{\text{sem}}(\Gamma).$$

For each finite compositional semantic diagram, the covering term retains its diagram, root, standing-effect, boundary, continuation, readiness, failure, and global-form values. This is not surjectivity onto arbitrary typed assignments of those values.

Proof. Induct on the compositional diagram. Atomic coverage supplies a certified generator for each semantic vertex, and the same retained interface witness licenses each grafting. Evaluation thus reproduces the original diagram and its actual test values. This induction is on finite syntax: physical continuation loops are certified loop atoms and are not cyclic term syntax. Repeated uses of one root retain that same root identity and their common-interface witnesses; they do not add unlicensed copying or independent source identities. $\square$

3.3 Complete contextual semantics

The four members of $\mathfrak{B}_B$ are peer response-root kinds. The full port-sort family $\mathfrak{S}_B$ also contains auxiliary structural sorts, such as product ledgers and $\text{Inc}_{H,EW}^{\text{pre}}$. They are output types of the already declared mechanisms, not additional carrier roots. Constructor profiles and tests are typed over $\mathfrak{S}_B$; the four-root response notation denotes their rooted part.

Definition 3.6 (Typed one-hole response contexts). *For port sorts b, c , let $\text{Ctx}_B(b; c)$ be the family of finite well-profiled response diagrams with one input hole of sort b and output sort c . The other inputs are fixed standing-semantic data in the same target and global-form fibre. The family contains the identity hole, every admitted unary update, and a hole in each individual input of every many-input constructor; it is closed under context substitution. Each context retains the intrinsic structural parameters and common-interface requirements of its constructors, without source-proof names or raw presentation pointers. A fixed mixed parameter retains its actual attachment and full pointed restrictions as in Definition 11.3; it is not replaced by an existential assertion that some attachment exists. Its evaluation $C[\xi]$ is partial: it is defined exactly when all required kind, root, boundary, continuation, global-form, and preservation witnesses are present. Thus a well-profiled context need not be executable at every input.*

Let $\mathcal{T}_{B,c}$ be the fixed target tests on output sort c . For a possibly undefined value, write

$$\hat{t}(C[\xi]) = \begin{cases} (\text{defined}, t(C[\xi])), & C[\xi] \text{ and } t(C[\xi]) \text{ are defined,} \\ \perp_{\text{undef}}, & \text{otherwise.} \end{cases}$$

The symbol $\perp_{\text{undef}}$ is distinct from every test value, including zero, a proved obstruction, and a conditional-readiness value. Tests also include the declared domain predicates for contexts and for individual probes, so context-definedness and probe-definedness are separately retained. For a source term s of sort b , define

$$\text{Sig}_B(s) = (\hat{t}(C[\text{ev}_B(s)]))_{c \in \mathfrak{S}_B, C \in \text{Ctx}_B(b;c), t \in \mathcal{T}_{B,c}}.$$

This includes kind and root, mechanism outputs, standing effect, guards, readiness, boundary trace, continuation, individual failure tests, and Γ . The same evaluation convention applies to the interface-only comparison diagrams used below. Source-paper names and file hashes remain provenance metadata rather than standing coordinates.

Unary contexts with matching input and output type form the update monoid $\mathcal{U}_B^{\text{upd}}$. That monoid is only a subfamily of the complete context system. In particular, comparison of two charge responses includes their use with fixed weak and Higgs inputs in one mixed context; an isolated unary signature would not suffice for that comparison.

Definition 3.7 (Failure on formed evaluation domains). *For $\xi = \text{ev}_B(s)$, let $\mathcal{F}_B(\xi)$ consist of the target failure tests whose scientific evaluation domains are formed at ξ . Domain formation is part of the retained structural record, not the height or length of the source term. Define*

$$\text{Fail}_B(s) = \text{pass} \iff \forall f \in \mathcal{F}_B(\xi), \quad f(\xi) = 0_f,$$

where 0_f is the neutral value of that particular obstruction test. Otherwise the failure record retains the actual nonneutral tests and their semantic obstruction classes; the proofs certifying them are not extra diagnostic coordinates. This aggregate is derived from the individual test and domain coordinates; it does not define those coordinates.

An anomaly, vacuum, metric, confinement, pole, or detector test whose domain is not formed remains an explicit unexecuted obligation, rather than a passing or failing test. Once a domain is formed its test cannot be hidden by shortening or padding the source expression. Passing means that every currently formed common-ledger failure test is neutral. It does not mean that every counterfactual future continuation or target extension must pass.

Definition 3.8 (Semantic-use algebra). *The target-relative semantic-use algebra $\mathcal{U}_B(\Gamma)$ consists of the typed contextual tests $\hat{t} \circ C[-]$, their declared domain tests, and finite lawful compositions with admitted output maps. Equivalently it is generated by*

$$\mathcal{T}_B, \quad \text{Ctx}_B, \quad \text{definedness, restriction, continuation, and failure evaluation.}$$

A partial use has a well-defined descended domain; its totalized form retains $\perp_{\text{undef}}$. Undefinedness is not coerced into either an obstruction or a passing result.

Definition 3.9 (Complete-signature relation). *For $s, s' \in \mathcal{S}_B(\Gamma)$, set*

$$s \sim_{\text{pres}} s' \iff \text{Sig}_B(s) = \text{Sig}_B(s').$$

This is the semantic kernel relation itself. It is not merely the congruence generated by a hand-picked list of familiar presentation moves.

Lemma 3.10 (Named skins lie in the semantic kernel). *Gauge and local-presentation changes, basis and generator relabellings, normalization rescalings that preserve witnessed ratios, field-chart changes, lawful rebracketing, and source-record renaming preserve Sig_B whenever their source nonpromotion conditions hold. Hence every such move lies in $\sim_{\text{pres}}$.*

Proof. Each lawful move carries its full source-witness ledger by the stipulated intertwining maps, preserving primitive test values and their domains. Theorem 11.5 proves that transport of the incident presentation maps leaves each fixed intrinsic operation unchanged, including two-sided domain preservation with fixed sibling inputs. This is stronger than simultaneous covariance of all inputs. Induction on one-hole contexts therefore preserves definedness, boundary trace, admitted continuation, and failure values. Fixed-domain operator identity is representation invariant, and a representative-preserving transformation changes no admitted consequence functional [4, Thm. 12.1 and Lem. 15.1]. Therefore every coordinate of the future-behavior signature agrees. $\square$

Theorem 3.11 (Complete semantic-use factorization). *Every totalized $F \in \mathcal{U}_B(\Gamma)$ is constant on $\sim_{\text{pres}}$ -classes. For a partial use both its domain and its values descend. There is consequently a unique function*

$$\overline{F} : \mathcal{S}_B(\Gamma) / \sim_{\text{pres}} \longrightarrow Y_F \quad \text{with} \quad F = \overline{F} \circ q_B.$$

Conversely, if an alleged same-scope standing discriminator fails to factor, there are $s \sim_{\text{pres}} s'$ with $F(s) \neq F(s')$; the pair is a finite certificate that F changes the target interface.

Proof. Every basic test, its domain, and its application after every finite one-hole context occur in Sig_B . Context substitution represents further lawful composition, including fixed auxiliary inputs to mixed constructors. Equality of complete signatures therefore preserves both definedness and value after every licensed composite. The quotient universal property gives existence and uniqueness. The converse is the finite witness for nonfactorization in the fixed-domain theorem [4, Lemmas 4.1–4.2 and Thm. 4.2]. $\square$

3.4 Source deletion and semantic recoverability

Definition 3.12 (Exact presentation recovery under source deletion). *A specified deletion Δ declares the removed generator instances and the removed operation instances, including their structural parameters. Let $\mathcal{S}_B^{-\Delta}$ be the retained-signature term algebra generated by the retained generators under every retained licensed constructor on its original exact domain. Removing a generator symbol does not silently remove a distinct nullary constructor, a source-package projection, or a retained operation that can produce the same value. Conversely, such a constructor cannot silently retain a structural parameter that Δ expressly removes. The exact-presentation image is reduced precisely when*

$$\text{im}(\text{ev}_B \mid_{\mathcal{S}_B^{-\Delta}}) \subsetneq \text{im}(\text{ev}_B).$$

A proof requires an actual admitted semantic witness outside the reclosed image; deletion of one citation or one mnemonic symbol is insufficient. This is a statement about the evaluated presentation image, not yet about distinguishability by the fixed target uses.

Definition 3.13 (Target-relative source recoverability). *Keep the original target tests and their complete contexts fixed. For each output sort b , write*

$$V_{B,b} = \text{im}(\text{Sig}_B \mid_{\mathcal{S}_{B,b}}), \quad V_{B,b}^{-\Delta} = \text{im}(\text{Sig}_B \mid_{\mathcal{S}_{B,b}^{-\Delta}}).$$

The specified deletion is observationally recoverable when $V_{B,b}^{-\Delta} = V_{B,b}$ for every b ; it loses a standing use when the inclusion is proper for some b . This definition includes context and probe definedness, null outcomes, and diagnostic values. The contexts are observation tests, not additional generators available to the reclosed source algebra.

Theorem 3.14 (Contextual reconstruction under a specified deletion). *Suppose every removed generator has a retained same-sort source expression with equal complete signature. Suppose also that every removed operation has a retained same-profile expression whose definedness and output signature agree on every well-profiled input tuple, including tuples on which the original operation is undefined. Then the deletion is observationally recoverable. An explicit choice of these replacements defines a translation T_Δ of all original source terms satisfying*

$$T_\Delta(s) \in \mathcal{S}_B^{-\Delta}, \quad \text{Sig}_B(T_\Delta(s)) = \text{Sig}_B(s).$$

No new target anchor, witness value, or source assumption may be inserted by a replacement.

Proof. Translate a term by induction on its finite constructor tree. Use the specified retained expression at a removed leaf. At an internal node, translate its inputs and then use either its retained constructor or the specified operation replacement. Complete contextual equality permits substitution in each input separately: place the corresponding one-hole context around that input while fixing the others. The independent domain tests preserve executability, and arbitrary enclosing contexts preserve every output test. Repeating this argument over the inputs gives the claimed signature equality. All translated expressions use retained operations and parameters, so belong to the fully reclosed retained algebra. Every original signature therefore has a retained representative; the reverse inclusion follows because retained terms are original terms. $\square$

Equality of exact-presentation images implies observational recoverability. A proper subimage, however, need not lose a standing distinction: a retained presentation may have the same complete signature as a removed one. Observational loss requires an actual signature value outside $V_{B,b}^{-\Delta}$, not merely an absent source label or diagram shape.

Lemma 3.15 (A sufficient separation test). *Suppose a semantic predicate P holds of every retained generator value and is preserved by every retained constructor. If an actual original generator value ξ satisfies $\neg P(\xi)$, then ξ is absent from the reclosed image after deletion.*

Proof. Induction over the retained syntax shows that every retained term evaluates to a value satisfying P . Therefore no retained term evaluates to ξ . A port-absence invariant is one instance only when *all* remaining constructors preserve that absence; a constructor that creates the port invalidates the argument. $\square$

The lemma proves exact-presentation separation as stated. It proves observational separation when P is a predicate of the complete signature and its preservation holds for the full retained closure. A primitive-profile count that an admitted use cannot detect satisfies neither requirement merely by being a syntactic invariant.

Theorem 3.16 (Signature deletion and semantic recoverability). *Let $X = \text{im Sig}_B$, and let $p_J : X \rightarrow X_J$ delete a specified set of signature coordinates. The following are equivalent:*

- (i) *Every deleted coordinate, including its definedness status, is recoverable from the retained coordinates on X .*
- (ii) *p_J is injective.*
- (iii) *There is a reconstruction $r : p_J(X) \rightarrow X$ with $r \circ p_J = \text{id}_X$.*

If these conditions fail, a pair of actual source terms with equal retained signatures and different complete signatures witnesses the loss. If they hold, every admitted use still factors through the reduced signature.

Proof. Coordinatewise recovery gives equality of complete signatures whenever the retained coordinates agree, hence injectivity. An injective map onto its image has a unique inverse, yielding r ; its coordinates recover every deleted field. Failure of injectivity supplies two signature values in X , hence two source terms, separated by a deleted coordinate. Under injectivity, compose complete-signature factorization with r . $\square$

Signature coordinates need not be independent: the root may determine its kind, Γ is constant in a fixed global-form fibre, and contextual tests may contain duplicate information. Essentiality requires an actual original signature, including definedness, outside the fully reclosed retained image; recoverability requires a reconstruction. An incomplete submitted record alone establishes neither separation nor essentiality. Deleting a fixed target anchor changes the problem. For a production with no demonstrated inhabitant, a deletion claim remains a conditional separation statement and does not establish that the production is live.

Scope 3.17 (Source coverage and semantic factorization). *Source coverage and complete contextual factorization hold for the independently defined standing-semantic response class. Source and signature deletions are governed by reclosure and semantic recoverability; essentiality requires an individually witnessed loss. These results are independent of activity reduction and response cardinality.*

Counterexample criterion 3.18 (Source and semantic-use failure conditions). *The result fails if a semantic candidate is defined by source-tag membership; if a generator omits its theorem instantiation or antecedent witness; if a standing-semantic atom lacks its positive, obstruction, or conditional certificate; if an unavailable antecedent is recoded as a proved failure; if a many-input diagram is replaced by an untyped word; if theorem ancestry is included in the standing signature; if $\sim_{\text{pres}}$ is only a non-complete list of familiar moves; or if a deletion claimed essential is recoverable by the closed remaining algebra or by the retained signature coordinates.*

4 Activity reduction and the semantic quotient

4.1 Reduction before identification

Activity measures standing behavior independently of presentation size. For each port sort and fixed anchor interface, write $\mathbf{0}_b$ for the interface-only comparison diagram: it retains that interface but asserts no additional response, obstruction, or readiness witness. It is a null *presentation* for comparison, not a claim that the nondegenerate physical target or its carrier is absent.

Definition 4.1 (Activity predicate). *For a source term s over the fixed interface of sort b , set*

$$\text{Act}_B(s) = 1 \iff \text{Sig}_B(s) \neq \text{Sig}_B(\mathbf{0}_b).$$

The comparison uses all typed one-hole contexts and their domain tests. Thus a presently silent use remains active if a licensed context can distinguish it from the interface-only diagram. Numerical zero is not nullity: an actual neutral-obstruction witness or a readiness witness has a defined evaluation domain and can differ from the absence of that witness.

Definition 4.2 (Positive and failure strata). *Let*

$$\begin{aligned} \mathcal{S}_{B,\text{act}}^+(\Gamma) &= \{s : \text{Act}_B(s) = 1, \text{Fail}_B(s) = \text{pass}\}, \\ \mathcal{S}_{B,\text{act}}^\times(\Gamma) &= \{s : \text{Act}_B(s) = 1, \text{Fail}_B(s) \neq \text{pass}\}. \end{aligned}$$

Then

$$\text{Act}(\mathcal{S}_B)(\Gamma) = \mathcal{S}_{B,\text{act}}^+(\Gamma) \sqcup \mathcal{S}_{B,\text{act}}^\times(\Gamma).$$

The second stratum retains actual current-domain obstruction witnesses. The first does not certify unexecuted downstream tests or every possible future continuation.

Lemma 4.3 (Activity is semantic). *If $s \sim_{\text{pres}} s'$, then*

$$\text{Act}_B(s) = \text{Act}_B(s'), \quad \text{Fail}_B(s) = \text{Fail}_B(s').$$

Hence an active term is not identified with a null term, and a currently passing term is not identified with one carrying a certified current failure.

Proof. The terms have the same anchor interface and complete signature, so their comparisons with the same $\mathbf{0}_b$ agree. The individual failure values and their formed-domain predicates are signature coordinates; hence their derived aggregate failure records agree as well. Syntactic term height and the presence of redundant vertices play no role. $\square$

A deletion from a finite diagram is *semantically inert* when it leaves a well-typed source diagram over the same interface and preserves the complete contextual signature. A sufficient certificate is that the deleted data have no dependency into any retained test or operation-domain port, and that this noninterference persists under every licensed restriction, continuation, and grafting. Being disconnected in one drawing alone does not supply this certificate.

Let $\text{Core}(s)$ be any diagram obtained by repeated certified inert deletions, and let $\text{Disc}(s)$ record the deleted presentation data. Every strict deletion lowers the finite number of vertices, so this reduction terminates. A unique reduced syntax is neither assumed nor needed.

Proposition 4.4 (Activity reduction preserves every licensed use). *Replacing s by $\text{Core}(s)$ preserves the definedness and value of every $F \in \mathcal{U}_B(\Gamma)$, as well as activity and the failure stratum. All such reductions give the same complete-signature class. If deleting a vertex changes any licensed contextual use or its domain, that deletion is not permitted.*

Proof. Each certified step preserves Sig_B . Finite composition of the steps does likewise, and semantic-use factorization preserves every use. Lemma 4.3 preserves activity and failure. Two reduction sequences have the signature of their common input, hence the same quotient class. A changed contextual use contradicts the definition of an inert deletion. $\square$

4.2 The quotient on objects and operations

Definition 4.5 (Activity-reduced response quotient). *For each peer root kind $b \in \mathfrak{B}_B$ and stratum $\epsilon \in \{+, \times\}$, define*

$$\mathcal{Q}_{B,b}^\epsilon(\Gamma) := \mathcal{S}_{B,\text{act},b}^\epsilon(\Gamma) / \sim_{\text{pres}}, \quad q_{B,b}^\epsilon : \mathcal{S}_{B,\text{act},b}^\epsilon \twoheadrightarrow \mathcal{Q}_{B,b}^\epsilon.$$

Set

$$\mathcal{Q}_B^\epsilon(\Gamma) = \coprod_{b \in \mathfrak{B}_B} \mathcal{Q}_{B,b}^\epsilon(\Gamma), \quad \mathcal{Q}_B(\Gamma) = \mathcal{Q}_B^+(\Gamma) \sqcup \mathcal{Q}_B^\times(\Gamma).$$

The same quotient construction is used for an auxiliary port sort whenever a mixed constructor has that output type. The full active port quotient is

$$\mathcal{Q}_B^{\text{port}}(\Gamma) := \coprod_{a \in \mathfrak{S}_B} (\mathcal{Q}_{B,a}^+(\Gamma) \sqcup \mathcal{Q}_{B,a}^\times(\Gamma)).$$

Its distinguished root part is exactly $\mathcal{Q}_B(\Gamma)$. An auxiliary fibre records an incidence or interface, not another peer carrier. Write $q_B^{\text{port}} : \text{Act}(\mathcal{S}_B) \twoheadrightarrow \mathcal{Q}_B^{\text{port}}$ for the sortwise projection; its root restrictions are the already defined $q_{B,b}^\epsilon$. No component cardinality enters the definition.

The active part is not assumed closed under every constructor: for example, a lawful composite may have only interface-level behavior. Consequently $\mathcal{Q}_B^{\text{port}}$ carries *partial* operations. An operation on active inputs is retained exactly when its source operation is defined and its output is active. The full source quotient, including any semantically null classes, remains available for evaluating composites that pass through a null intermediate presentation. Removing null presentations from the active response inventory does not give them a physical response status.

For a fixed operation profile, define $\omega \sim_{\text{op}} \omega'$ when every same-profile semantic input tuple gives the same operation-definedness value and, when defined, complete-signature-equivalent outputs. Write

$$\overline{\mathcal{O}}_B(b_1, \dots, b_n; b) := \mathcal{O}_B(b_1, \dots, b_n; b) / \sim_{\text{op}}.$$

This relation compares operations, not source-proof names.

Theorem 4.6 (Complete congruence). *Suppose $s_i \sim_{\text{pres}} s'_i$ for $1 \leq i \leq n$. Then $\omega(s_1, \dots, s_n)$ is defined if and only if $\omega(s'_1, \dots, s'_n)$ is defined. When defined, their complete signatures agree. Their output activity and failure strata therefore agree, and*

$$\overline{\omega}([s_1], \dots, [s_n]) := [\omega(s_1, \dots, s_n)]$$

is well-defined on its active-output domain. The assertion also remains true when ω is replaced by an equivalent operation. Thus objects, operation domains, and many-input composition descend to the partial typed response system.

Proof. Replace inputs one at a time. With the other inputs fixed, the expression

$$C_i[-] = \omega(s'_1, \dots, s'_{i-1}, [-], s_{i+1}, \dots, s_n)$$

is one of the well-profiled one-hole contexts of Definition 3.6. Its domain test is a signature coordinate. Therefore $s_i \sim_{\text{pres}} s'_i$ preserves the definedness of this replacement. For any later context $D[-]$ and

target test t , the composed context $D[C_i[-]]$ is again in the complete context family. Equality of input signatures consequently preserves $\hat{t}(D[C_i[-]])$. After all replacements, every output-context coordinate agrees. This proves both definedness equivalence and equality of complete output signatures.

Activity and failure preservation follow from Lemma 4.3. Replacing the operation itself uses $\sim_{\text{op}}$, followed by the same contextual argument. Identities, associative grafting on common defined domains, and lawful input permutations descend because they already hold for the licensed source operations. No assumption that unary signatures determine mixed behavior is needed. $\square$

Corollary 4.7 (All semantic evaluators descend). *Kind, root, mechanism, standing effect, boundary, continuation, current failure, guard, readiness, global form, and later signature-defined eligibility predicates have unique representative-independent values and domains on the corresponding quotient fibres.*

Proof. Apply Theorem 3.11 to the totalized evaluator and its domain test, then restrict to the activity strata. $\square$

Theorem 4.8 (Universal response quotient). *Let X be a partial many-sorted target algebra on the same port profiles, and let $F : \text{Act}(\mathcal{S}_B) \rightarrow X$ be constant on complete-signature classes and preserve every licensed operation whose output is active. There is a unique map $\bar{F} : \mathcal{Q}_B^{\text{port}} \rightarrow X$ satisfying $F = \bar{F} \circ q_B^{\text{port}}$ and preserving those partial operations.*

Proof. On each sort and stratum the quotient universal property supplies the unique factor. The congruence theorem makes operation domains representative-independent. Applying the constructor-preservation identity for F to representatives shows that the factors preserve each defined active-output operation. Surjectivity of the sortwise quotient maps gives uniqueness. $\square$

4.3 Structural properties of the quotient

Theorem 4.9 (Activity and quotient finality). *For every $\Gamma \in \mathfrak{G}_{\text{adm}}$:*

- (i) *semantic-null presentations are excluded from the active response inventory, and only certified inert presentation data are deleted;*
- (ii) *every active positive and failure presentation occurs in exactly one complete-signature class;*
- (iii) *every source constructor, including product and Higgs many-input operations, has representative-independent definedness and descends on its active-output domain; and*
- (iv) *every same-target standing use and its domain factor uniquely through the corresponding quotient.*

None of these assertions inspects $|\mathcal{Q}_{B,b}^e|$, $|\pi_0 \mathcal{Q}_B|$, a live residue, or an assignment orbit.

Proof. The first three claims follow from Definition 4.1, Proposition 4.4, Definition 4.5, and Theorem 4.6. The final claim is Theorem 3.11 restricted to the quotient fibres. The named cardinalities are not arguments of any definition or proof. $\square$

Scope 4.10 (Contextual quotient structure). *The activity-reduced contextual quotient retains partial operation domains and mixed composition. Its operations are restricted to their active-output domains, while the full source quotient remains available for null intermediate values. These properties are independent of component cardinalities.*

Counterexample criterion 4.11 (Activity and quotient failure conditions). *Theorem 4.9 fails if numerical zero is confused with absence of a witness; if drawing-level disconnectedness licenses deletion despite a contextual effect; if source padding changes the current failure tests; if a mixed context or its*

domain test is omitted; if only objects but not operation domains descend; if active outputs are assumed for every composite; or if a component count is consulted before the live assignment object exists.

5 Well-definedness and quotient-level eligibility

5.1 Well-definedness of the response comparison

Activation certifies that the response comparison is well-defined independently of compatibility or occupancy. It is distinct from the existence of a physical boson.

Definition 5.1 (BCRS-I activation certificate). *For a fixed Γ , let*

$$\mathbf{A}_B(\Gamma) = (\text{TargetFinal}, \text{AntecedentsInstantiated}, \text{ExternalSemanticClass}, \text{SourceCoverage}, \\ \text{UseComplete}, \text{ActivityComplete}, \text{KernelCongruence}, \text{OperationDescent}, \\ \text{ContinuationComplete}, \text{FailureComplete}, \text{MutationSoundness}).$$

The value is Positive iff all eleven fields are proved in the same target and global-form fibre. The last field requires the reclosure and recoverability criterion and a valid witness for each deletion verdict actually asserted. It does not assert that every production is inhabited, every production is indispensable, or every signature coordinate is independent. A missing field returns NotReady; later compatibility, cardinality, or physical success cannot fill it.

Theorem 5.2 (Well-definedness of the response comparison). *For every $\Gamma \in \mathfrak{G}_{\text{adm}}$,*

$$\mathbf{A}_B(\Gamma) = \text{Positive}.$$

Hence $\mathcal{Q}_B^+(\Gamma)$ and $\mathcal{Q}_B^\times(\Gamma)$ are the source-covered, activity-reduced root fibres of the contextual quotient $\mathcal{Q}_B^{\text{port}}$, with representative-independent partial constructor domains and all target tests, continuation values, and failure diagnoses defined on their declared domains before eligibility or compatibility is evaluated. Unexecuted downstream obligations remain explicitly conditional.

Proof. Target finality is Theorem 1.2; full source antecedents are instantiated by Theorem 1.7; and the registry-independent semantic class is Theorem 2.7. Source coverage and semantic-use factorization are Theorems 3.5 and 3.11. Activity preservation, complete-kernel congruence, and operation descent are proved in Section 4. Continuation and failure are coordinates of Sig_B , so they descend by Corollary 4.7. Finally, Theorem 3.16 gives the exact signature recoverability criterion, and Lemma 3.15 gives a sufficient source-deletion separator after reclosure. No unwitnessed claim that every listed field is independently essential enters the activation certificate. Every cited result precedes eligibility and compatibility in (1.1). $\square$

Corollary 5.3 (No downstream activation repair). *If a required field of $\mathbf{A}_B(\Gamma)$ remains unproved, no product match, role assignment, observed particle display, vacuum reduction, pole theorem, confinement result, or numerical fit restores Positive in the same target fibre.*

Proof. The missing field is an earlier obligation in (1.1). A same-target proof, including an equivalent source certificate, can discharge it; source-proof names are not semantic coordinates. A genuinely new test changes the target, whereas treating a later successful display as proof of the missing earlier obligation reverses the dependency. The latter is not a discharge of that obligation. $\square$

5.2 Eligibility is defined on quotient classes

Eligibility uses the role index R_g^0 fixed by the three imported gauge roots in Definition 2.1.

Definition 5.4 (Quotient-level pre-eligibility). *For $X \in \mathcal{Q}_{B,b}^+(\Gamma)$ and $r \in R_g^0$, write*

$$E_B^0(X, r)$$

iff all of the following hold:

- (i) *the sort b and root coordinate of X map to r under $\mathbf{rt}_g$;*
- (ii) *the complete signature of X contains every response, support, guard, boundary, continuation, and readiness port required by the kindwise packet for r ;*
- (iii) *every such coordinate is an admitted structural use at this target; its source certificate is supplied by Theorem 3.5 and is not an additional provenance coordinate in the eligibility predicate; and*
- (iv) *no failure coordinate of X is incompatible with occupation of r .*

The predicate is evaluated directly on complete-signature quotient classes; no further application of q_B or $\sim_{\text{pres}}$ is involved.

Lemma 5.5 (Representative-independent eligibility). *For an active positive source presentation s and a role r ,*

$$E_B^0(q_B s, r)$$

depends only on $[s]_{\text{pres}}$. Equivalently, if $[s]_{\text{pres}} = [s']_{\text{pres}}$, then

$$E_B^0([s]_{\text{pres}}, r) \iff E_B^0([s']_{\text{pres}}, r).$$

Proof. Each eligibility clause is a descended coordinate or predicate of Sig_B . Equality in $\mathcal{Q}_B$ is equality of complete signatures, so all four truth values agree. $\square$

Lemma 5.6 (No role occupancy from the index alone). *The facts that R_g^0 has three tags and that the imported carrier root set has three gauge members imply neither:*

$$\forall r \exists X E_B^0(X, r), \quad \forall r \exists! X E_B^0(X, r), \quad \text{nor} \quad \exists (X_r)_r J_B((X_r)_r).$$

Proof. The first statement requires the three incidence packets, the second also requires exclusion of duplicate eligible classes, and the third requires one simultaneous common-ledger witness. None is part of a finite index set. Role-Occupancy Closure explicitly separates role demand, eligibility, compatibility, live residue, assignments, and assignment orbits [8]. $\square$

5.3 Prerequisites for joint compatibility

Joint compatibility requires the electromagnetic-readiness, color, weak, and Higgs packets in addition to the semantic class, source algebra, activity quotient, and pre-eligibility predicate. These packets support the simultaneous predicate J_B , the live residue, the lawful assignment object, and its automorphism quotient. Accordingly, the cardinalities

$$|\mathcal{Q}_{B,b}^+|, \quad |\pi_0 \mathcal{Q}_B|, \quad |\text{Live}_r|, \quad |\text{Assign}_B / \text{Aut}_B|$$

belong to the subsequent joint-compatibility analysis, not to activation or pre-eligibility.

Scope 5.7 (Well-definedness and eligibility). *The positive activation certificate establishes the well-definedness of quotient-level pre-eligibility. Component and occupancy counts require the kindwise packets and one common compatibility ledger.*

Counterexample criterion 5.8 (Activation and eligibility failure conditions). *Activation fails if an antecedent is merely named rather than instantiated; if source coverage is relative only to registry members or admits arbitrary typed values without structural witnesses; if continuation or failure is appended after quotienting; if deletion is claimed essential without reclosing the remaining algebra or without testing semantic recoverability; if eligibility is applied to the wrong domain; if the three role tags are treated as three live occupants; or if any count is inspected before the assignment orbit is formed.*

6 Representative nonpromotion and network schemas

The semantic quotient identifies presentations with the same complete contextual behavior. A display coordinate contributes independent standing only through a source-certified distinction. We establish this nonpromotion result and define the product-network and higher-arity schemas whose dispositions depend on an actual common assembly.

6.1 Displays and source-certified standing

Definition 6.1 (Display family). *Let $\mathfrak{D}_B(\Gamma)$ be the typed family of gauge potentials, local field components, Lie-algebra bases and indices, paired normalization coordinates, polarization bases, propagator displays, gauge-fixing variables, Goldstone coordinates, scalar charts, particle and resonance names, and detector labels that can occur in descriptions of the fixed BCRS-I target. Membership in $\mathfrak{D}_B(\Gamma)$ records a possible representation of structural data; it is not a source certificate.*

Definition 6.2 (Independent standing and representation). *For $d \in \mathfrak{D}_B(\Gamma)$, write $\text{SrcStand}_B(d)$ when an instantiated interface $\mathcal{E}_i = (j_i, w_i, \rho_i, \text{NoProm}_i)$ supplies a source term s_d whose complete signature contains a standing-effect, boundary, continuation, or failure coordinate that is not obtained by forgetting d from an already certified term.*

For $X \in \mathcal{Q}_B(\Gamma)$, write $\text{Rep}_B(d, X)$ when there is an active source term s with $q_B(s) = X$ and a display realization of s using d , related to the display-free realization by an isomorphism that commutes with every port of Sig_B . Thus Rep_B is a partial relation; no map from all displays to response classes is presumed.

Theorem 6.3 (No representative promotion). *Each submitted use of $d \in \mathfrak{D}_B(\Gamma)$ has one of the following priority-disjoint logical statuses.*

- (i) *If $\text{SrcStand}_B(d)$, then d is not a mere display. Its certified term enters the source algebra and is subjected to activity reduction and the semantic quotient. Its standing content is evaluated in the fixed target; no new class follows from visual distinctness. A proposed coordinate outside that target instead requires a successor problem.*
- (ii) *If $\neg \text{SrcStand}_B(d)$ and $\text{Rep}_B(d, X)$ for some X , then each particular witness of $\text{Rep}_B(d, X)$ descends to its associated existing class X . The same display convention may represent several different classes; no function from the bare display d to a unique class is asserted.*
- (iii) *If $\neg \text{SrcStand}_B(d)$ and no such X exists, then d supplies no active response object.*

This is a partition of standing claims, not a decision procedure for finding missing evidence. In particular, no display-only distinction creates an additional member of $\mathcal{Q}_B^+(\Gamma)$ or $\mathcal{Q}_B^\times(\Gamma)$.

Proof. First test the source condition. If it holds, the standing claim is carried by the instantiated theorem interface, not by the display, and the ordinary source–activity–quotient chain applies. If it does not hold, test the partial representation relation. A represented display commutes with all signature ports, hence its realizing term is equivalent under $\sim_{\text{pres}}$ to the display-free realization and has the same quotient class. A submitted display satisfying neither condition supplies no certified response witness distinguishing it from the interface-only comparison of Definition 4.1. The ordered tests are exhaustive and disjoint. In none of the three cases is a new quotient class inferred from the display datum alone. $\square$

Corollary 6.4 (Dimension and multiplicity firewall). *Lie-algebra dimension, an adjoint-index range, representation multiplicity, the dimension of a later polarization space, and the number of nonzero coordinate components are not response-role cardinalities.*

Proof. Each is either an internal coordinate of a certified mechanism, a representation index, or a downstream projection datum. If it represents a fixed response it falls under Theorem 6.3(ii); if it carries a genuinely new certified standing coordinate it falls under (i) and must pass through the source algebra and quotient. Counting its displayed values never bypasses that alternative. $\square$

6.2 Display transformations and their semantic action

Definition 6.5 (Primitive display mutation). *A primitive display mutation is one of the following typed transformations, with the fixed root and global-form fibre and with an invertible, source-licensed intertwiner of the full instantiated structural ledger. The intertwiner carries admitted probes and their domains, commutes with primitive witness evaluation and restriction, and carries each common-interface preservation witness to the corresponding witness. Preserving the names or codomains of ports alone is insufficient. The transformation has one of these forms:*

- (a) *change of gauge chart or local trivialization induced by the fixed transition groupoid;*
- (b) *Lie-basis change or index relabeling that intertwines the bracket, representation action, and adjoint action;*
- (c) *paired generator/normalization reparametrization that leaves the represented action and holonomy fixed;*
- (d) *change of coordinates inside an already fixed conditional readiness space, without executing a polarization map;*
- (e) *gauge-fixing, Goldstone, or scalar-chart reparametrization over the same certified root and mechanism fibre; or*
- (f) *renaming of a particle, resonance, or detector display, which has no action on the BCRS-I source term.*

Let Δ_B^{disp} be the typed closure of these partial isomorphisms under lawful composition and inverse. Their primitive intertwining laws do not assume invariance under arbitrary future contexts; that consequence is proved below.

Lemma 6.6 (Primitive invariance). *For every primitive mutation δ and every source term s on which it is defined,*

$$\text{Sig}_B(\delta s) = \text{Sig}_B(s).$$

Proof. For (a), the transition and transport interfaces are natural under the fixed groupoid action, so root incidence, loop residue, boundary restriction, and continuation commute with the chart change [20, 21]. For (b), an intertwining basis isomorphism preserves the bracket, representation response,

adjoint response, anomaly expression, and every composite formed from them; this is precisely the representation-sensitive but presentation-invariant content retained from the gauge and pre-Standard-Model interfaces [23, 28, 30]. In (c), the represented action, loop evaluation, charge witnesses, anomaly aggregation, and shared-interface tests are carried by the paired intertwiner. An unpaired coupling change or a change of a witnessed ratio is not a member of this family. In (d), the conditional space and readiness map are fixed and only their coordinates change; no later polarization value is produced. In (e), the cited variables are charts on an already typed mechanism fibre; the root, order/deformation/fixation maps, and all interface ports commute with the reparametrization [40, 44]. Finally, (f) is outside the domain of every BCRS-I source evaluator, so its induced action on s is the identity. These source laws preserve each primitive test and its domain. To cover the complete signature, induct on a one-hole context. The identity hole has the primitive invariance just proved. At each grafting, naturality of the intertwiner carries the common-interface witnesses and preserves the constructor's definedness. The induction hypothesis preserves each retained input, so the primitive output evaluation and any final test agree. Thus all contextual signature coordinates, including undefinedness, are preserved. $\square$

Theorem 6.7 (Strong mutation invariance). *Every $\delta \in \Delta_B^{\text{disp}}$ induces the identity on its domain in $\mathcal{Q}_B^+(\Gamma) \sqcup \mathcal{Q}_B^\times(\Gamma)$. Any later construction defined functorially from complete signatures, rigid role tags, and common-interface predicates receives the induced identity on quotient data wherever δ is licensed; the represented construction consequently has the same isomorphism class.*

Proof. Lemma 6.6 is preserved under composition and inverse, so $\text{Sig}_B(\delta s) = \text{Sig}_B(s)$ for every $\delta \in \Delta_B^{\text{disp}}$. Hence $q_B(\delta s) = q_B(s)$. Substitution of identical quotient classes into a later signature-defined construction supplies the stated canonical automorphism. This conclusion uses the primitive source intertwiners and context induction. It is not inferred from unchanged field names, a common root alone, or matching codomains. $\square$

Proposition 6.8 (Negative controls). *A transformation that changes a certified charge or representation class, the color or weak root, the Higgs identity, the global-form datum Γ , a boundary trace, a continuation value, a failure value, or a mechanism tag is not in Δ_B^{disp} . Nor are physical vacuum execution, pole extraction, confinement success, numerical fitting, or detector response display mutations of the BCRS-I object.*

Proof. Every datum in the first sentence is an explicit coordinate of either $\tau_B(\Gamma)$ or Sig_B ; changing it therefore changes the target or the complete signature. The operations in the second sentence belong to successor targets in the dependency order (1.1). They do not define endomorphisms of the BCRS-I source algebra. These controls separate genuine semantic change from the six invariant presentation changes. $\square$

6.3 Network and higher-arity schemas

Let $X_Q, X_{\text{col}}, X_{\text{wk}}$ be quotient classes satisfying the kindwise pre-eligibility predicates of Definition 5.4. Pre-eligibility alone does not supply a simultaneous compatibility witness.

Definition 6.9 (Gauge product-network schema). *The gauge product-network schema on these three classes is*

$$\text{NetSch}_g(X_Q, X_{\text{col}}, X_{\text{wk}}) = (X_Q, X_{\text{col}}, X_{\text{wk}}; \mathcal{I}_{Q_{\text{col}}}^{\text{decl}}, \mathcal{I}_{Q_{\text{wk}}}^{\text{decl}}, \mathcal{I}_{\text{colwk}}^{\text{decl}}; \partial, \text{Cont}, \text{Fail}, \Gamma).$$

Each $\mathcal{I}_{ij}^{\text{decl}}$ is the typed domain on which a later common-interface comparison may be attempted. The schema asserts neither that a pairwise match exists nor that separately obtained pairwise witnesses glue to one triple witness. It contains exactly the three displayed response vertices.

Definition 6.10 (Higher-arity proposal schema). *A higher-arity proposal is a source-certified active quotient object or diagram with at least two response inputs, together with its complete signature and declared factorization interfaces. The definition does not require it either to factor through NetSch_g or to be independent of that schema. Its disposition is therefore not encoded in its name.*

Lemma 6.11 (No standing-bearing predicate coercion). *The BCRS-I source grammar contains no source-certified, standing-bearing coercion*

$$\{\text{witnesses of a compatibility predicate}\} \longrightarrow \mathcal{Q}_B^+(\Gamma)$$

whose value would be an additional gauge-role occupant.

Proof. Every generator with response codomain is indexed by an instantiated source interface and outputs a rooted effect/continuation signature (Definition 3.1). A compatibility witness instead has a dependent proposition as codomain and compares already formed quotient classes. No generator in the complete source-use algebra has that proposition as its domain and a new response root as its codomain. Thus no source-certified coercion of the displayed type exists. This is a statement about the typed grammar, not a denial that arbitrary underlying sets can be injected into one another. $\square$

Theorem 6.12 (Network schemas and assembly requirements). *The product-network and higher-arity proposal schemas are well-defined after source, signature, quotient, and pre-eligibility formation. Neither schema is a fourth gauge-role occupant. A coupled-network disposition requires an actual common ledger. Higher-arity claims require the verified factorization, separation, or obstruction witnesses specified in Definition 12.40. Fine gauge-assignment uniqueness is not a prerequisite for these assembly-relative constructions.*

Proof. Definitions 6.9 and 6.10 use only objects available at the current dependency rank. Lemma 6.11 excludes promotion of a later truth-valued match into a response vertex. Factorization through the three-vertex schema, independence from it, and obstruction are properties of one simultaneous ledger and the declared comparison interfaces. The schema alone supplies neither that ledger nor a factorization or separation witness. Section 12 constructs the assembly-relative dispositions without assuming a unique gauge assignment. $\square$

Theorem 6.13 (Fixed-authority no backflow). *At fixed $\tau_B(\Gamma)$ and $\mathbb{I}_B(\Gamma)$, product notation, compatibility success, physical vacuum realization, polarization, pole extraction, confinement, numerical fitting, and detector response cannot create, delete, or identify a source, semantic use, active class, or role demand at an earlier dependency rank.*

Proof. An invariant presentation change acts trivially by Theorem 6.7. A change to a target or signature coordinate is caught by Proposition 6.8 and belongs to a re-instantiated or successor problem. A later operation is not an endomorphism of the earlier object. These exhaust the typed ways in which the listed data could be proposed as inputs. $\square$

Scope 6.14 (Representation and network scope). *The nonpromotion theorem applies to the declared display transformations and distinguishes them from genuine structural changes. The three-vertex product-network and higher-arity schemas require an actual common assembly and verified comparison witnesses for their disposition; product notation alone supplies no additional carrier.*

Counterexample criterion 6.15 (Promotion and network failure conditions). *The result fails if a potential, generator, index, polarization vector, propagator, gauge-fixing field, Goldstone coordinate, scalar chart, or pole label acquires standing without a certified source term; if an unpaired coupling or global-form change is called a display mutation; if a compatibility proposition is coerced into a response occupant; if marginal pairwise witnesses from different Γ -fibres are spliced; or if a product or higher-arity disposition is asserted without its common assembly and required comparison witnesses.*

7 Charge response and structural residual transport

The charge-response incidence packet is defined at the fixed premetric target. Its tag r_Q identifies the response position, independently of simultaneous occupation, exhaustion of the response arena, or cardinality. Those questions require the separate packet loci to be joined over one common witness ledger.

7.1 Exact source scope and inherited antecedents

Several gauge results used below assume a fixed realized physical interior. This source hypothesis is retained in full; it denotes neither a smooth metric background nor a particle projection.

Definition 7.1 (Target-relative restriction of an imported theorem). *Write an imported theorem as a dependent implication*

$$T : H_T \Longrightarrow C_T,$$

where H_T is its full hypothesis record and C_T its conclusion record. The target-relative restriction $\text{pr}_{\text{pre}} T$ retains exactly the coordinates of C_T whose structural reduct is typed in $\mathcal{R}_B^{\text{sem}}(\Gamma)$, together with the complete dependency map from H_T . If a retained coordinate depends on a later vacuum, metric, spectral, or particle antecedent h , the restriction retains the conditional arrow $h \rightarrow c$, not an inhabited value of c .

Thus restriction may forget a display coordinate, but it cannot erase an antecedent or turn a conditional result into an unconditional one. Every retained coordinate is evaluated by the same target-relative semantic map and descends through the appropriate $q_{B,b}^\epsilon$ only after it occurs in a certified active source term of sort b and stratum ϵ .

Definition 7.2 (Fixed-interior gauge antecedent ledger). *For a domain D and fixed global-form datum Γ , let $\text{GAnte}_{\leq 5}(D, \Gamma)$ denote the following source-exact antecedent ledger.*

- (i) *D is a fixed admissible realized physical interior obtained from an inhabited RPICert_D , with induced RPI_D , and the instantiated A1 interface $\mathcal{E}_{A1} = (j_{A1}, w_{A1}, \rho_{A1}, \text{NoProm}_{A1})$ lies in the common import fibre $\mathbb{I}_B(\Gamma)$. Every source-specific A1 field is read from the full witness w_{A1} , and Γ is not varied during the argument.*
- (ii) *Local presentations are admitted on the continuation site and their transition arrows satisfy the A1 predicate Preserve_R , including boundary fixation, restriction, and composition admissibility.*
- (iii) *Along admitted typed continuation morphisms, the A2 hypotheses hold: field-carrier mediation, particle-persistence identity, quotient comparison, bounded-transport witnesses, and boundary-trace fixation are preserved. The resulting transition groupoid layer and connection role are retained.*
- (iv) *Every loop datum used below is an admitted endpoint-closed finite continuation loop with an admitted endpoint-closing datum. A nonidentity loop output is retained only when it passes the A3 boundary-detectability, quotient, transport, role, endpoint, and same-domain tests.*
- (v) *The representation-bearing candidate has the A4 hypothesis that particle-persistence classes may be distinguished by admitted representation-sensitive boundary or classification witnesses.*
- (vi) *Every admitted anomaly probe is carried to one common anomaly-obstruction interface. A5 supplies the failure rule $\text{Anom}_{\partial D}(\alpha) \not\equiv_{\partial} 0_{\partial} \Rightarrow \text{Fail}$; it does not supply a local passing witness for each packet.*

These are the actual domains of Gauge A1–A5, in dependency order [20, Thm. 7.1] [21, Thm. 7.1] [22, Thm. 11.1] [23, Thm. 11.1] [24, Thm. 10.1]. The field-carrier and particle-persistence conditions in

clauses (iii) and (v) are retained source hypotheses at the role-ledger level; they are not reconstructed here from a particle display.

For a source term s_Q and its evaluated semantic response $\xi_Q \in \mathcal{R}_B^{\text{sem}}(\Gamma)$, write $\text{Ante}_Q(s_Q, \xi_Q; D, \Gamma)$ when $\text{GAnte}_{\leq 5}(D, \Gamma)$ holds and the exact PSM4 inputs are present: the A4 ledger $\text{ChWit}_{\partial D}$, a witness-preserving algebraic representation object $\mathcal{A}_{\partial D}$, a source-level PSM2 product-compatibility status $\text{ProdComp}_{\partial D}(I)$, a PSM1 Lie specialization, and an Abelian-refinement package satisfying all eleven clauses of PSM4 Theorem 14.1. Those clauses retain only witness-bearing Abelian response, quotient normalization skin, and mark anomaly and vacuum compatibility as downstream guards [29, Thms. 6.1, 7.1, and 14.1]. The evaluated response ξ_Q must carry the same-domain boundary and continuation trace and a witnessed typed incidence to r_Q . The source term is required to satisfy, in the same fibre,

$$s_Q \in \mathcal{S}_B(\Gamma) \text{ of sort } Q, \quad \text{ev}_B(s_Q) = \xi_Q, \quad \text{Act}_B(s_Q) = 1, \quad \text{Fail}_B(s_Q) = \text{pass},$$

Thus the whole certified failure ledger passes; absence of one incidence-specific failure is not used as a substitute. No vacuum baseline, residual ledger, photon datum, or pole datum is an antecedent of Ante_Q .

The antecedent conjunction is required to have one common instantiation h_Q over the same (D, Γ) . Thus the A1 output used by A2, the A2 output used by A3, the A4 ledger used by PSM4, and the evaluated source, boundary, continuation, incidence, and global failure fields are restrictions of one composable hypothesis witness. Separately inhabited theorem domains do not satisfy Ante_Q .

The charge-response chain licensed by these antecedents is

$$\text{Gpd}_{\partial}(D) \rightsquigarrow \nabla_{\partial D} \rightsquigarrow \Omega_{\partial D} \rightsquigarrow (\text{Rep}_{\partial D}, \text{ChWit}_{\partial D}) \rightsquigarrow \text{AbRef}_{\partial D}. \quad (7.1)$$

The arrows in (7.1) are dependencies, not additional carriers. Gauge A4 identifies charge equality with equality of all admitted response witnesses [23, Thm. 13.1]. PSM4 then licenses Abelian coordinates only as witness-preserving specializations and quotients every normalization change that leaves the witness ledger invariant.

Gauge A6 supplies a dependent residual interface. Its Theorem 10.1 requires an anomaly-neutral gauge context, a non-null support profile, and a full gauge-vacuum compatibility ledger; its Theorem 12.1 additionally requires an admissible vacuum-mediated reduction before a residual response ledger is produced. These are structural antecedents: the support profile of A6 Definition 5.2 contains no scalar energy, condensate, vacuum expectation value, Hilbert-state vacuum, or field configuration space. A6's residual ledger can therefore be evaluated at structural scope when those antecedents are supplied, without identifying a physical residual photon [25, Def. 5.2; Thms. 10.1 and 12.1; Cor. 13.5]. We denote the dependent interface by ResPort_Q . The base charge packet retains its exact domain without choosing an inhabitant. The residual extension below specifies this family and its common-source instance.

Definition 7.3 (Charge/EM-readiness incidence packet). *For $X_Q \in \mathcal{Q}_{B,Q}^+(\Gamma)$, define*

$$\mathcal{P}_Q(X_Q) = (X_Q; \text{Gpd}_{\partial}, \nabla_{\partial D}, \Omega_{\partial D}, \text{Rep}_{\partial D}, \text{ChWit}_{\partial D}, \text{AbRef}_{\partial D}; \text{AnomPort}_Q, \text{ResPort}_Q; \partial_Q, \text{Cont}_Q, \text{Fail}_Q).$$

It is a complete charge/EM-readiness packet when:

- (i) X_Q has kind Q , the source chain (7.1), and the typed incidence $X_Q \rightarrow r_Q$;
- (ii) every structural entry is the exact descended conclusion of its source theorem, and normalization-only distinctions are quotiented;
- (iii) AnomPort_Q is the restriction of the common A5 obstruction interface in the same (D, Γ) -fibre, while ResPort_Q retains the exact antecedent-dependent structural residual interface, without a selected argument;

- (iv) its boundary and continuation records are present, and its descended failure value is $\text{Fail}_Q = \text{pass}$.
- (v) there exist $s_Q \in \mathcal{S}_{B,\text{act},Q}^+(\Gamma)$, $\xi_Q = \text{ev}_B(s_Q)$, and one composable witness h_Q over the fixed (D, Γ) -fibre such that $q_{B,Q}^+(s_Q) = X_Q$, and every displayed mechanism, guard port, boundary, continuation, incidence, and failure field is the corresponding restriction of that same h_Q . Its theorem-interface projections instantiate exactly $\mathbf{GAnte}_{\leq 5}$ and the $\mathbf{PSM}_4$ antecedents above.

The packet contains no packet-local anomaly verdict, photon label, polarization, propagator, coupling value, vacuum coordinate, residual-sector occupant, or pole condition.

Define the packet locus and packet map by

$$\text{Pkt}_Q(D, \Gamma) := \{X \in \mathcal{Q}_{B,Q}^+(\Gamma) : \mathcal{P}_Q(X) \text{ is complete and } E_B^0(X, r_Q)\}, \quad \mathbf{I}_Q(D, \Gamma) := \mathcal{P}_Q|_{\text{Pkt}_Q(D, \Gamma)}.$$

Thus $\mathbf{I}_Q(D, \Gamma)$ is defined on all qualified classes in $\text{Pkt}_Q(D, \Gamma)$, without selecting a distinguished occupant.

7.2 The packet theorem

Lemma 7.4 (Exact charge-packet formation). *Let (s_Q, ξ_Q) satisfy $\mathbf{Ante}_Q(s_Q, \xi_Q; D, \Gamma)$. Then*

$$s_Q \in \mathcal{S}_{B,\text{act},Q}^+(\Gamma), \quad X_Q := q_{B,Q}^+(s_Q) \in \text{Pkt}_Q(D, \Gamma),$$

and the composite of the $A1$ – $A4$ and $\mathbf{PSM}_4$ certificates is the complete packet $\mathcal{P}_Q(X_Q)$. $A5$ contributes the common obstruction interface and its failure rule; $A6$ contributes the dependent residual interface, without an argument being selected by the base charge packet. In particular, $E_B^0(X_Q, r_Q)$.

Proof. $A1$ – $A3$ give the transition, transport, and admitted loop-residue entries. $A4$ gives the representation response and charge-witness quotient. The exact $\mathbf{PSM}_4$ antecedent package gives the witnessed Abelian refinement and removes normalization skin without solving either downstream guard. $A5$ supplies the obstruction map, not a manufactured zero. The certified active source term supplies the same-target boundary and continuation trace and the global failure value **pass**. $A6$ leaves the residual conclusion dependent on its non-null-support, compatibility, and reduction antecedents. These fields have matching typed interfaces and therefore form the asserted packet without a vacuum or particle representative. The common witness h_Q , witnessed role incidence, instantiated-interface supply, boundary/continuation trace, and global passing failure value prove the four clauses of E_B^0 . None is obtained merely by listing source generators. $\square$

Lemma 7.5 (Presentation-independent packet descent). *If $s, s' \in \mathcal{S}_{B,\text{act},Q}^+(\Gamma)$ are two source presentations of the same certified charge-response use and $s \sim_{\text{pres}} s'$, then $\mathcal{P}_Q(q_{B,Q}^+s)$ is complete if and only if $\mathcal{P}_Q(q_{B,Q}^+s')$ is complete. In that case the two packets agree as dependent records over $\mathcal{R}_B^{\text{sem}}(\Gamma)$.*

Proof. Every structural packet field and every eligibility value is a mechanism, incidence, guard-port, boundary, continuation, failure, or typed-open-port coordinate of the complete signature. Presentation equivalence preserves all of them, and semantic-use factorization makes their projections through $q_{B,Q}^+$ unique. The requirement that a coordinate be supplied by an instantiated interface is retained, but a paper name or provenance record is not a signature coordinate. A gauge, basis, normalization, or field display therefore cannot change packet completeness. $\square$

Theorem 7.6 (Charge/EM incidence-packet theorem). *For fixed D, Γ ,*

$$\text{Pkt}_Q(D, \Gamma) \neq \emptyset \iff \exists (s_Q, \xi_Q) \mathbf{Ante}_Q(s_Q, \xi_Q; D, \Gamma).$$

More precisely, every pair on the right determines the class $X_Q = q_{B,Q}^+(s_Q) \in \text{Pkt}_Q(D, \Gamma)$, and equivalent positive source presentations determine the same class. Conversely, every class in the packet locus has a positive certified source presentation satisfying the displayed antecedents. The theorem does not assert that those antecedents are inhabited in every abstract Γ -fibre.

Proof. The forward construction is Lemma 7.4. Its active same-domain evaluated response, boundary and continuation trace, witnessed role incidence, and global passing failure value place the term in $\mathcal{S}_{B,\text{act},Q}^+$ and give the four clauses of Definition 5.4. Presentation independence is Lemma 7.5. Conversely, completeness of a packet in Pkt_Q recovers instantiated A1–A4/PSM4 interfaces supplying the required semantic fields, the fixed (D, Γ) -fibre, active source/response record, and boundary, continuation, guard-port, and global failure fields required by Ante_Q . No later port is used in either direction. $\square$

Corollary 7.7 (Antecedent dependence of charge-packet existence). *If no pair (s_Q, ξ_Q) satisfies $\text{Ante}_Q(s_Q, \xi_Q; D, \Gamma)$, then $\text{Pkt}_Q(D, \Gamma) = \emptyset$. Source/use completeness and every later compatibility or occupation predicate remain predicates on already formed uses or classes; none may manufacture a charge packet in that fibre.*

Proof. The emptiness statement is the contrapositive direction of Theorem 7.6. A later predicate can restrict or relate existing classes but has no constructor whose codomain is the empty packet locus. $\square$

Corollary 7.8 (No photon by charge-response promotion). *No lawful source-display mutation turns an element of $\text{Pkt}_Q(D, \Gamma)$ into a photon or splits it into additional response-role occupants. Vacuum, metric, polarization, and pole data do not act on this packet locus at all.*

Proof. A lawful source-display mutation that fixes the complete signature fixes the packet. A change of a witnessed field is a different source use or a successor target. Later vacuum and particle data are outside the domain of the packet map, so neither class of operation constructs a photon. $\square$

7.3 Residual charge transport on the structural support ledger

The charge packet and its residual extension have different domains. Ante_Q suffices for the first; the second requires an actual reduction with its preservation witnesses. An evaluated structural support relation is distinct from a metric vacuum, a vacuum expectation value, or a pole condition, none of which supplies the missing reduction.

Definition 7.9 (Residual structural antecedent). *Let h_Q be a composable witness of $\text{Ante}_Q(s_Q, \xi_Q; D, \Gamma)$. The dependent record $\text{RAnt}_Q(h_Q)$ consists of the following data, all over the same (D, Γ) , boundary, and continuation ledger.*

- (i) *The complete A6 context Z_D , its non-null support profile V_D , and a passing gauge-vacuum compatibility ledger $\text{GVac}_{\partial D}$, restricting to the gauge and charge data of h_Q . In particular, anomaly neutrality is the verdict on this actual shared ledger, not an inference from the A5 failure rule alone.*
- (ii) *An actual reduction $\text{br}_{\partial D}$ with all the A6 admissibility witnesses: vacuum mediation, boundary detectability, quotient stability, same-domain preservation, anomaly neutrality, transport and loop-residue compatibility, particle-persistence preservation, lawful charge-witness preservation or reduction, deformation-baseline compatibility, and absence of hidden gauge or representative selection.*
- (iii) *The PSM1–PSM5 inputs and an instantiated $\text{ResRef}_{\partial D}$ satisfying all thirteen clauses of PSM6 Theorem 15.1. These include induction from the A6 residual ledger; probe-distinguishability after quotienting; transition, transport, and loop inheritance or lawful reduction; representation, charge,*

anomaly, product-status, and Lie-specialization preservation; non-null support; local finite witness-presentability; deformation-baseline preservation; no independent selector coordinates; and witness-minimal quotient normalization.

- (iv) *The SR6 survivor certificate for that same residual candidate, including every predicate of SR6 Definition 9.1 and the shared residual pullback of its Definition 10.3. Its SR2–SR5 projections, support, reduction, admitted probes, and residual response records therefore agree on one witness ledger; the residual charge projection inherits the SR4 charge witness used by h_Q .*

All source dependency maps are part of the record. No equality of distinct BCRS response signatures, photon label, physical vacuum value, or numerical coupling is a field of $\mathbf{RAnt}_Q(h_Q)$.

The full refinement and survivor records are necessary here. PSM6 compatibility alone is not SR6 survivability, and an SR6 residual label without a compatible reduction probe has no residual standing [31, Thm. 15.1] [37, Defs. 10.1–10.3; Thms. 12.1, 19.1 and 26.1].

Definition 7.10 (Structural residual charge packet). *For $h \in \mathbf{RAnt}_Q(h_Q)$, first evaluate $A6$'s actual residual ledger $\text{Res}_{\partial D}(\text{br}_{\partial D})$, and then the PSM6 normal form certified by h . Let*

$$\begin{aligned} \mathcal{R}_Q(h) := \text{pr}_{Q,\text{pre}}(\text{Res}_{\partial D}, \text{Gpd}_{\partial D}^{\text{res}}, \nabla_{\partial D}^{\text{res}}, \Omega_{\partial D}^{\text{res}}, \text{Rep}_{\partial D}^{\text{res}}, \text{ChWit}_{\partial D}^{\text{res}}, \\ \text{ResAnom}_{\partial D}, \text{ResProd}_{\partial D}, \text{VacSupp}, \text{Red}, \text{ProbeRedCompat}, \mu_D^0, \partial, \text{Cont}). \end{aligned} \quad (7.2)$$

Here ResProd retains the actual independent, coupled, or quotient-collapsed residual product status. The projection $\text{pr}_{Q,\text{pre}}$ retains the charge-bearing restriction and every support, reduction, probe, transition, and transport dependency needed for its standing. It does not discard the adjoining representation, anomaly, or product witnesses on which that charge restriction depends. Its complete source antecedent remains attached as in Definition 7.1.

The dependent residual interface and extended packet are

$$\text{ResPort}_Q(h_Q) : \mathbf{RAnt}_Q(h_Q) \longrightarrow \mathbf{RStr}_Q(h_Q), \quad h \longmapsto \mathcal{R}_Q(h),$$

$$\widehat{\mathcal{P}}_Q(X_Q; h) := (\mathcal{P}_Q(X_Q), \mathcal{R}_Q(h)), \quad X_Q = q_{B,Q}^+(s_Q).$$

The codomain $\mathbf{RStr}_Q(h_Q)$ is the structural record type in (7.2); it is not a new carrier sort. The base packet does not imply that $\mathbf{RAnt}_Q(h_Q)$ is inhabited.

The residual packet is an evaluated response ledger on the specified non-null structural support profile. It requires no metric execution, chosen vacuum vector, Higgs potential, or residual electromagnetic identification.

Theorem 7.11 (Source-induced residual charge comparison). *For every $h \in \mathbf{RAnt}_Q(h_Q)$, the extended packet is defined. Its charge restriction carries the residual transport-comparison assignment*

$$\nabla_Q^{\text{res},h} : \mathbf{K}_{\text{res}}(h) \longrightarrow \mathbf{Comp}_{Q,\partial}^{\text{res}}(h), \quad \kappa \longmapsto \text{pr}_{Q,\text{pre}}[\Theta_{\kappa,h}^{\text{res}}]_{\partial,\text{res}}. \quad (7.3)$$

Here $\mathbf{K}_{\text{res}}(h)$ is exactly the admitted residual continuation domain in h , and the codomain consists of its boundary-fixed residual comparison classes with their typed endpoints. This assignment has identity normalization and restriction compatibility, preserves the reduced charge-witness and boundary ledgers under the admitted comparison, and inherits weak path-composition compatibility wherever the source admits the concatenated path and both comparison expressions preserve the same restricted role-data tuple. Any loop-residue reduction is recorded by the same admitted reduction witness. None of these assertions supplies strict functoriality, path independence, flatness, or a differential transport equation.

Proof. A6 Theorem 12.1 evaluates the actual admissible reduction in h and produces its residual response ledger. PSM6 Theorem 15.1 applies because h contains its complete refinement record, not only a vacuum-support claim. Its residual sector components are the surviving or lawfully reduced transition, transport, loop, representation, charge, and anomaly ledgers of Definition 4.1. Consequently the charge-bearing component and all its dependencies in (7.2) exist [31, Def. 4.1; Thms. 8.1, 11.1 and 15.1].

For each admitted κ , take the comparison class already carried by that residual transport ledger and apply its charge restriction. The reduced charge witness is induced by the same reduction and is quotient-stable; SR6 charge inheritance and the shared residual pullback ensure that it is compared on the same support and continuation ledger, not against a separately selected residual charge convention [37, Thms. 10.4 and 19.1]. This is the assignment (7.3). Identity, restriction, boundary, and comparison preservation are the inherited A1–A2 properties of the PSM6 residual transport ledger. The inherited composition law is only A2’s weak comparison-class law on its exact admitted domain [21, Def. 3.7; Prop. 6.10 and Rem. 6.11]. Projection preserves that law and does not enlarge its domain. PSM6 Section 8.3 requires a witnessed residual ledger for every lawful loop-residue trivialization or transformation, so projection cannot erase the corresponding witness. Thus all retained comparisons are actual source values, with exactly the claimed preservation properties. $\square$

Lemma 7.12 (Descent of residual evaluation). *Two lawful presentations of the same complete charge-residual source diagram determine the same structural residual packet and comparison assignment. More generally, the residual construction factors through the appropriate response and auxiliary-port quotients whenever the complete signatures of all its inputs agree. Equality of the base charge class alone is not asserted to identify residual packets obtained from different admitted reductions.*

Proof. Residual reduction, normal-form extraction, charge restriction, and every admitted comparison are structural operations in the semantic-use domain of Sections 2 and 3. Their domains retain support, reduction, refinement, and survivor antecedents. All-instance source coverage supplies the evaluated source terms for this diagram, and contextual congruence preserves both operation definedness and every retained value. The complete-signature factorization therefore descends the entire operation with its dependencies. Source redescription of charge normalization, residual notation, or a vacuum representative changes none of these witnesses. In contrast, a different witnessed reduction may change a residual comparison or response and is not identified merely because its base charge packet agrees. $\square$

Theorem 7.13 (An actual common source supplies residual transport). *Let Ξ be a full common Standard Model source realization in the sense of Definition 10.18. Its charge restriction has a composable witness $h_{Q,\Xi}$, and its residual restriction determines*

$$h_{\text{res},\Xi} \in \text{RAnt}_Q(h_{Q,\Xi}).$$

Hence the family of structural residual charge packets is nonempty on this source-realized fibre, and $\widehat{\mathcal{P}}_Q(X_{Q,\Xi}; h_{\text{res},\Xi})$ carries actual residual charge comparisons. The construction uses neither a BCRS compatible-assignment theorem nor a BCRS response count.

Proof. Use the charge restriction of the full source, as in Lemma 10.19. Before forming any BCRS joint assignment, retain the independent SR7 source survivor $Z_\Xi = \pi_{\text{SR7}}(\Xi)$. By SR7 Definition 5.1 and its shared-ledger typing, its residual projection is an actual SR6 survivor whose inherited product, representation, charge, and anomaly maps agree on one common witness ledger. It is not merely a point of a separately inhabited residual family [38, Def. 5.1, §5.1; Thm. 15.1].

The source dependency maps of this SR6 projection retain the actual A6 support and admissible reduction, the PSM6 normal-form arena, and the full SR6 survivor certificate. SR6’s defining arena,

equation (14), places this actual survivor in the PSM6 residual normal form. The forward direction of PSM6 Theorem 15.1 therefore supplies a complete thirteen-clause refinement witness; the inference uses the actual normal-form point, not only necessary conditions for survivability [37, Eq. (14); Defs. 7.1, 9.1 and 10.3; Thm. 10.4] [31, Thm. 15.1]. Extract those records, together with their maps to the same charge restriction. They give respectively clauses (i)–(iv) of Definition 7.9. The actual passing anomaly and support verdicts are inherited from this source’s SR5–SR6 ledger; they are not manufactured by applying a necessity statement to an arbitrary candidate. This extraction defines $h_{\text{res}, \Xi}$. Theorem 7.11 then evaluates it. The independent SR7–SM-2 source precedes every BCRS assignment, so no compatibility or cardinality conclusion is used to obtain the witness. $\square$

The physical residual electromagnetic carrier is a further interpretation of this structural charge-transport restriction. SM-2 relates its photon carrier to charge response together with unbroken electromagnetic transport [13, Def. 16.1 and §16.2]. Retaining the structural restriction of that full source does not select a photon polarization, pole, propagator, or coupling value. Those physical realization data must retain their additional antecedents. In particular, the present positive residual construction neither creates an additional gauge root nor proves that all admitted residual reductions have the same fine response signature.

Scope 7.14 (Charge-response scope). *The charge/EM-readiness packet locus and its presentation-independent packet map are formed exactly relative to the fixed-interior and PSM4 antecedents. The structural residual extension and its actual charge-transport comparisons are constructed on their full A6–PSM6–SR6 domain; the common Standard Model source supplies an instance. No packet-local anomaly pass is inferred from A5 alone. Neither unconditional existence in every global-form fibre nor a simultaneous live occupation, component count, polarization multiplicity, or photon count is asserted here.*

Counterexample criterion 7.15 (Charge/EM packet failure conditions). *Theorem 7.6 fails if the fixed-realized-interior hypotheses of A1–A5 are dropped; if charge identity is finer or coarser than the admitted witness ledger; if a normalization convention changes standing without a witness; if A5 is used as a packet-local anomaly-pass theorem; if the PSM4 anomaly or vacuum guard is reported as solved without the common residual antecedent; if the A6 residual port is evaluated without non-null structural support, the full compatibility ledger, and an admissible reduction; if strict functoriality is inferred from weak residual comparison compatibility; if distinct reductions are identified from base charge equality alone; if a photon, polarization, propagator, or coupling value is used as a source; or if packet formation is identified with simultaneous live occupation.*

8 Color transport and the extended-support sector layer

The color packet combines two incident structures of different types: the boundary-fixed transition–transport–loop response of a non-Abelian use, and the theorem-scope extended-support sector layer over a fixed local Yang–Mills object. The sector layer is neither a bounded-region local observable nor an additional response-role occupant. Both structures are retained without assigning primitive standing to a Lie generator, adjoint or sector label, or confinement claim.

8.1 Exact color and Yang–Mills antecedents

For a source term s_{col} and its evaluated semantic response $\xi_{\text{col}} \in \mathcal{R}_B^{\text{sem}}(\Gamma)$, write $\text{Ante}_{\text{col}}(s_{\text{col}}, \xi_{\text{col}}; D, \Gamma, G_\Gamma)$ when the following conditions hold.

- (i) $\text{GAnte}_{\leq 5}(D, \Gamma)$ holds, and ξ_{col} is a same-domain use with a boundary-detectable, quotient-stable, transport-witnessed non-Abelian loop or representation response.

- (ii) The PSM3 source domain is present: the A4 response ledger, a PSM1 Lie-type specialization, a lawful source-level PSM2 product-or-coupled compatibility status, and a pre-linear response family carrying an algebraic-refinement package satisfying all twelve clauses of PSM3 Theorem 13.1. In particular, actions, invariant subcarriers, sums, tensors, conjugates, handed splittings, dimensions, and multiplicities are retained only when they realize admitted response witnesses [28, Thm. 13.1].
- (iii) G_Γ is a compact connected simple gauge group representing the fixed theorem-scope global-form datum used in this packet. The Yang–Mills source supplies the bounded-region local gauge-invariant net $\mathcal{A}_{\text{loc}}(G_\Gamma)$, its fixed vacuum theorem package, and the frozen extended-support class $\mathcal{E}(G_\Gamma)$.
- (iv) The theorem-scope sector construction is evaluated over that same fixed local object and the same Γ ; data from another global-form fibre are not spliced into the packet.
- (v) The use carries its same-target boundary, continuation, common anomaly-obstruction port, product-interface port, and witnessed typed incidence to r_{col} .

The source term must also satisfy

$$s_{\text{col}} \in \mathcal{S}_B(\Gamma) \text{ of sort } C, \quad \text{ev}_B(s_{\text{col}}) = \xi_{\text{col}}, \quad \text{Act}_B(s_{\text{col}}) = 1, \quad \text{Fail}_B(s_{\text{col}}) = \text{pass}.$$

The last equality is the global failure-ledger value, not merely the absence of a color-specific failure. The PSM2 status in clause (ii) is an antecedent of the imported PSM3 specialization. In a coupled branch it retains the actual mixed transition, transport, and loop-residue ledger with the stability conditions of PSM2, Definition 13.1 and Propositions 13.2–13.3; it does not replace that ledger by the independent-product predicate of Theorem 11.2 [27]. It is not the BCRS simultaneous predicate J_B , which is formed only after all packet loci and the Higgs interface exist.

All clauses are required to share one hypothesis witness h_{col} over (D, Γ, G_Γ) . In particular, the gauge outputs entering PSM3, the PSM3 response use, the Yang–Mills fixed-local/sector incidence, and the evaluated source, boundary, continuation, role-incidence, and global failure fields must lie in one typed fibre product. A collection of separately instantiated source theorems is not a color antecedent.

The full Yang–Mills source object contains realized operator-algebraic, vacuum, and Hamiltonian data:

$$\mathcal{B}_{\text{YM}}^{\text{full}}(G_\Gamma) = (\mathcal{A}_{\text{loc}}(G_\Gamma), \pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}}, H_{\text{vac}}).$$

It is source data, not a premetric packet coordinate. The same distinction applies to the source support class and realized sector class. Define

$$\mathcal{E}^{\text{pre}}(G_\Gamma) := \text{pr}_{\text{pre}} \mathcal{E}(G_\Gamma), \quad \mathcal{S}_{\text{ext}}^{\text{pre}}(G_\Gamma) := \text{pr}_{\text{pre}} \mathcal{S}_{\text{ext}}(G_\Gamma).$$

Here pr_{pre} is the target-relative restriction of Definition 7.1, applied to the entire witnessed support–sector diagram, not an identification of its represented objects with premetric ones. It retains support types and their label-role data, the source support/deformation comparisons, the induced sector assignment, the source global-form comparison tests, and the incidence to the unchanged local root. Source hypotheses and the conditional sector-to-completion interface remain attached. Embedded spacetime supports, Hilbert-space representations, vacuum restrictions, and unitary intertwiners stay in the source realization, not in these reduced records. The retained local-root package is

$$\mathcal{B}_{\text{YM}}^{\text{pre}}(G_\Gamma) := (\text{LocalRootId}(G_\Gamma), \text{ExtSuppType}(\mathcal{E}^{\text{pre}}(G_\Gamma)), \sim_{\text{supp/def}}, \text{Inc}_{\text{ext} \rightarrow \text{loc}}),$$

namely fixed-local identity, extended-support type, support/deformation equivalence, and the reduced incidence of the sector layer over the unchanged local root. The comparison and assignment maps just specified are retained in that incidence record. No vacuum vector, Hamiltonian, spectral projection, mass gap, or time-translation datum is transported into the premetric signature.

Under the exact source hypotheses in clauses (iii)–(iv), the Yang–Mills endpoint proves:

- (a) no nontrivial member of $\mathcal{E}(G_\Gamma)$ belongs to a bounded-region local algebra;
- (b) a canonical theorem-scope sector class $\mathcal{S}_{\text{ext}}(G_\Gamma)$ exists over the fixed quasi-local algebra and respects support and deformation equivalence;
- (c) this sector class distinguishes the frozen theorem-scope global-form data invisible to the strictly local net; and
- (d) the bounded-region local net, distinguished vacuum representation, and predecessor local theorem content remain unchanged.

These are precisely parts (A)–(E) of the main theorem and their source proofs [47, Thm. II.1(A)–(E); Thms. V.12, VI.12, VII.8, and VIII.2]. The further sector-to-completion theorems prove that sector representation alone does not imply same-theory completion eligibility [47, Thms. VIII.41–VIII.43]. BCRS-I therefore retains the restricted local-root/sector incidence and not the later spectral bridge.

The gauge spine has the exact antecedents listed in Definition 7.2. PSM3 licenses the representation-response specialization; A5 supplies only the common anomaly-failure test. Neither source turns representation dimension, adjoint-index range, or a generator basis into a response-role cardinality.

Definition 8.1 (Color-response incidence packet). *For $X_{\text{col}} \in \mathcal{Q}_{B,C}^+(\Gamma)$, define*

$$\begin{aligned} \mathcal{P}_{\text{col}}(X_{\text{col}}) = & (X_{\text{col}}; \text{Gpd}_\partial, \nabla_{\partial D}, \Omega_{\partial D}, \text{RepResp}_{\text{col}}; \\ & \mathcal{B}_{\text{YM}}^{\text{pre}}(G_\Gamma), \mathcal{E}^{\text{pre}}(G_\Gamma), \mathcal{S}_{\text{ext}}^{\text{pre}}(G_\Gamma); \text{AnomPort}_{\text{col}}, \text{ProdPort}_{\text{col}}; \\ & \partial_{\text{col}}, \text{Cont}_{\text{col}}, \text{Fail}_{\text{col}}). \end{aligned}$$

It is a complete color incidence packet when:

- (i) X_{col} has kind C, a witnessed non-Abelian loop/representation response, and the typed incidence $X_{\text{col}} \rightarrow r_{\text{col}}$;
- (ii) its transition, transport, loop-residue, and representation-response fields are exact descended uses; basis, generator, component, and multiplicity distinctions survive only when witness bearing;
- (iii) $\mathcal{S}_{\text{ext}}^{\text{pre}}(G_\Gamma)$ is the structural reduct of the theorem-scope sector class over the same restricted local root $\mathcal{B}_{\text{YM}}^{\text{pre}}(G_\Gamma)$, in the same Γ -fibre, modulo the source support/deformation equivalence;
- (iv) the anomaly field is the restriction of the common A5 obstruction interface and the product field retains the source-level PSM2 status and its actual mixed ledger when coupled, neither being reported as the later BCRS joint verdict; and
- (v) its boundary and continuation records are present and its descended failure value is $\text{Fail}_{\text{col}} = \text{pass}$.
- (vi) there exist $s_{\text{col}} \in \mathcal{S}_{B,\text{act},C}^+(\Gamma)$, $\xi_{\text{col}} = \text{ev}_B(s_{\text{col}})$, and one composable witness h_{col} over the fixed (D, Γ, G_Γ) -fibre such that $q_{B,C}^+(s_{\text{col}}) = X_{\text{col}}$, and every displayed gauge, PSM, Yang–Mills, port, boundary, continuation, incidence, and failure field is the corresponding restriction of that same h_{col} .

No full realized Yang–Mills object, generator enumeration, confinement status, free-gluon state, or spectral conclusion is a packet field.

Define

$$\text{Pkt}_{\text{col}}(D, \Gamma; G_\Gamma) := \{X \in \mathcal{Q}_{B,C}^+(\Gamma) : \mathcal{P}_{\text{col}}(X) \text{ is complete,} \\ E_B^0(X, r_{\text{col}})\}.$$

$$\mathbf{I}_{\text{col}}(D, \Gamma; G_\Gamma) := \mathcal{P}_{\text{col}}|_{\text{Pkt}_{\text{col}}(D, \Gamma; G_\Gamma)}.$$

Thus $\mathbf{I}_{\text{col}}(D, \Gamma; G_\Gamma)$ is the packet map on $\text{Pkt}_{\text{col}}(D, \Gamma; G_\Gamma)$. It selects no generator, sector representative, or unique occupant.

8.2 The color packet theorem

Lemma 8.2 (Extended support without role promotion). *Within a complete color packet,*

$$(\mathcal{B}_{\text{YM}}^{\text{pre}}(G_\Gamma), \mathcal{S}_{\text{ext}}^{\text{pre}}(G_\Gamma))$$

is a required retained local-root/sector-layer coordinate. It is not a second top-level color response. Omitting this content without an equivalent reconstruction drops a positive Yang–Mills source conclusion; no information-theoretic independence of the displayed coordinates is asserted.

Proof. The Yang–Mills source distinguishes two incident types: the unchanged fixed local theorem object and a sector class constructed over it. Its local-net boundary excludes nontrivial extended-support objects from bounded local algebras; its compatibility theorem forbids replacement or enlargement of the fixed local root. Target-relative restriction preserves this type distinction while omitting the realized vacuum and spectral coordinates. The sector class is therefore structural support incident to the color packet, not a peer response role. Omitting this content without reconstructing it omits part (C) of the Yang–Mills main theorem; promoting it changes its codomain. $\square$

Lemma 8.3 (Exact color-packet formation). *Let $(s_{\text{col}}, \xi_{\text{col}})$ satisfy $\text{Ante}_{\text{col}}(s_{\text{col}}, \xi_{\text{col}}; D, \Gamma, G_\Gamma)$. Then*

$$s_{\text{col}} \in \mathcal{S}_{B, \text{act}, C}^+(\Gamma), \quad X_{\text{col}} := q_{B, C}^+(s_{\text{col}}) \in \text{Pkt}_{\text{col}}(D, \Gamma; G_\Gamma).$$

The exact A1–A5, PSM2–PSM3, and Yang–Mills certificates form the complete packet $\mathcal{P}_{\text{col}}(X_{\text{col}})$; in particular $E_B^0(X_{\text{col}}, r_{\text{col}})$.

Proof. A1–A4 supply the transition, transport, admissible loop-residue, and representation-response chain. PSM3 retains only witness-realizing algebraic response structure, with the source-level PSM2 status kept as an interface rather than identified with J_B . A5 contributes the common obstruction map and its failure consequence, not a packet-local passing verdict. The Yang–Mills theorem supplies the positive extended-support sector class over the unchanged fixed local object, and target-relative restriction retains exactly its local-root/sector incidence. The certified active source term supplies the matching boundary and continuation trace and the global failure value **pass**. All fields remain in one (D, Γ, G_Γ) -fibre, so they form the stated packet. No confinement or spectral value is introduced. The common witness h_{col} , non-Abelian role witness, instantiated-interface supply, boundary/continuation trace, and global passing failure value prove the four clauses of E_B^0 ; source enumeration alone establishes none of them. $\square$

Lemma 8.4 (Presentation-independent color descent). *Positive color-sorted source presentations equivalent under $\sim_{\text{pres}}$ determine the same complete color packet under $q_{B, C}^+$. Within its Yang–Mills structural restriction, support-equivalent or deformation-equivalent source representatives have the same reduced support–sector data, modulo the source equivalence and canonical transport retained by that restriction.*

Proof. Equality of complete signatures preserves every color-packet coordinate, so the packet map factors through $q_{B, C}^+$. The Yang–Mills source equivalences preserve the fixed local root and give the same sector class, or its canonical deformation transport, over the same global-form datum [47, Thm. VI.12(ii)–(iv)]. Their declared structural restrictions therefore agree. This source equivalence does not identify unrelated response or continuation data outside that restriction. $\square$

Theorem 8.5 (Color incidence-packet theorem). *For fixed D, Γ, G_Γ ,*

$$\text{Pkt}_{\text{col}}(D, \Gamma; G_\Gamma) \neq \emptyset \iff \exists (s_{\text{col}}, \xi_{\text{col}}) \text{Ante}_{\text{col}}(s_{\text{col}}, \xi_{\text{col}}; D, \Gamma, G_\Gamma).$$

Every pair on the right determines a qualified packet class, and $\sim_{\text{pres}}$ -equivalent positive color-sorted presentations determine the same class. Conversely, every class in the packet locus recovers a positive same-fibre source/response pair satisfying the displayed antecedents.

Proof. The forward construction is Lemma 8.3; the kind, incidence, instantiated-interface supply, boundary, continuation, and failure fields give E_B^0 . Lemma 8.4 removes presentation choice. Conversely, completeness of an element of Pkt_{col} records the positive source/response pair and instantiated gauge/PSM interfaces, compact connected simple G_Γ , fixed local Yang–Mills source object, frozen extended-support class, same global-form fibre, and active boundary/continuation/failure data required by Ante_{col} , including the global passing failure value. $\square$

Corollary 8.6 (Antecedent dependence of color-packet existence). *If no pair $(s_{\text{col}}, \xi_{\text{col}})$ satisfies the exact color antecedent, then $\text{Pkt}_{\text{col}}(D, \Gamma; G_\Gamma) = \emptyset$. Neither source/use completeness nor later compatibility, confinement, or particle language can create a member of that empty locus.*

Proof. This is the contrapositive direction of Theorem 8.5. The later maps consume qualified packet classes; none has a packet-construction codomain. $\square$

Corollary 8.7 (No generator, confinement, or gluon promotion). *A lawful change of Lie basis, generator numbering, adjoint index, local chart, line or surface representative, or sector representative cannot create an additional color packet. Confinement and asymptotic-gluon data lie outside the packet map and cannot establish color-packet standing.*

Proof. The listed source-display changes either preserve every witnessed signature coordinate and hence act trivially on the quotient, or change the source use and target. The extended-support sector class remains incident to the same restricted local root in either description. The Yang–Mills theorems used here supply no map from sector status to confinement, free-particle status, or a generator-count object. $\square$

Scope 8.8 (Color-response scope). *The color packet locus and packet map include the proved extended-support Yang–Mills sector layer in its exact restricted role. They are presentation-independent relative to the stated fixed-interior, PSM3, and Yang–Mills antecedents. No packet-local anomaly verdict, unconditional existence in every global-form fibre, joint occupation, generator count, sector-class count, confinement claim, or free-gluon state has been asserted.*

Counterexample criterion 8.9 (Color packet failure conditions). *Theorem 8.5 fails if the A1–A5 fixed-interior hypotheses are dropped; if PSM3 algebraic notation is retained without response witnesses; if the compact connected simple group or predecessor local theorem package is absent; if Γ and G_Γ are mismatched; if the full vacuum/Hamiltonian Yang–Mills object is inserted as a premetric coordinate; if a nontrivial extended-support object is inserted into a bounded local algebra; if the sector layer changes the fixed local theorem object; if its source support or deformation equivalence changes the retained Yang–Mills support-sector subrecord; if anomaly failure is ignored; if sector representation is promoted directly to completion; or if a generator range, confinement label, or asymptotic particle claim is used to establish packet standing.*

9 The pre-vacuum weak-response packet

Before vacuum reduction, the weak packet is distinguished by a witnessed chiral representation response carried by the common transition–transport–loop structure. The $W^\pm$ and Z^0 projections require later vacuum, Higgs, metric, polarization, and pole data; they are not antecedents of the present packet.

9.1 Exact weak-response antecedents

For a source term s_{wk} and its evaluated semantic response $\xi_{\text{wk}} \in \mathcal{R}_B^{\text{sem}}(\Gamma)$, write $\text{Ante}_{\text{wk}}(s_{\text{wk}}, \xi_{\text{wk}}; D, \Gamma)$ when:

- (i) $\mathbf{GAnte}_{\leq 5}(D, \Gamma)$ holds and ξ_{wk} is a same-domain representation-response use.
- (ii) A source-level PSM2 sector family has a lawful same-domain $\mathbf{ProdComp}_{\partial D}$ status, either independent-product or coupled. Both branches retain sector standing, co-scope, quotient-distinctness, admissible transition and transport comparison, field-carrier and particle-persistence preservation, no hidden representative or parameter standing, and the exact anomaly and vacuum guard status. In the independent-product branch every mixed transition, transport, and loop residue is quotient-trivial. In the coupled branch the actual ledger

$$\mathbf{Coup}_{\partial D}(i, j) = (\mathbf{MixTrans}, \mathbf{MixConn}, \mathbf{MixRes})_{\partial D}(i, j)$$

is retained: its mixed witnesses are boundary-detectable, quotient-stable, same-domain, role-preserving, locally refinable, transport-coherent, and finitely representable along admitted continuation chains. Independent-product notation is licensed only in the former branch; a lawful coupled witness is not erased or reclassified as failure [27, Thms. 10.2, 11.2, and 12.1; Def. 13.1 and Props. 13.2–13.3].

- (iii) The A4 ledger, PSM1 Lie specialization, PSM2 status, and pre-linear response family admit a PSM3 algebraic-refinement package satisfying Theorem 13.1, and that package contains a chiral splitter whose two handed response sectors are distinguished by admitted standing witnesses and remain stable under transport, loop response, boundary extraction, and local reduction [28, Thms. 10.3 and 13.1(10)].
- (iv) The use carries the common anomaly-obstruction interface, its same-target boundary and continuation trace, and a witnessed typed incidence to r_{wk} .

The source term must also satisfy

$$s_{\text{wk}} \in \mathcal{S}_B(\Gamma) \text{ of sort } \mathbf{EW}^{\text{pre}}, \quad \mathbf{ev}_B(s_{\text{wk}}) = \xi_{\text{wk}}, \quad \mathbf{Act}_B(s_{\text{wk}}) = 1, \quad \mathbf{Fail}_B(s_{\text{wk}}) = \text{pass}.$$

This is a passing value of the whole certified failure ledger, not merely the absence of a weak-incidence failure. The PSM2 status in clause (ii) is the source antecedent needed to license the representation specialization: PSM3 requires this compatibility status, not the stronger independent-product predicate. It is not the later BCRS simultaneous predicate J_B , and it does not establish coexistence of the three packet loci.

The conjunction must be inhabited by one common witness h_{wk} over (D, Γ) . The A1–A4 outputs, PSM2 status, PSM3 algebraic refinement, witnessed chiral splitter, and evaluated source, boundary, continuation, role-incidence, and global failure fields must therefore be jointly composable in one source fibre. Separately instantiated theorem certificates do not satisfy $\mathbf{Ante}_{\text{wk}}$.

The retained source chain is

$$\mathbf{Gpd}_{\partial}(D) \rightsquigarrow \nabla_{\partial D} \rightsquigarrow \Omega_{\partial D} \rightsquigarrow \mathbf{Rep}_{\partial D} \rightsquigarrow \mathbf{Chir}_{\partial D}. \quad (9.1)$$

The final arrow is not a left/right label assignment. PSM3 licenses chirality only when the split is witnessed in the exact sense of clause (iii). Gauge A4 supplies the underlying representation-response ledger, while A5 supplies the common non-neutrality failure rule [23, Thm. 11.1] [24, Thm. 10.1]. Neither theorem supplies a packet-local anomaly pass.

Two later interfaces must be kept open. First, the A6 vacuum interface is a conditional theorem with the exact domain

$$(\mathbf{Anom}_{\partial D} = 0_{\partial}, V_D \neq 0, \mathbf{GVac}_{\partial D}, \mathbf{AdmBreak}_{\partial D}).$$

Without those data, neither a vacuum-compatible residual ledger nor a mass response is licensed by this interface [25, Thms. 10.1 and 12.1; Cors. 13.1 and 13.5]. Second, the Higgs interface records only the type of the incidence constructed in Section 10. No evaluation of $g_{EW}^{\text{pre}} H$ occurs in the present section. Denote the two unexecuted interfaces by $\mathbf{VacPort}_{\text{wk}}$ and $\mathbf{HPort}_{\text{wk}}$.

Definition 9.1 (Pre-vacuum weak incidence packet). *For $X_{\text{wk}} \in \mathcal{Q}_{B, \text{EW}^{\text{pre}}}^+(\Gamma)$, define*

$$\begin{aligned} \mathcal{P}_{\text{wk}}^{\text{pre}}(X_{\text{wk}}) = & (X_{\text{wk}}; \text{Gpd}_{\partial}, \nabla_{\partial D}, \Omega_{\partial D}, \text{Rep}_{\partial D}, \text{Chir}_{\partial D}; \\ & \text{AnomPort}_{\text{wk}}, \text{ProdPort}_{\text{wk}}, \text{VacPort}_{\text{wk}}, \text{HPort}_{\text{wk}}; \\ & \partial_{\text{wk}}, \text{Cont}_{\text{wk}}, \text{Fail}_{\text{wk}}). \end{aligned}$$

It is a complete pre-vacuum weak packet when:

- (i) X_{wk} has kind EW^{pre} , the witnessed chain (9.1), and the typed incidence $X_{\text{wk}} \rightarrow r_{\text{wk}}$;
- (ii) the gauge-spine and representation entries are exact descended uses, and the chiral split is distinguished by admitted response witnesses, not by a basis or handedness label;
- (iii) the anomaly field is the restriction of the common A5 obstruction interface and the product field retains the source-level PSM2 status, including its full coupled ledger when that branch applies, neither being reported as a BCRS joint verdict;
- (iv) $\text{VacPort}_{\text{wk}}$ and HPort_{wk} are open conditional ports: neither a vacuum value nor a Higgs-mediated mass-response output is retained as an executed field;
- (v) its boundary and continuation records are present and its descended failure value is $\text{Fail}_{\text{wk}} = \text{pass}$.
- (vi) there exist $s_{\text{wk}} \in \mathcal{S}_{B, \text{act}, \text{EW}^{\text{pre}}}^+(\Gamma)$, $\xi_{\text{wk}} = \text{ev}_B(s_{\text{wk}})$, and one composable witness h_{wk} over the fixed (D, Γ) -fibre such that $q_{B, \text{EW}^{\text{pre}}}^+(s_{\text{wk}}) = X_{\text{wk}}$, and every displayed gauge, PSM, guard-port, boundary, continuation, incidence, and failure field is the corresponding restriction of that same h_{wk} .

No weak generator, mixing angle, polarization, Goldstone coordinate, mass, width, decay channel, $W^\pm$, or Z^0 label is a packet field.

Define

$$\text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma) := \{X \in \mathcal{Q}_{B, \text{EW}^{\text{pre}}}^+(\Gamma) : \mathcal{P}_{\text{wk}}^{\text{pre}}(X) \text{ is complete,} \\ E_B^0(X, r_{\text{wk}})\}.$$

$$\mathbf{I}_{\text{wk}}^{\text{pre}}(D, \Gamma) := \mathcal{P}_{\text{wk}}^{\text{pre}}|_{\text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma)}.$$

The map $\mathbf{I}_{\text{wk}}^{\text{pre}}(D, \Gamma)$ is defined on $\text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma)$, without selecting a distinguished weak-boson object.

9.2 The weak packet theorem

Lemma 9.2 (Exact pre-vacuum packet formation). *Let $(s_{\text{wk}}, \xi_{\text{wk}})$ satisfy $\text{Ante}_{\text{wk}}(s_{\text{wk}}, \xi_{\text{wk}}; D, \Gamma)$. Then*

$$s_{\text{wk}} \in \mathcal{S}_{B, \text{act}, \text{EW}^{\text{pre}}}^+(\Gamma), \quad X_{\text{wk}} := q_{B, \text{EW}^{\text{pre}}}^+(s_{\text{wk}}) \in \text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma).$$

The exact A1–A5 and PSM2–PSM3 certificates form the complete packet $\mathcal{P}_{\text{wk}}^{\text{pre}}(X_{\text{wk}})$, while the A6 and Higgs interfaces remain unexecuted ports. In particular, $E_B^0(X_{\text{wk}}, r_{\text{wk}})$.

Proof. Gauge A1–A3 supply the transition, transport, and admitted loop-residue entries. A4 supplies representation response. The exact PSM2 status supplies the source product-or-coupled interface and preserves every lawful mixed witness. PSM3 retains the chiral field only because the two handed response sectors are witness-distinguishable. A5 supplies the common obstruction map and non-neutrality failure rule, not a local zero. The certified active source term supplies its boundary and continuation trace and the global failure value **pass**. A6 supplies only the type of the later vacuum-mediated reduction, and the Higgs incidence remains a reserved codomain. The fields therefore form one pre-vacuum packet without executing a vacuum, Higgs, or particle map. Their common witness h_{wk} , witnessed chiral incidence, instantiated-interface supply, boundary/continuation trace, and global passing failure value prove the four clauses of E_B^0 . No conclusion follows from source enumeration alone. $\square$

Lemma 9.3 (Independence from later vacuum, Higgs, and pole data). *A later vacuum-compatible residual ledger, Higgs incidence, or physical weak-boson projection cannot create, split, merge, or repair an element of $\text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma)$.*

Proof. A6 makes anomaly neutrality, a non-null vacuum, compatibility, and admissible reduction antecedents of a residual ledger. The Higgs incidence and pole data are later still. Such data therefore do not act on the pre-vacuum packet locus. If appended as identity-bearing fields, they define a successor target; if used to repair a missing source, boundary, continuation, or failure field, they violate the no-backfill theorem. No case changes the packet in the fixed target. $\square$

Lemma 9.4 (Presentation-independent weak descent). *Two positive EW^{pre} -sorted source presentations of the same certified weak-response use that are equivalent under $\sim_{\text{pres}}$ determine the same complete pre-vacuum packet under $q_{B, \text{EW}^{\text{pre}}}^+$.*

Proof. The witnessed-chirality, product-port, anomaly-port, boundary, continuation, and failure fields are coordinates of the complete signature. Presentation equivalence preserves them, and the packet map therefore factors through $q_{B, \text{EW}^{\text{pre}}}^+$. Instantiated-interface supply is required, but the name or provenance of a supplying paper is not a signature coordinate. The open later ports are types, not representative values. $\square$

Theorem 9.5 (Pre-vacuum weak incidence-packet theorem). *For fixed D, Γ ,*

$$\text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma) \neq \emptyset \iff \exists(s_{\text{wk}}, \xi_{\text{wk}}) \text{ Ante}_{\text{wk}}(s_{\text{wk}}, \xi_{\text{wk}}; D, \Gamma).$$

Every pair on the right determines a qualified packet class, and $\sim_{\text{pres}}$ -equivalent positive EW^{pre} -sorted presentations determine the same class. Conversely, every class in the packet locus recovers a positive same-fibre source/response pair satisfying the displayed antecedents. No vacuum, Higgs, or particle antecedent is supplied by this theorem.

Proof. Lemma 9.2 forms a complete packet. Its kind, witnessed response incidence, instantiated-interface supply, boundary and continuation trace, and global passing failure value give E_B^0 . Lemma 9.4 removes presentation choice. Conversely, completeness of an element of $\text{Pkt}_{\text{wk}}^{\text{pre}}$ records instantiated gauge, PSM2, and PSM3 interfaces supplying the required semantic fields, witnessed chirality, fixed (D, Γ) -fibre, positive source/response record, open later ports, and boundary/continuation/global-failure fields required by Ante_{wk} . $\square$

Corollary 9.6 (Antecedent dependence of weak-packet existence). *If no pair $(s_{\text{wk}}, \xi_{\text{wk}})$ satisfies the exact weak antecedent, then $\text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma) = \emptyset$. No later vacuum, Higgs, compatibility, or occupation construction may be used to populate the empty pre-vacuum locus.*

Proof. Theorem 9.5 gives the emptiness implication. Later maps are typed on existing packet classes and cannot supply a missing source witness without changing the target and theorem order. $\square$

Corollary 9.7 (No W/Z or generator promotion). *No lawful weak-response source-display mutation creates an additional pre-vacuum packet or establishes simultaneous live occupation. A mixing coordinate, polarization, Goldstone description, pole, or W/Z name lies outside the packet map and cannot act as a packet constructor.*

Proof. A source-display mutation either preserves the complete signature and acts trivially on the quotient, or changes the source use or target. Later vacuum, Higgs, metric, and particle data do not act on the pre-vacuum locus. None supplies the common compatibility witness required for live occupation. $\square$

Scope 9.8 (Weak-response scope). *The pre-vacuum weak packet locus and packet map are formed exactly relative to the fixed-interior, PSM2, PSM3, and witnessed-chirality antecedents. The anomaly, vacuum, and Higgs fields remain common or open interfaces at their proper theorem ranks. No unconditional existence in every global-form fibre, joint occupation, response cardinality, internal representation count, or physical W/Z mode has been asserted.*

Counterexample criterion 9.9 (Weak packet failure conditions). *Theorem 9.5 fails if the A1–A5 fixed-interior hypotheses are dropped; if chirality is asserted by a handed label without an admitted response witness; if the source-level PSM2 status is identified with the later BCRS common-ledger predicate; if independent-product status is imposed on a lawful coupled ledger or a standing mixed witness is suppressed; if an anomaly pass is asserted locally; if a vacuum, vev, executed Higgs incidence, Goldstone coordinate, mass, width, polarization, or W/Z display is used to form the packet; if $g_{EW\ H}^{\text{pre}}$ is evaluated before the formation of the mixed Higgs domain; if an open port is reported as an executed map; or if packet qualification is identified with simultaneous live occupation.*

10 One Higgs identity and premetric electroweak incidence

The Higgs contribution has two distinct levels: the same-scope carrier identity occupying the bridge/fixation role, and the typed standing uses incident on that identity. The source theorems give one carrier and three mechanism types, without executing a metric Higgs–electroweak realization. Throughout this section D is the fixed realized-interior source object of Section 1, and $\Gamma \in \mathfrak{G}_{\text{src}}$ is fixed. Arguments D, Γ , and where relevant G_Γ , are suppressed only after their common-fibre typing has been stated.

10.1 Same-scope Higgs identity

Carrier identity and complete response identity have different comparison targets. The inherited Higgs identity theorem fixes the former. The complete-signature quotient $\mathcal{Q}_B$ retains the latter, including every admitted response, boundary, continuation, readiness, and failure coordinate. We retain an explicit projection between these levels.

Definition 10.1 (Higgs identity quotient). *In the fixed source fibre, let $\mathcal{P}_H^{\text{id}}(\Gamma)$ be the source-certified same-scope Higgs identity-ground presentations satisfying the neutral identity-ground criterion: preservation of D_H , the full role bundle H_{role} , the admissibility boundary, identity-preserving redescription, and identity-role occupation without representative or selector promotion. This criterion does not select C_H by definition.*

Let $\sim_H^{\text{id}}$ be the inherited identity-preserving redescription relation: it preserves the target, carrier occupation, full identity-role bundle, representative/route status assignments, and admissibility boundary [40, Criterion 3.5 and Def. 5.2]. Define

$$\text{Hld}_B(\Gamma) := \mathcal{P}_H^{\text{id}}(\Gamma) / \sim_H^{\text{id}}, \quad \eta_H := [C_H]_{\text{id}}, \quad C_H = \mathcal{H}_{\text{br}/\text{fix}}.$$

This is the imported carrier-identity quotient. It is not defined as a subset of $\mathcal{Q}_B^+(\Gamma)$, and does not replace that finer response quotient.

Theorem 10.2 (One-Higgs identity theorem). *For every admitted source fibre in which the fixed Higgs carrier package is received,*

$$\text{Hld}_B(\Gamma) = \{\eta_H\}. \tag{10.1}$$

Every admitted same-scope Higgs identity ground is $\sim_H^{\text{id}}$ -equivalent to $C_H = \mathcal{H}_{\text{br}/\text{fix}}$.

Proof. The bridge-witness closure certificate and carrier-admissibility theorem supply the admitted carrier C_H [39, Thms. 7.4 and 7.6]; its occupation supplies a Higgs identity ground [40, Thm. 8.1]. Thus η_H exists. For any $P \in \mathcal{P}_H^{\text{id}}(\Gamma)$, the Higgs identity normal-form theorem gives carrier-identical status. The same-scope replacement lemma gives role equivalence to C_H , and the role-equivalence-to-identity-redescription lemma gives $P \sim_H^{\text{id}} C_H$ [40, Thm. 5.6, Lems. 6.4–6.5, and Thm. 8.1]. Hence every identity class is η_H . Domain exhaustion excludes a further primitive Higgs identity ground while retaining the registered downstream statuses [44, Thms. 8.3 and 9.1]. No equality of complete BCRS response signatures is used. $\square$

Let $\mathcal{Q}_{B,H}^+(\Gamma) \subseteq \mathcal{Q}_B^+(\Gamma)$ be the fine classes whose certified source roots have admitted Higgs identity-ground presentations. A root is read from the source restriction of a term, not chosen by inspecting its later response.

Lemma 10.3 (Higgs root projection). *The source-root restriction induces a well-defined map*

$$\text{rt}_H^{\text{car}} : \mathcal{Q}_{B,H}^+(\Gamma) \longrightarrow \text{Hld}_B(\Gamma), \quad [s]_{\text{pres}} \longmapsto [\text{root}(s)]_{\text{id}} = \eta_H.$$

No injectivity is asserted. Its constancy expresses one carrier identity, not a collapse of the full response fibre.

Proof. Root and kind are complete-signature coordinates, and the source restriction certifies the identity of each admitted root. Equal complete signatures therefore induce the same source-root identity class, independently of representative choice. Theorem 10.2 identifies every value with η_H . The converse implication from equal root identity to equal complete signatures is neither assumed nor needed. $\square$

The shared-carrier restriction preserves η_H . It identifies neither distinct fine response classes nor mixed attachments, and supplies no metric realization.

10.2 The target-relative Higgs mechanism reduct

The Higgs carrier package is independently defined; its attachment to particular BCRS gauge packets is a separate question. Write the full source package schematically as

$$\mathcal{W}_H^{\text{full}} = (D_H, C_H, W_D^{\text{br}}, Q_D^{\text{ord}}, Q_D^{\text{def}}, N_D^{\text{res}}, \text{GVac}_D, \text{DefMed}, \text{MassResp}, \text{IdCont}, \text{RedStab}).$$

This is the fixed carrier package supplied by the Higgs bridge and identity sources. In particular, the closure certificate identifies its carrier as $C_H = \mathcal{H}_{\text{br}/\text{fix}}$, and the carrier-admissibility theorem admits that carrier in the fixed Higgs domain [39, Thms. 7.4 and 7.6]. The MGN3 theorem has a different logical direction: it gives necessary survivability conditions for any Higgs-like bridge over its SR7-based comparison arena and expressly does not assert that a nontrivial bridge exists in every admissible arena [19, Thm. 21.1]. We use MGN3 only for that necessary-condition and anti-selector boundary. Existence and identity of the fixed carrier used here come from the independently imported Higgs package, not from MGN3 or from SR7.

Definition 10.4 (Premetric target-relative Higgs reduct). *The target-relative reduct*

$$\rho_H^{\text{pre}} : \mathcal{W}_H^{\text{full}} \longrightarrow \overline{\mathcal{W}}_H^{\text{pre}}$$

retains

$$\overline{\mathcal{W}}_H^{\text{pre}} = (C_H; u_H^{\text{ord}}, u_H^{\text{def}}, u_H^{\text{fix}}; \partial_H, \text{Cont}_H, \text{Fail}_H; \text{VacPort}_{H,EW}, \text{MetricPort}_{H,EW}, \text{RespPort}_{H,EW}).$$

Here u_H^{ord} is the descended use that receives and preserves admitted gauge/representation order, u_H^{def} is the descended order-to-deformation mediation use, and u_H^{fix} is the descended residual-normal-form and identity-fixation use. The final three entries are the types and conditional dependency arrows of the vacuum, metric, and physical-response obligations. Their source values are not transported as executed BCRS data. The reduct also forgets scalar coordinates, vev and potential presentations, masses, poles, mixing data, and numerical scale representatives. It neither chooses a BCRS gauge packet nor supplies a mixed Higgs–electroweak attachment.

For a descended Higgs use u in this reduct, the mechanism-type evaluator has the displayed values

$$\text{mtype}_H(u_H^{\text{ord}}) = \text{Ord}_H, \quad \text{mtype}_H(u_H^{\text{def}}) = \text{Def}_H, \quad \text{mtype}_H(u_H^{\text{fix}}) = \text{Fix}_H.$$

Let $\mathcal{U}_H^{\text{pre}}(C_H; \Gamma)$ instead denote all positive, standing-bearing, target-relative Higgs semantic-use classes rooted at C_H that occur in the reduct of the fixed carrier package. The three displayed assignments specify known values of the evaluator; they do not restrict its image by definition. A mechanism use here is an individual typed incidence of the reduct; a composite with several outputs is evaluated through its component incidences. Boundary, continuation, failure, and readiness records accompany those incidences rather than being counted as additional mechanism outputs. Put

$$\text{Mech}_H(C_H; \Gamma) := \text{im} \left(\text{mtype}_H \mid_{\mathcal{U}_H^{\text{pre}}(C_H; \Gamma)} \right).$$

The root map is

$$\pi_\mu : \text{Mech}_H(C_H; \Gamma) \longrightarrow \text{Hld}_B(\Gamma), \quad \pi_\mu(\text{mtype}_H(u)) = [\text{root}(u)]_{\text{id}} = \eta_H.$$

Theorem 10.5 (Target-relative Higgs mechanism fibration). *For the fixed imported Higgs package,*

$$\pi_\mu : \{\text{Ord}_H, \text{Def}_H, \text{Fix}_H\} \longrightarrow \{\eta_H\}, \quad \pi_\mu^{-1}(\eta_H) = \{\text{Ord}_H, \text{Def}_H, \text{Fix}_H\}. \quad (10.2)$$

The three entries are pairwise distinct mechanism types over one Higgs identity. This statement is a reduct theorem for the independently formed carrier package; it is not a claim that every charge/weak packet pair admits a Higgs attachment. This is not the fibre of $\text{rt}_H^{\text{car}} : \mathcal{Q}_{B,H}^+ \rightarrow \text{Hld}_B$; no cardinality for that fine response fibre follows.

Proof. The bridge-witness closure package contains an admitted order quotient and witness class, a reduced mediation map to deformation response, and the residual-normal-form/fixation data of the one carrier. The Higgs domain and identity results retain those three uses while excluding scalar, vev, potential, scale, vacuum-route, excitation, and pole data from the mechanism fibre [39, Thms. 7.4–7.6 and 8.4] [40, Thms. 5.6 and 8.1] [44, Thms. 8.3 and 9.1]. Hence all three displayed types occur. For exhaustion within this reduct, inspect the retained source fields. Gauge inheritance supplies the order incidence; the reduced mediation map supplies the deformation incidence; and residual-normal-form preservation together with identity continuity supplies fixation. Vacuum support and mass-response readiness are the retained dependency ports, while standing preservation and redescription stability govern these incidences. The remaining boundary, continuation, and failure fields are their tests and guards. Thus every mechanism-output incidence of the displayed reduct has one of the three stated types; this is a typed reduct calculation, not an inference from primitive carrier exhaustion. They are distinct because reception/preservation of order, mediation of deformation, and fixation have different output sorts in the complete signature. All three retain the same semantic root C_H , whose identity class is unique by Theorem 10.2. No converse to the necessary MGN3 filter and no SR7 joint-survivor conclusion is used. $\square$

Theorem 10.6 (Gauge–Higgs noncollapse). *Let $r \in \mathbb{R}_g^0$, and let $x \in \mathcal{Q}_B^+(\Gamma)$ carry a complete kindwise gauge packet for r . For every $y \in \mathcal{Q}_{B,H}^+(\Gamma)$, one has $x \neq y$. No lawful redescription identifies a gauge mechanism in that packet with an element of $\text{Mech}_H(C_H)$. A mixed incidence may meet both kinds without identifying them.*

Proof. Gauge packet roots and Higgs bridge/fixation roots have different rigid kind tags. Their response and mechanism output sorts are retained complete-signature coordinates. Lawful redescription preserves those coordinates. Thus the fine classes remain distinct; equality with the identity-level object η_H is not a well-typed comparison. $\square$

10.3 Pointwise packet inhabitation from direct source interfaces

Each sector antecedent is one internally composable dependent witness, as required by Sections 7–9. At this point the sectors share their fixed target anchors, not a simultaneous response ledger.

Definition 10.7 (Pointwise sector-antecedent record). *Let $A_B^{\text{pt}}(D, \Gamma; G_\Gamma)$ consist of triples*

$$(h_Q, h_{\text{col}}, h_{\text{wk}})$$

such that h_Q , h_{col} , and h_{wk} are respectively single composable witnesses for Ante_Q , Ante_{col} , and Ante_{wk} . Their restrictions agree on

$$(D, K_D, \text{RPI}_D, \partial D, \Gamma)$$

and on the types of the shared transition, transport, loop, product, anomaly, continuation, and failure ports. Equality of product, anomaly, vacuum, or Higgs values across the three witnesses is not part of A_B^{pt} .

Lemma 10.8 (Existence of common-fibre component antecedents). *For every admitted Γ and the fixed G_Γ in the imported Yang–Mills interface,*

$$A_B^{\text{pt}}(D, \Gamma; G_\Gamma) \neq \emptyset.$$

Equivalently, there are certified source/response pairs (s_Q, ξ_Q) , $(s_{\text{col}}, \xi_{\text{col}})$, and $(s_{\text{wk}}, \xi_{\text{wk}})$ satisfying the three exact sector antecedents in the same fixed source fibre. The proof uses direct component restrictions; no BCRS joint-realization or fine response-count conclusion is a premise.

Proof. Choose the conservative import witness supplied by Theorem 1.7. For the charge component take one full PSM4 instantiation together with its dependent A1–A5, PSM1, and source-level PSM2 antecedent projections. For the color component take one full PSM3 instantiation and the Yang–Mills local-root/extended-support restriction in the same (D, Γ, G_Γ) -fibre. For the weak component take one full PSM2–PSM3 instantiation containing the witnessed chiral splitter. These are the direct source interfaces imported in Theorem 1.7. The prior SM-2 and SR7 architecture may also be received with its own dependencies; its independent source status does not create a circular BCRS dependency.

Within each component, compose the certified A1–A5, PSM, boundary, continuation, role-incidence, and failure generators along the dependency maps already contained in that one full antecedent record. This produces finite root-connected terms $s_Q, s_{\text{col}}, s_{\text{wk}}$ of sorts $Q, C, \text{EW}^{\text{pre}}$, and set $\xi_b = \text{ev}_B(s_b)$. Activity follows from the witnessed root incidence of each term. The global failure value is evaluated, not inferred from source enumeration: every registered failure coordinate of each finite term is supplied by its same component witness and has passing value, while its anomaly, product, residual, vacuum, and Higgs data that are not decided at this rank remain explicitly typed open ports. Thus $\text{Fail}_B(s_b) = \text{pass}$ without asserting aggregate anomaly neutrality or a non-null vacuum.

The three internally composable witnesses therefore satisfy their exact sector antecedents. Their anchor restrictions agree because they were taken from the one conservative import fibre. Nothing in the construction equates their open guard values or combines them into the later predicate J_B , proving the stated pointwise result. $\square$

Corollary 10.9 (Common-fibre packet inhabitation). *For every admitted Γ , the loci*

$$\text{Pkt}_Q(D, \Gamma), \quad \text{Pkt}_{\text{col}}(D, \Gamma; G_\Gamma), \quad \text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma)$$

are each inhabited. Their witnesses share the fixed source-fibre anchors, but the conclusion does not assert a common response ledger, $J_B = 1$, a live occupation, or a Higgs attachment.

Proof. Apply Theorems 7.6, 8.5, and 9.5 to the three exact pairs supplied by Lemma 10.8. The respective descent clauses remove presentation choice. No cross-sector compatibility conclusion is used. $\square$

With the fixed arguments displayed above, abbreviate the packet evaluations by

$$\mathbf{I}_Q(x) := \mathbf{I}_Q(D, \Gamma)(x), \quad \mathbf{I}_{\text{col}}(x) := \mathbf{I}_{\text{col}}(D, \Gamma; G_\Gamma)(x), \quad \mathbf{I}_{\text{wk}}^{\text{pre}}(x) := \mathbf{I}_{\text{wk}}^{\text{pre}}(D, \Gamma)(x).$$

These abbreviations denote the maps of Theorems 7.6–9.5; they select no distinguished packet occupants.

10.4 The mixed Higgs–electroweak readiness domain

The partial map reserved by FCGS-III remains

$$\beta_{H,EW} : D_{H,EW} \dashrightarrow \mathcal{P}_{H,EW}^{\text{metric}}.$$

Its domain requires metric licence, executed vacuum/fixation data, and physical-response authority. The premetric readiness map below retains these requirements as conditional interfaces.

Definition 10.10 (Fine Higgs reduct packets). *Let $\text{Pkt}_H^{\text{pre}}(D, \Gamma)$ be the locus of classes $h = [s_H]_{\text{pres}} \in \mathcal{Q}_{B,H}^+(\Gamma)$ admitting a positive certified presentation of the retained Higgs package $\rho_H^{\text{pre}}(\mathcal{W}_H^{\text{full}})$, with its order, deformation, fixation, boundary, continuation, readiness, and failure records. Membership retains the complete signature; it does not identify two different signatures merely because both have root C_H . The restricted root map has $\text{rt}_H^{\text{car}}(h) = \eta_H$.*

Definition 10.11 (Common premetric interface base and anchor maps). *Let $\mathbf{B}_{H,EW}^{\text{pre}}(\Gamma)$ be the object of fixed common interface records*

$$b = (\tau_B(\Gamma), D, K_D, \text{RPl}_D, \Gamma; \text{BdryType}, \text{ContType}, \text{FailType}).$$

The Higgs reduct and the two packet maps have anchor morphisms

$$\begin{aligned} a_H &: \text{Pkt}_H^{\text{pre}}(D, \Gamma) \rightarrow \mathbf{B}_{H,EW}^{\text{pre}}(\Gamma), \\ a_Q &: \text{Pkt}_Q(D, \Gamma) \rightarrow \mathbf{B}_{H,EW}^{\text{pre}}(\Gamma), \\ a_{\text{wk}} &: \text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma) \rightarrow \mathbf{B}_{H,EW}^{\text{pre}}(\Gamma). \end{aligned}$$

obtained by restricting their target, domain, kernel/interior, global-form, boundary-type, continuation-type, and failure-type fields. Their fibre product

$$\mathbf{F}_{H,EW}^{\text{pre}} := \text{Pkt}_H^{\text{pre}}(D, \Gamma) \times_{\mathbf{B}_{H,EW}^{\text{pre}}(\Gamma)} \text{Pkt}_Q(D, \Gamma) \times_{\mathbf{B}_{H,EW}^{\text{pre}}(\Gamma)} \text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma)$$

asserts only type and anchor agreement. It does not assert that a mixed incidence exists.

Definition 10.12 (The single mixed incidence schema). *The source registry contains one mixed schema*

$$g_{EW\ H}^{\text{pre}} : (H, Q, \text{EW}^{\text{pre}}; \text{Inc}_{H,EW}^{\text{pre}}). \quad (10.3)$$

For an exact certified antecedent λ , interpreted as the intrinsic joint instance it certifies, write $g_{EW\ H}^{\text{pre}}[\lambda]$ for its instance. Its charge and weak restrictions are the two typed input legs of this one many-input generator. There are no separate g_{QH} and g_{WH} schemas.

Definition 10.13 (Exact source antecedent for mixed readiness). *Write*

$$\text{Ante}_{H,EW}^{\text{rd}}(s_H, s_Q, s_{\text{wk}}, \lambda; D, \Gamma)$$

when all of the following hold on one mixed-incidence source witness:

- (i) s_H is a positive certified presentation of $\rho_H^{\text{pre}}(\mathcal{W}_H^{\text{full}})$, rooted at the independently formed C_H ;
- (ii) $(s_Q, \text{ev}_B(s_Q))$ and $(s_{\text{wk}}, \text{ev}_B(s_{\text{wk}}))$ satisfy respectively Ante_Q and Ante_{wk} ;
- (iii) the three quotient classes have one point in $\mathbb{F}_{H,EW}^{\text{pre}}$;
- (iv) λ is an instantiated source certificate for the single generator (10.3), and its three input restrictions are exactly the evaluated intrinsic Higgs, charge, and weak uses with their shared-interface comparison. Raw names may be re-presented as in Theorem 11.4; mere isomorphism of separately chosen marginal carriers is insufficient; and
- (v) the generator output retains the common anomaly and vacuum-readiness ports and records metric, vacuum-execution, and physical-response requirements as unexecuted obligations. It supplies no aggregate anomaly verdict, vacuum value, mass, pole, or particle projection.

An *SR7* survivor without clause (iv), or an *MGN3* candidate satisfying only necessary filters, does not satisfy this predicate.

For $(h, x_Q, x_{\text{wk}}) \in \mathbb{F}_{H,EW}^{\text{pre}}$, let $\text{Att}_{H,EW}^{\text{pre}}(h, x_Q, x_{\text{wk}})$ be the presentation quotient of all witnesses λ occurring in a source antecedent of Definition 10.13 with quotient inputs (h, x_Q, x_{wk}) . Define the readiness domain as the dependent sum

$$D_{H,EW}^{\text{rd}}(\Gamma) := \sum_{(h, x_Q, x_{\text{wk}}) \in \mathbb{F}_{H,EW}^{\text{pre}}} \text{Att}_{H,EW}^{\text{pre}}(h, x_Q, x_{\text{wk}}). \quad (10.4)$$

Thus a point is $d = (h, x_Q, x_{\text{wk}}, [\lambda])$. The evident projections p_H, p_Q, p_{wk} recover its three inputs, while p_{att} recovers the one mixed-incidence class. Unlike a bare fibre product, this domain cannot be inhabited without an attachment certificate.

The Higgs input h is a fine response packet, while its carrier identity is the separately projected coordinate $\text{rt}_H^{\text{car}}(p_H(d)) = \eta_H$. The attachment class retains the exact input restrictions. Replacing h by its identity alone would discard information required to evaluate the mixed generator and is not performed.

For any source antecedent representing d , define the evaluated mixed incidence map by

$$\epsilon_{H,EW}^{\text{pre}}(d) := [\text{ev}_B(g_{EW\ H}^{\text{pre}}[\lambda](s_H, s_Q, s_{\text{wk}}))]_{\text{pres}}.$$

The representative-independence lemma below proves that this expression depends only on d .

Definition 10.14 (Premetric incidence record and readiness map). *For $d = (h, x_Q, x_{\text{wk}}, [\lambda])$, let $\mathcal{I}_{H,EW}^{\text{pre}}(d)$ be the complete record*

$$\left(\begin{array}{l} h, \mathbf{I}_Q(x_Q), \mathbf{I}_{\text{wk}}^{\text{pre}}(x_{\text{wk}}); \epsilon_{H,EW}^{\text{pre}}(d); \\ \text{Ord}_H, \text{Def}_H, \text{Fix}_H; \text{AnomPort}_{Q,\text{wk}}, \text{VacPort}_{H,EW}; \\ \partial, \text{Cont}, \text{Fail}; \text{Need}_{\text{metric}}, \text{Need}_{\text{vac}}, \text{Need}_{\text{resp}} \end{array} \right).$$

Let $\mathcal{I}_{H,EW}^{\text{pre}}(\Gamma)$ be its presentation quotient. The premetric readiness map is

$$\beta_{H,EW}^{\text{rd}} : D_{H,EW}^{\text{rd}}(\Gamma) \longrightarrow \mathcal{I}_{H,EW}^{\text{pre}}(\Gamma), \quad d \longmapsto [\mathcal{I}_{H,EW}^{\text{pre}}(d)]_{\text{pres}}. \quad (10.5)$$

Lemma 10.15 (Representative independence of mixed readiness). *The map $\beta_{H,EW}^{\text{rd}}$ is well defined and is natural under every licensed same-target restriction of its already formed source witness.*

Proof. The three anchor maps use complete-signature coordinates. The mixed output is the evaluation of one registered many-input semantic use. Presentation equivalence preserves the former, while universal semantic-use factorization and congruence preserve the latter and all three typed input restrictions. The open obligations are typed coordinates and are preserved as open obligations. Therefore changing source presentation does not change the quotient record. Naturality follows from the same restriction identities. No later maximal-interface or codescent construction is used. $\square$

Theorem 10.16 (Exact formation of premetric Higgs–electroweak readiness). *For fixed D, Γ , the following are equivalent:*

- (i) $D_{H,EW}^{\text{rd}}(\Gamma) \neq \emptyset$;
- (ii) *there exist $s_H, s_Q, s_{\text{wk}}, \lambda$ satisfying $\text{Ante}_{H,EW}^{\text{rd}}(s_H, s_Q, s_{\text{wk}}, \lambda; D, \Gamma)$.*

Whenever these conditions hold, $\beta_{H,EW}^{\text{rd}}$ is a total, source-formed, quotient-independent readiness specification on its exact domain, and every value is Ready at the premetric incidence boundary. The theorem does not assert $D_{H,EW}^{\text{rd}} \subseteq D_{H,EW}$, construct a map to $\mathcal{P}_{H,EW}^{\text{metric}}$, or infer a mixed attachment from packet nonemptiness, MGN3 survivability, or an SR7 survivor alone.

Proof. An exact source antecedent descends by the three quotient maps and the congruence of the single mixed constructor. Clauses (i)–(iii) of Definition 10.13 give a point of $\mathcal{F}_{H,EW}^{\text{pre}}$; clause (iv) gives its attachment fibre; and clause (v) gives the typed open obligations. Hence it determines a point of $D_{H,EW}^{\text{rd}}$. Lemma 10.15 removes all presentation choice and proves the positive readiness statement.

Conversely, a point of the dependent sum (10.4) contains an attachment class by definition. Choose any certified representative λ . The definition of its fibre recovers positive source presentations of the fixed Higgs reduct and the exact charge and weak antecedents, with the three inputs equal to the restrictions of the one mixed generator. These data satisfy every clause of $\text{Ante}_{H,EW}^{\text{rd}}$. Thus (i) and (ii) are equivalent.

The reserved metric domain requires executed vacuum/fixation, metric licence, and physical-response authority. Here those fields remain Need-coordinates. The absence of a comparison to the metric domain is therefore a typing fact, not an unproved negative physical claim. $\square$

Corollary 10.17 (Mixed attachment is separate data). *The singleton Higgs identity, the three mechanism types, and pointwise Higgs, charge, and weak packet data specify no mixed attachment by themselves. To obtain a point of $D_{H,EW}^{\text{rd}}$, clause (iv) of Definition 10.13 must also be supplied. An SR7 gauge-skeleton survivor has no Higgs attachment coordinate; a full Higgs source witness can provide that additional data.*

Proof. Even when the fine packet data give a point of $\mathcal{F}_{H,EW}^{\text{pre}}$, a point of the readiness domain additionally contains an element of the dependent attachment fibre. Omitting that fibre does not construct a mixed incidence. The conclusion follows by the definition of $D_{H,EW}^{\text{rd}}$. $\square$

10.5 A common Higgs–electroweak source witness

The Higgs bridge source provides more than a carrier name. Its neutral target retains one gauge/representation structure, residual/vacuum support, and deformation-response target, and its same-scope condition preserves those same ledgers [39, Defs. 4.1 and 4.5]. Its bridge certificate contains the joint source tuple

$$\mathcal{W}_H = (W_D^{\text{br}}, Q_D^{\text{ord}}, Q_D^{\text{def}}, N_D^{\text{res}}, \text{GVac}_D, \text{DefMed}, \text{MassResp}).$$

The Standard Model instantiation places that bridge within the electroweak structural tuple, preserving gauge inheritance, residual normal form, mediation, and mass-response readiness [39, Def. 7.3, Thm. 7.4 and Table 7; Thm. 7.6]. Restriction of this one tuple is stronger than separate existence of a Higgs identity and two gauge packets.

Definition 10.18 (A source-realized electroweak fibre). *An electroweak source realization Ξ is a full instantiated Standard Model gauge source together with the Higgs witness package occurring in that same source’s electroweak structural tuple. All source antecedents and dependency maps are retained. The gauge source receives the independent A1–A6, PSM1–PSM6, and SR1–SR7 chain and the full SM-2 source premises; its Higgs restriction is the source instantiation in the Higgs bridge closure theorem. In particular, it retains the projection $\pi_{\text{SR7}}(\Xi) = Z_\Xi$ to the minimal SR7 survivor used in the imported Standard Model source, rather than an arbitrary independently chosen gauge-arena element [38, Thm. 19.1] [13, §A2.8].*

The source-realized fibre is the image (D_Ξ, Γ_Ξ) of this one source under its certified target map. This refers to the full common instantiation, not merely to a list of individually inhabited source families. It does not assert that every formal value of Γ is realized. A BCRS mixed generator, a BCRS assignment, and a BCRS response-identity conclusion are not fields of Ξ .

The full SM-2 common source stack supplies this joint instantiation: its gauge dependencies retain the minimal SR7 survivor, while its electroweak restriction receives the Higgs bridge/fixation package [13, §§18.2, A1.1–A1.2 and A2.8]. The Higgs bridge theorem supplies the Higgs restriction of this stack, not the later gauge stack by itself. This use does not invert the dependency order: SR7 is an independent prior theorem, not a conclusion obtained from a BCRS assignment, and SM-2 receives that source result rather than supplying a premise for its proof. The common-source conclusion below is asserted on the image of this full instantiation, not uniformly on the larger formal family.

Lemma 10.19 (Common-source charge and weak restrictions). *Let Ξ be a source realization as in Definition 10.18. Its premetric charge and weak restrictions have certified presentations s_Q, s_{wk} satisfying Ante_Q and Ante_{wk} in (D_Ξ, Γ_Ξ) . Their evaluated uses are the actual charge and weak restrictions of the gauge/representation order inherited by $\mathcal{W}_H$; they need not be replaced by separately chosen isomorphic packets.*

Proof. Retain the gauge-order part of the one source tuple before applying any BCRS quotient. Its A1–A5 dependency maps supply the fixed interior, transition, transport, admitted loop, representation, charge-witness, and common anomaly-obstruction data of Definition 7.2. PSM1 supplies the admitted Lie specialization. The PSM2 classification is evaluated on this same sector family. Its independent-product branch retains quotient-trivial mixed residues; its lawful coupled branch retains the actual mixed-sector ledger. Both are same-domain compatibility statuses; a coupled status is not rewritten as product independence [27, Thm. 10.2; §13 and Props. 13.2–13.3].

For the charge restriction, PSM3–PSM4 receive the inherited representation and charge witness data. The complete Abelian-refinement certificate has the eleven fields of PSM4 Theorem 14.1 on this ledger. For the weak restriction, SM-2’s fixed weak-chiral response supplies a witness-bearing chiral

distinction. PSM3 Theorem 13.1, including its chirality clause, certifies that response realization rather than inserting a handedness label. All-instance source certification applies to these particular restrictions: the resulting generators and their composites evaluate to the same structural uses and preserve boundary, continuation, and restriction maps. Their root incidences witness activity.

The positive-preservation tests defined on this admitted source witness have passing values. Conditional ports retain their exact antecedents and are not assigned passing values merely because the source is listed. In particular, structural residual transport is executed only on the complete reduction domain of Theorem 7.13; metric, physical-vacuum, and particle-realization requirements remain separate. Nor are the diagnostic outcomes of counterfactual continuation tests replaced by passing values. This establishes its failure-pass clauses without requiring all future diagnostic outcomes to be positive.

The two composable antecedent records therefore have the required source dependencies and actual diagnostic data. Applying Theorems 7.6 and 9.5 forms their packet classes. Because certification applies to every lawful source instance, it certifies the restrictions already supplied by $\mathcal{W}_H$; no different member of either packet locus is substituted. $\square$

Theorem 10.20 (Source-relative mixed-readiness existence). *Every source realization Ξ determines an exact mixed-readiness antecedent*

$$\text{Ante}_{H,EW}^{\text{rd}}(s_H, s_Q, s_{\text{wk}}, \lambda_{\Xi}; D_{\Xi}, \Gamma_{\Xi}),$$

and therefore a point

$$d_{\Xi} = (h_{\Xi}, x_{Q,\Xi}, x_{\text{wk},\Xi}, [\lambda_{\Xi}]) \in D_{H,EW}^{\text{rd}}(\Gamma_{\Xi}).$$

Here $h_{\Xi} = [s_H]_{\text{pres}} \in \text{Pkt}_H^{\text{pre}}(D_{\Xi}, \Gamma_{\Xi})$ is the fine Higgs input, and $\text{rt}_H^{\text{car}}(h_{\Xi}) = \eta_H$ is its carrier identity. The three input uses and their mixed incidence restrict one common source ledger. Consequently the readiness domain is nonempty on this source-realized fibre. The theorem neither includes every charge/weak packet pair in that domain nor identifies all of its response signatures.

Proof. Apply all-instance source certification to the retained Higgs restriction $\rho_H^{\text{pre}}(\mathcal{W}_H^{\text{full}})$, obtaining a certified term s_H with that evaluated reduct. Take s_Q, s_{wk} from Lemma 10.19. Their fine quotient classes retain their full input records. All three inherit their domain, kernel, realized-interior record, global form, boundary, and continuation types from Ξ , so their anchor restrictions agree.

The witness W_D^{br} , its order quotient, and its reduced DefMed map already relate the inherited gauge/representation order to deformation-response standing. Retain the graph of this actual mediation together with its source-certified Higgs, charge, and weak restrictions, then restrict this common diagram to the BCRS premetric coordinates. Call the resulting source restriction λ_{Ξ} . Any remaining order data are fixed ambient data of this instance; they are not discarded by asserting that mediation factors through the charge and weak projections alone.

The three input legs of λ_{Ξ} are exactly the evaluated uses of s_H, s_Q, s_{wk} , because they were obtained from those same source projections. These are equalities of structural data, not equalities of proof terms or an incidence inferred from vertices. All-instance certification records the existing admitted diagram as $g_{EW,H}^{\text{pre}}[\lambda_{\Xi}]$, the single mixed schema with its full fixed source context. In particular, it supplies clause (iv) of Definition 10.13, rather than assuming that clause from the Higgs identity quotient.

Finally N_D^{res} , GVac_D , and MassResp retain their source dependencies. The target restriction carries their premetric interfaces and conditional arrows, not an executed vacuum value, numerical mass, metric response, or physical pole. These are the obligations of clause (v). The Higgs reduct, the two exact gauge antecedents, and common anchor establish clauses (i)–(iii). Hence the entire source antecedent holds, and Theorem 10.16 supplies d_{Ξ} .

The common ledger is the restriction of Ξ containing the three fine uses, the mediation incidence, and their retained boundary, continuation, and failure records. This establishes mixed readiness on one actual source witness. The remaining compatibility clauses and uniqueness require separate arguments. $\square$

The Higgs identity quotient therefore has one carrier class, and the source electroweak tuple supplies an actual mixed incidence with a nonempty readiness domain on its own fibre. Equality of all compatible response signatures is a distinct question requiring a complete-signature argument.

Proposition 10.21 (Three incidence maps remain distinct). *The fermionic structural incidence β_F , the premetric bosonic readiness specification $\beta_{H,EW}^{\text{rd}}$, and the reserved partial metric descent $\beta_{H,EW}$ are pairwise distinct typed maps. Equality of their shared carrier coordinate C_H identifies none of their domains, codomains, or proof obligations.*

Proof. The map β_F has a fermionic structural-incidence codomain. The map $\beta_{H,EW}^{\text{rd}}$ is total on the exact dependent domain containing a certified mixed attachment and has a premetric readiness codomain. The map $\beta_{H,EW}$ is partial on a metric-licensed, vacuum/fixation-authorized domain and has a physical-response codomain. These types are distinct. $\square$

Scope 10.22 (Higgs identity and readiness formation). *The same-scope Higgs identity and its target-relative mechanism reduct are fixed: $C_H = \mathcal{H}_{\text{br}/\text{fix}}$ is the unique identity root, and $\text{Ord}_H, \text{Def}_H, \text{Fix}_H$ are three distinct uses over that root. The mixed readiness map is formed exactly on $D_{H,EW}^{\text{rd}}$. The full common Standard Model source, including its Higgs restriction, proves this domain nonempty on its source-realized electroweak fibre, retaining the fine Higgs input separately from η_H . This existence is not inferred from the SR7 base class alone or pointwise packet existence. Metric descent, physical vacuum execution, and pole realization remain outside the target.*

Counterexample criterion 10.23 (Higgs identity and mixed-readiness failure conditions). *The conclusions of Theorems 10.2–10.16 fail if MGN3’s necessary filters are used as an existence theorem; if the fixed Higgs carrier package is inferred from SR7 rather than imported through its own certificate; if order, deformation, or fixation is counted as another identity; if the two restrictions of $g_{EW,H}^{\text{pre}}$ are replaced by separate generators; if packet nonemptiness is promoted to mixed-incidence existence; if a source-level PSM2 interface is identified with J_B ; if a prior coarse carrier certificate is promoted to a fine response singleton; if source enumeration is used instead of a global failure-pass evaluation; if the readiness map relies on later maximal interfaces; or if premetric readiness is promoted to an executed vacuum, metric, particle, mass, or pole projection.*

11 Intrinsic source descent and fixed-input covariance

Source covariance concerns an entire witnessed structure. A response comparison, however, may replace only one input of a mixed operation while holding its other inputs fixed. The bridge between these statements is the distinction between a structural operation and a presentation of that operation. We give the construction explicitly, including its partial domain. It neither identifies distinct physical attachments nor requires that every Higgs input have the same response.

11.1 Standing records and their presentations

The common realization ledger of SR1 is

$$W = (W_{\partial}, W_{\text{quot}}, W_{\text{trans}}, W_{\text{loop}}, W_{\text{field}}, W_{\text{part}}, W_{\text{rep}}, W_{\text{ch}}, W_{\text{anom}}, W_{\text{vac}}, W_{\text{res}}, W_{\text{def}}).$$

Every standing-bearing assertion of a component is witnessed through the corresponding typed entry and invariant under lawful redescription preserving this ledger. Realization equivalence is a finite zigzag preserving the full ledger up to skin, not equality of a short list of sector labels [32, Defs. 4.3–4.4 and 4.7]. Product equivalence retains this underlying realization comparison [33, Def. 5.2(xi)]. Representation,

charge, anomaly, and residual specializations retain their full dependent witness records [34, Def. 7.1][35, Def. 7.1] [36, Def. 7.1][37, Def. 8.1].

For the Higgs bridge the retained intrinsic tuple is

$$\mathcal{B}_H = (W_{\text{br}}/\sim_{\text{br}}, \Sigma_{\text{br}}, \beta_{\text{red}}, \rho_{\text{br}}).$$

The tuple includes the actual bridge-witness classes and reduced mediation map. It is not merely the name C_H , nor a sampled output of β_{red} . Its inherited gauge, order, deformation, residual, and guard data remain attached [19, Defs. 3.2–3.6, 6.2, and 7.1]. The existence of the fixed canonical Higgs package is supplied by the independent Higgs bridge theorem, not by a necessary-condition filter [39, Thms. 7.4 and 7.6].

Definition 11.1 (Intrinsic pointed source record). *A pointed source record is an actual finite witnessed diagram with its declared input and output ports, common anchor, incidence maps, boundary, continuation, guard, and conditional-domain records. Its intrinsic value is its class under the independently licensed, full-witness-preserving presentation comparisons of the participating sources. A comparison must preserve the whole pointed diagram, including every relative attachment, rather than only its separate marginal objects.*

Only a source-proved skin comparison generates this relation. An internal transport or a changed loop residue is not skin merely because it is an arrow of the diagram. A changed anchor, source-standing value, failure witness, or relative attachment is likewise not a presentation change. This source relation is fixed before, and without reference to, the BCRS signature relation $\sim_{\text{pres}}$. For a normalization or compression step, the source comparison must include relative descent of every incident map, admission predicate, and partial composite of this pointed record. A quotient of an isolated input object without those dependent descent data is not a comparison of the record. In particular, no arbitrary incoming map is presumed to lift through a noninvertible quotient.

Definition 2.2 uses these intrinsic structural values. Proofs of admission and displayed representatives can be retained when executing a diagram, but their names are not standing coordinates. The intrinsic source quotient removes only independently identified skin; the later BCRS quotient asks the different question of indistinguishability under the complete target-relative observation family.

11.2 The joint mixed record

A mixed presentation consists of one shared source ledger J , full Higgs, charge, and weak restrictions, and an actual mixed incidence:

$$L = (J; H_{\text{full}}, Q_{\text{full}}, W_{\text{full}}; p_H, p_Q, p_W; \lambda_{\text{str}}, o; \mathcal{D}).$$

Here $\mathcal{D}$ retains the boundary, continuation, guards, and exact conditional families. In particular, λ_{str} contains its actual bridge-witness class, order input, reduced-mediation incidence, and all shared maps. Let $[L]_{\text{src}}$ be its intrinsic pointed class. Restriction defines

$$\Phi_{\text{mix}}([L]_{\text{src}}) = ([p_H L]_{\text{src}}, [p_Q L]_{\text{src}}, [p_W L]_{\text{src}}; [\lambda_{\text{str}}]_{\text{src}}, [o]_{\text{src}}, [\mathcal{D}]_{\text{src}}). \quad (11.1)$$

The entries on the right retain their common incidence. This is a restriction of a joint quotient, not a freely assembled product of quotients of independently chosen margins.

Lemma 11.2 (Joint restriction descent). *The map Φ_{mix} is well-defined and preserves every declared source-standing restriction, including the attachment and conditional domains.*

Proof. For a generating lawful comparison, each field of the SR1 ledger and each incident comparison square is preserved. Restriction carries that comparison to the corresponding pointed port. On the Higgs restriction, the bridge reduction preserves the complete tuple $\mathcal{B}_H$, with its inherited ledger. Hence each entry of (11.1), and the maps relating the entries, has the same intrinsic value. Composition along a finite zigzag gives the assertion. A changed bridge signature is not identified by bridge reduction [19, Thms. 17.4 and 18.1–18.4]. No equality of BCRS signatures or response cardinality enters this proof. $\square$

Definition 11.3 (Fixed intrinsic mixed instance). *An intrinsic instance $\bar{\lambda}$ is the full source class of one admitted mixed-incidence record. It retains its actual attachment, output, and three pointed input restrictions. The operation $g[\bar{\lambda}](h, q, w)$ is defined exactly when the inputs match those intrinsic restrictions with the required shared-interface comparison. Its value is the retained output of that instance. A representative execution includes the witnessed presentation comparisons used for gluing. It does not search for another attachment over the same roots.*

Exact matching here is stronger than matching root names. It is also different from equality of raw names for two presentations of the same port. A gluing path that changes a standing loop, relative incidence, or guard is a change of structural data, not another execution of the same fixed instance.

Theorem 11.4 (Fixed-sibling source-domain reflection). *Let q, q' be presentations of the same full pointed source-standing charge port. For every fixed intrinsic mixed instance $\bar{\lambda}$ and fixed Higgs and weak inputs h, w ,*

$$(h, q, w) \in \text{dom } g[\bar{\lambda}] \iff (h, q', w) \in \text{dom } g[\bar{\lambda}].$$

Where defined, the intrinsic outputs, actual attachment, guards, and conditional domains agree. The analogous assertion holds for the weak slot. Neither the Higgs input nor the intrinsic operation is varied.

Proof. For an invertible presentation comparison $u : q' \rightarrow q$, re-present the charge leg of the admitted joint diagram. An incident outgoing map m becomes $m \circ u$; an incoming map f becomes $u^{-1} \circ f$; an endomorphism T becomes $u^{-1} T u$. Thus every incident square commutes. The other two input presentations are unchanged. This is transport of the presentation of the same diagram, not a change to its physical relative data. Its intrinsic attachment and output are unchanged by Lemma 11.2.

The reduced mediation is checked on its actual input. If a, a' are the corresponding order-witness presentations, gauge inheritance and order-quotient preservation give $\pi_{\text{ord}}(a') = \pi_{\text{ord}}(a)$. The fixed partial map β_{red} therefore has the same domain membership and value on both. Independently, transport of the same bridge witness preserves its class in $W_{\text{br}}/\sim_{\text{br}}$. This last assertion is not inferred from equality of mediation outputs and does not assume injectivity of β_{red} .

For a lawful normalization or compression comparison, use its certified common standing record together with the relative descent required in Definition 11.1. At this descended level the incident restrictions, domain predicates, and partial composites are the same. Reversing such a step uses equality at that common intrinsic record, not a raw inverse or an unproved lifting of incident maps. Applying these comparisons along the supplied finite zigzag re-presents the same intrinsic instance at q' . Reversing the zigzag proves the converse domain implication. Arrow congruence, restriction-compatible faithful descent, and fixed-comparison quotient domains are precisely the corresponding finite-comparison conditions in the covariant-differentiation construction [52, Def. 4.4 and App. A.3]. The argument for the weak slot changes only the pointed leg. $\square$

This theorem does not assert invariance under an arbitrary physical variation of one input. For example, in a bilinear display $z = h^T M q$, the presentation change $q' = U q$, $M' = M U^{-1}$ preserves z with h fixed. Changing q while fixing the physical attachment M need not do so. Only the former describes the same intrinsic mixed instance.

11.3 Conservation of the observation algebra

For precision, let π be the intrinsic presentation map on a probe's source type, and let π_o be the corresponding output map. A partial probe t descends with its domain when

$$\text{dom } t = \pi^{-1}(\text{dom } \bar{t}), \quad \pi_o \circ t = \bar{t} \circ \pi \quad \text{on } \text{dom } t. \quad (11.2)$$

For an operation use the presentation map π_{rel} of its *joint* input record, including its fixed intrinsic parameter and relative gluing data. The corresponding equations are

$$\text{dom } g = \pi_{\text{rel}}^{-1}(\text{dom } \bar{g}), \quad \pi_o \circ g = \bar{g} \circ \pi_{\text{rel}}.$$

These are not equations on a product of unrelated marginal quotients. For a fixed intrinsic instance they restrict to each fixed-sibling slice. Saturation of the domains is supplied by preservation of the actual source admission predicate, which is a standing field of the pointed record. A raw certificate-pointer execution test does not have this property and is not substituted for that admission predicate.

Theorem 11.5 (Source-standing conservation). *Every independently admitted source-standing observation of the fixed target factors through intrinsic source descent, with both its partial domain and its value preserved. The assertion holds for all port sorts and all finite licensed contexts with fixed intrinsic parameters and fixed sibling inputs. Consequently source skin comparisons lie in the complete BCRS signature kernel.*

Proof. For a primitive gauge test, SR1 ledger factorization and the specialized preservation records give invariance of the actual test and its admission predicate. Higgs tests retain the full bridge tuple and actual attachment, not only carrier identity. Extended color tests preserve the fixed local root, support class, and actual extended-to-local incidence. A negative test preserves its failed witness law; a conditional test preserves its dependent antecedent and implication without executing an absent input. Reversing a lawful source comparison gives domain reflection, not just agreement where two executions happen to exist. Thus its admission domain is saturated under the source relation. Set $\text{dom } \bar{t} = \pi(\text{dom } t)$; saturation gives the first equation in (11.2), and invariance of the retained output gives the second. The identical construction on joint records gives the operation equations above.

For a many-input vertex re-present the replaced leg and its incident maps; all fixed intrinsic parameters remain unchanged. The mixed case is Theorem 11.4; the same commuting-square argument uses the declared preservation laws at the other licensed vertices. Induction on a finite context now preserves its domain and, when defined, its output. Composing with a primitive probe preserves probe-definedness and value. These are exactly the totalized coordinates of Sig_B . Factorization through the source quotient follows from its universal property, uniquely on its image. $\square$

The argument establishes the fixed-parameter invariance required by Lemma 3.10. The semantic-use, activity, congruence, and compatibility constructions therefore apply to these intrinsic records with every source-standing distinction retained. Every actual finite witnessed diagram has a source certificate by Theorem 3.5; its intrinsic evaluation retains every declared witness and removes only presentation information.

A raw check that a certificate stores exactly a particular written input name is an execution check, not a standing observation. If q and q' are two names for one intrinsic port, such a check can distinguish them even though the underlying source law does not. It is excluded by the independently imposed redescription law, not because of a desired response count. In contrast, a genuinely different intrinsic input can fail the exact domain of $g[\bar{\lambda}]$; that distinction is retained. Thus source covariance closes the presentation issue without establishing that different marked response values are equal.

12 Simultaneous compatibility, role occupancy, and codescent

Simultaneous realization requires the three gauge packets and the Higgs interface to belong to one compatible source ledger. We define the exact eligibility and compatibility relations, the live response residue, and the lawful assignment action. The SR7 comparison identifies the common gauge-realization structure; its fixed-extension fibre retains the actual response assignments and requires their additional source attachments. Different attachment values remain in the family of common ledgers, rather than being identified by the gauge-assignment quotient.

The complete SM chain determines the carrier-root architecture. Section 13 strengthens this to rigidity of the complete structured gauge-law reduct. SCIC gives the separate exact signature criterion for uniqueness of the marked response assignment; the finer response cardinality follows only when that criterion holds. Every simultaneous ledger also supports a common-interface codescent construction, independently of uniqueness of the possible assignments.

Throughout, $\text{Rel}_B(\Gamma)$ denotes the exact source-word relations used to form $\mathcal{U}_B(\Gamma)$, while E_B below denotes role eligibility. These relations have different domains and perform different work. The fixed arguments D, Γ, G_Γ are the same as in the preceding section.

12.1 Exact role eligibility

For $r \in \mathbf{R}_g^0 = \{r_Q, r_{\text{col}}, r_{\text{wk}}\}$, define its qualified packet locus and exported packet evaluation by

$$\text{Pkt}_r = \begin{cases} \text{Pkt}_Q(D, \Gamma), & r = r_Q, \\ \text{Pkt}_{\text{col}}(D, \Gamma; G_\Gamma), & r = r_{\text{col}}, \\ \text{Pkt}_{\text{wk}}^{\text{pre}}(D, \Gamma), & r = r_{\text{wk}}, \end{cases}$$

$$\mathbf{I}_r(x) = \begin{cases} \mathbf{I}_Q(D, \Gamma)(x), & r = r_Q, \\ \mathbf{I}_{\text{col}}(D, \Gamma; G_\Gamma)(x), & r = r_{\text{col}}, \\ \mathbf{I}_{\text{wk}}^{\text{pre}}(D, \Gamma)(x), & r = r_{\text{wk}}. \end{cases}$$

Write $\text{Inc}_r(x)$ when $x \in \text{Pkt}_r$, when it admits a certified source lift, and when the root, target, global form, response use, boundary, continuation, and failure coordinates of $\mathbf{I}_r(x)$ have the types required by r . Source-lift existence certifies formation; proof ancestry is not a signature coordinate. Equivalently, the packet may be replaced by a source-preserving lawful isomorphic packet. This definition ranges over every packet root satisfying the kindwise incidence conditions; it does not select a preferred root or encode a uniqueness conclusion.

Definition 12.1 (Exact BCRS role eligibility). *The exact eligibility relation*

$$E_B \subseteq \mathcal{Q}_B^+(\Gamma) \times \mathbf{R}_g^0$$

is defined by

$$E_B(x, r) \iff E_B^0(x, r) \wedge \text{Inc}_r(x). \quad (12.1)$$

Its eligible target-relative residue is

$$\text{Residue}_g^{\text{elig}}(\Gamma) := \{x \in \mathcal{Q}_B^+(\Gamma) : \exists r \in \mathbf{R}_g^0 \ E_B(x, r)\}.$$

Theorem 12.2 (Eligibility exactness and completeness). *The relation E_B is presentation invariant and has exactly the positive edges licensed by the three kindwise incidence packets. It is neither enlarged by later compatibility nor narrowed by a particle, generator, polarization, pole, or confinement display.*

Proof. The pre-eligibility relation E_B^0 factors through the complete-signature quotient. Each predicate Inc_r is completeness of a dependent incidence packet whose source and use data have already descended through the same quotient, and this property is preserved by a lawful source-preserving isomorphism. Both conjuncts of (12.1) are therefore presentation invariant.

By definition, $E_B(x, r)$ holds exactly when $E_B^0(x, r)$ and the complete exported packet $\mathbf{I}_r(x)$ exist with the required typed coordinates. Each locus Pkt_r was itself defined by those two conditions, so equivalently

$$E_B(x, r) \iff x \in \text{Pkt}_r.$$

Theorems 7.6, 8.5, and 9.5 prove that every such member recovers an exact certified source use satisfying the corresponding antecedent, and that every such use descends to a member independently of presentation. This establishes exactness and completeness without appealing to an unproved global exhaustion principle. Compatibility and display data are absent from the definition and cannot modify it. $\square$

12.2 One simultaneous common-ledger predicate

For an eligible triple $\mathbf{x} = (x_Q, x_{\text{col}}, x_{\text{wk}})$, let $\omega_B(\mathbf{x}; \eta_H)$ be one witness ledger containing:

- (i) the common target and the single fixed Γ ;
- (ii) gauge order, transition, transport, curvature/holonomy, and the three role-incidence restrictions;
- (iii) Abelian charge normalization and residual charge response;
- (iv) aggregate anomaly-neutrality, global-form, product, and vacuum/residual guard values;
- (v) the unique Higgs identity, an actual point $d_\omega \in D_{H,EW}^{\text{rd}}$, and its $\beta_{H,EW}^{\text{rd}}(d_\omega)$ readiness record, retaining the fine Higgs input and mixed-incidence class; and
- (vi) common boundary, continuation, readiness, and failure records; and
- (vii) source-minimality, no-selector, and no-repair certificates.

The ledger is a dependent record. Separate witnesses for different entries cannot be spliced after the fact.

Definition 12.3 (Simultaneous BCRS compatibility). *For a rolewise eligible triple $\mathbf{x}$, define*

$$J_B(\mathbf{x}; \eta_H) = 1 \iff J_\Gamma \wedge J_{\text{gauge}} \wedge J_{\text{ch}} \wedge J_{\text{anom}} \wedge J_{\text{vac}} \wedge J_{\text{prod}} \wedge J_H \wedge J_\partial \wedge J_{\text{cont}} \wedge J_{\text{fail}} \wedge J_{\text{min}} \quad (12.2)$$

on one and the same ledger $\omega_B(\mathbf{x}; \eta_H)$. The clauses mean: fixed-global-form equality; common gauge transport and loop data; charge and residual-response agreement; anomaly neutrality; vacuum/residual guard agreement; product-sector co-instantiation; preservation of the one-Higgs identity and its premetric readiness interface; boundary agreement; continuation compatibility; and closure of every registered failure condition; and source-minimal, selector-free, no-repair formation. If no single ledger supports all clauses, $J_B = 0$, even when every pair admits a witness.

The anomaly clause is satisfied by a certified neutral obstruction value, not by deleting the anomaly coordinate. The Higgs clause uses $\beta_{H,EW}^{\text{rd}}$; it neither assumes nor executes the reserved metric map $\beta_{H,EW}$. More explicitly, J_H requires that the ledger retain an actual $d_\omega \in D_{H,EW}^{\text{rd}}(\Gamma)$ with

$$p_Q(d_\omega) = x_Q, \quad p_{\text{wk}}(d_\omega) = x_{\text{wk}}, \quad \text{rt}_H^{\text{car}}(p_H(d_\omega)) = \eta_H,$$

and that its recorded readiness value equal $\beta_{H,EW}^{\text{rd}}(d_\omega)$. The point retains its exact domain, fine Higgs input, and mixed-incidence class, together with the unexecuted vacuum, metric, and physical-response requirements. It is not recovered from the readiness value: no injectivity of $\beta_{H,EW}^{\text{rd}}$ is assumed.

Lemma 12.4 (Compatibility descent). *The value of J_B is invariant under presentation equivalence in every argument and under replacement of the common ledger by a lawfully isomorphic ledger. Consequently J_B is a predicate on quotient classes, not on chosen gauge or scalar representatives.*

Proof. Every clause of (12.2) is either a coordinate of the complete signature or an equality between evaluations of registered semantic uses. The former is fixed by presentation equivalence and the latter descends by the universal semantic-use factorization theorem. Congruence preserves all finite products and incidence constructors. A lawful ledger isomorphism preserves the same coordinates and equalities, so it preserves their conjunction. $\square$

Proposition 12.5 (Pointwise packets do not imply a joint packet). *Corollary 10.9 makes each rolewise packet locus inhabited in one fixed source fibre, but this fact and the SR1–SR6 projection-preservation results do not imply*

$$\exists \mathbf{x} \, J_B(\mathbf{x}; \eta_H) = 1.$$

Joint existence is equivalent to inhabitation of the source-preserving extension fibre defined below. The SR7 base theorem alone does not supply such an inhabitant.

Proof. Pointwise packet inhabitation supplies three eligible vertices and their shared interface types. It does not supply one value for every field of ω_B . In particular, SR7 Proposition 3.1 explicitly separates survival under each of the SR1–SR6 projections from joint realization. Treating the conjunction of marginal survivals as $J_B = 1$ would therefore be an invalid assembly inference. The common-ledger condition is therefore an additional mathematical requirement, discharged by the common-source construction below. $\square$

12.3 Live residue and lawful assignment orbits

Definition 12.6 (Compatible pre-assignments and live residue). *Let $\Sigma_g^{\text{pre}}(\Gamma)$ be the set of injections*

$$\tau : R_g^0 \hookrightarrow \text{Residue}_g^{\text{elig}}(\Gamma)$$

such that $E_B(\tau(r), r)$ for every $r \in R_g^0$ and

$$J_B(\tau(r_Q), \tau(r_{\text{col}}), \tau(r_{\text{wk}}); \eta_H) = 1.$$

The live target-relative residue is

$$\text{Residue}_g^{\text{live}}(\Gamma) := \bigcup_{\tau \in \Sigma_g^{\text{pre}}(\Gamma)} \text{im}(\tau). \quad (12.3)$$

Only an eligible class occurring in a simultaneous compatible complete pre-assignment is live.

Definition 12.7 (Certificate-lawful automorphisms). *Let $\mathcal{I}_{B,\Gamma}$ be the invariant tuple*

$$(\tau_B, \mathbb{I}_B, R_g^0, \text{Residue}_g^{\text{live}}, E_B, J_B, \text{Sig}_B, \mathcal{T}_B, \text{Ctx}_B, \mathcal{O}_B, \text{Dom}_B, \mathcal{U}_B^{\text{upd}}, \partial, \text{Cont}, \text{Fail}, \text{Hld}_B, \pi_\mu, \beta_{H,EW}^{\text{rd}}).$$

Here Dom_B denotes the exact definedness predicates for individual operations, one-hole contexts, and target tests. The full context family is retained, not merely its unary-update submonoid. The certificate-lawful group is

$$\Gamma_g^{\text{lawful}}(\Gamma) := \text{Aut}(\mathcal{I}_{B,\Gamma}). \quad (12.4)$$

Its elements preserve the rigid role tags and every entry of the invariant tuple. They may relate lawful presentations or indistinguishable incidence representatives; they may not erase a standing-bearing

distinction or exchange charge, color, and weak roles. Signature preservation is pointwise on the fixed target-test domains:

$$\text{Sig}_B(\alpha x) = \text{Sig}_B(x) \quad \text{for every } x \in \text{Residue}_g^{\text{live}}.$$

Thus neither a permutation of the admitted probes nor a relabelling of their value codomains counts as preservation of a signature coordinate. The same requirement fixes all context and operation domains and the distinct undefined marker $\perp_{\text{undef}}$. An unexecuted operation cannot become executed under a lawful action.

Lemma 12.8 (Lawful action on the response quotient). *Every certificate-lawful automorphism acts as the identity on the live response quotient and hence on compatible pre-assignments.*

Proof. The complete-signature map is injective on the quotient formed by equality of those signatures. Hence $\text{Sig}_B(\alpha x) = \text{Sig}_B(x)$ implies $\alpha x = x$. Rigid role tags are fixed, so postcomposition fixes every rolewise assignment. This does not require an automorphism to be the identity on its presentation or certificate representatives. $\square$

Definition 12.9 (Compatible assignments and their orbit). *Before totality is known, the compatible packet-orbit object is*

$$\tilde{\Sigma}_B(\Gamma) := \Sigma_g^{\text{pre}}(\Gamma) / \Gamma_g^{\text{lawful}}(\Gamma).$$

The compatible assignment set is

$$\Sigma_g(\Gamma) := \{\sigma \in \text{Bij}(\mathbf{R}_g^0, \text{Residue}_g^{\text{live}}) : E_B(\sigma(r), r) \text{ for every } r, \\ J_B(\sigma(r_Q), \sigma(r_{\text{col}}), \sigma(r_{\text{wk}}); \eta_H) = 1\}. \quad (12.5)$$

After totality has been proved, the BCRS gauge-occupancy object is the orbit set

$$\Sigma_B(\Gamma) := \Sigma_g(\Gamma) / \Gamma_g^{\text{lawful}}(\Gamma).$$

The preliminary orbit is defined even if the compatible pre-assignment set is empty. The use of bijections in Σ_g is not a cardinality premise. The live residue was first defined from all compatible injections. A bijection exists only if no excess live candidate and no unoccupied role remain after that formation.

12.4 Exact SR7 reduction and the fixed-extension fibre

The reduction to SR7 and the existence of a BCRS extension are distinct results. The unique load-minimal SR7 class does not by itself construct the three gauge packets, their common BCRS ledger, or the additional fixed attachments. The common-source construction below supplies these data without presuming uniqueness of their marked responses.

Definition 12.10 (Exact ten-clause SR7 regime record). *Let $\mathbf{A}_{\text{SR7}}(D, \Gamma)$ be the dependent record whose ambient field fixes D as a nondegenerate realized physical interior, its AASC kernel K_D , its complete Realized Physics role ladder, and the mass-pre-dynamical fixed-global-form comparison fibre at Γ . Its ten regime fields are exactly:*

- (i) the PSM1–PSM6 licensed arena;
- (ii) SR1 simultaneous survivor coupling;
- (iii) SR2 product-role survivability;
- (iv) SR3 representation-response survivability;

- (v) *SR4 Abelian charge-coordinate survivability;*
- (vi) *SR5 anomaly-ledger survivability;*
- (vii) *SR6 residual/vacuum-sector survivability;*
- (viii) *a certificate that no Higgs, Yukawa, mass-generation, weak-mixing, or empirical-fit assumption is used in the SR7 reduction;*
- (ix) *a certificate that no hidden parameter, coupling, scale, normalization, representative, gauge, regulator, branch, or vacuum selector is used; and*
- (x) *a certificate that no larger-domain rescue or undeclared repair is used.*

These are the ten clauses of SR7 Definition 7.1, not a compressed proxy for them. The eighth field does not deny the independently formed BCRS Higgs readiness attachment. It requires that this attachment be forgotten before SR7 standing is evaluated and that it perform no work in that evaluation.

Lemma 12.11 (Discharge of the exact SR7 regime). *For every admitted Γ , the conservative import object determines an inhabitant*

$$a_{\text{SR7}}(\Gamma) \in \mathbf{A}_{\text{SR7}}(D, \Gamma).$$

No BCRS packet, compatible assignment, extension witness, orbit, or response count is used in forming this inhabitant.

Proof. The realized-interior and kernel fields are supplied by Proposition 1.3. The PSM1–PSM6 and SR1–SR6 fields are the individually instantiated source interfaces retained in $\mathbb{I}_B(\Gamma)$ by Theorem 1.7; none is obtained from the SM-2 capstone or from a BCRS joint-assembly conclusion. Target finality fixes the domain, global form, nondegeneracy, and mass-pre-dynamical granularity. The target exclusion ledger supplies clauses (viii)–(x): the SR7 reduct uses no Higgs or mass-sector assumption, no hidden selector, and no repair or domain enlargement. Thus each of the ten fields is discharged directly at its own source interface. $\square$

Let $\mathbf{S}_{\text{SR7}}(D, \Gamma)$ be the joint-survivor object of SR7 Definition 5.1 and let

$$\mathbf{M}_{\text{SR7}}(\Gamma) := \mathbf{S}_{\text{SR7}}(D, \Gamma) / \sim_{\text{SM}}.$$

Write $\mathbf{M}_{\text{SR7}}^{\min}(\Gamma)$ for the subobject of classes which are load-minimal in the SR7 sense. Lemma 12.11 and SR7 Theorem 19.1 give a distinguished class $z_*(\Gamma)$ and prove

$$\mathbf{M}_{\text{SR7}}^{\min}(\Gamma) = \{z_*(\Gamma)\}. \quad (12.6)$$

This is a theorem about the SR7 gauge-realization skeleton. It says nothing yet about the fibre of BCRS extensions over that skeleton.

Definition 12.12 (Source-defined minimality of the common ledger). *For any prospective common ledger ω , let $L_B(\omega)$ be its structural reduct retaining the SR2–SR6 candidate subledgers and their common SR1 interfaces. This operation is defined on records before any compatibility verdict. It forgets the Higgs and additional Yang–Mills attachments. Let Min_{SR7} be the load-minimality predicate of the independently fixed SR7 comparison regime, extended by false outside its joint-survivor object. Thus $\text{Min}_{\text{SR7}}(y)$ holds precisely when $y \in \mathbf{S}_{\text{SR7}}(D, \Gamma)$ and no proper lawful deletion of standing-bearing load preserves all its survivor requirements. This is the existing source predicate on a prospective reduct, not a BCRS uniqueness condition. Then*

$$J_{\min}(\omega) = 1 \iff \text{Min}_{\text{SR7}}(L_B(\omega)) \wedge \text{NoSelector}(\omega) \wedge \text{NoRepair}(\omega). \quad (12.7)$$

This uses a source-defined deletion test, not a BCRS response count or a chosen list of surviving presentations. It asserts neither that a compatible ledger exists nor that two compatible ledgers are equivalent. Singletonhood of an assignment orbit is not a field of J_B .

Definition 12.13 (SR7 forgetful reduction). *Let $\tau \in \Sigma_g^{\text{pre}}(\Gamma)$, and let ω_τ be a ledger witnessing its J_B -equation. Define $F_B(\tau, \omega_\tau)$ by retaining, on that same ledger,*

- (a) *the product, representation-response, Abelian charge, anomaly, and residual/vacuum survivor sub-ledgers;*
- (b) *their common boundary, transport, loop-residue, deformation-baseline, no-selector, and no-repair data; and*
- (c) *the three SR7 product positions $(C_{\text{Ab}}, P_{\text{pers}}, P_{\text{resp}})$, read respectively from the charge, color, and premetric weak role restrictions.*

It forgets proof ancestry, BCRS-only packet presentation, the fixed Yang–Mills extended-support attachment beyond its retained persistence-role restriction, and the Higgs identity/readiness attachment. No standing-bearing SR7 coordinate is forgotten. The resulting orbit map is denoted

$$\Theta_B : \tilde{\mathfrak{S}}_B(\Gamma) \longrightarrow \mathbf{M}_{\text{SR7}}^{\min}(\Gamma).$$

No reverse map is defined here.

Theorem 12.14 (Soundness of the one-way SR7 reduction). *The reduction F_B takes every compatible BCRS pre-assignment to a load-minimal SR7 joint survivor, and Θ_B is well defined. Therefore*

$$\Theta_B(\tilde{\mathfrak{S}}_B(\Gamma)) \subseteq \{z_*(\Gamma)\}. \quad (12.8)$$

Equation (12.8) remains true when the source is empty. It is not a surjectivity, injectivity, existence, or orbit-uniqueness theorem.

Proof. Exact role eligibility supplies the kindwise gauge and refinement data; it does not by itself assert joint SR survival. The source-domain clause of $J_{\min}$ places $L_B(\omega)$ in $\mathbf{S}_{\text{SR7}}(D, \Gamma)$, and all of its witness maps agree because they are restrictions of the one J_B -ledger. Definition 12.10 supplies the ten-clause regime data. In particular, the Higgs/readiness fields and the extra Yang–Mills attachment are removed rather than used as SR7 premises. The retained object is therefore a point of the fixed-domain SR7 pullback, not a tuple assembled from marginal witnesses.

The minimality clause is exactly $\text{Min}_{\text{SR7}}(L_B(\omega))$, so the reduct is load-minimal without a BCRS uniqueness premise. Any other ledger witnessing the same assignment, and any certificate-lawful redescription, yields another load-minimal SR7 survivor. Equation (12.6) identifies their SR7 classes with z_* . Thus the class of the reduct is independent of these choices and defines Θ_B . This argument does not identify the full BCRS ledgers with one another. $\square$

Fix a representative Z_* of $z_*(\Gamma)$. The two attachment interfaces used below are independent of SR7:

- $\eta_{\text{YM}}^{\text{fix}}$ is the restriction to the fixed local Yang–Mills theorem object together with its distinct theorem-scope extended-support sector layer, supplied by $\mathbb{I}_B(\Gamma)$ and the Yang–Mills endpoint theorem [47];
- η_H^{fix} is the one Higgs identity, its order, deformation, and fixation interfaces, and the independently formed family $\beta_{H,EW}^{\text{rd}} : D_{H,EW}^{\text{rd}} \rightarrow \mathcal{I}_{H,EW}^{\text{pre}}$ of Theorems 10.2–10.16 [39, 40, 44].

Neither attachment is extracted from an SR7 representative, and neither executes confinement, a vacuum choice, a metric response, or a particle projection. Fixing the Higgs readiness family does not choose a point of its domain. An extension must still supply an actual mixed witness whose input restrictions are its own charge, weak, and Higgs presentations.

Definition 12.15 (Raw gauge-assignment fixed-extension fibre). *The raw BCRS gauge-assignment fixed-extension fibre over Z_* is the following subobject of the compatible pre-assignment set:*

$$\widehat{\text{Ext}}_B(Z_*) := \{\tau \in \Sigma_g^{\text{pre}}(\Gamma) : \exists \omega_\tau, \kappa_\tau, \eta_{\text{YM},\tau}^{\text{fix}}, \eta_{H,\tau}^{\text{fix}} \text{ satisfying clauses (i)–(v) below}\}.$$

The clauses are:

- (i) τ is the displayed rolewise eligible injection into the already formed BCRS quotient;
- (ii) ω_τ is one common ledger witnessing every clause of $J_B(\tau(r_Q), \tau(r_{\text{col}}), \tau(r_{\text{wk}}); \eta_H) = 1$, with $J_{\min}$ interpreted by Definition 12.12;
- (iii) $\kappa_\tau : F_B(\tau, \omega_\tau) \sim_{\text{SM}} Z_*$ is an SR7 realization equivalence;
- (iv) $\eta_{\text{YM},\tau}^{\text{fix}}$ identifies the color packet's local and extended-support restrictions with the independently fixed Yang–Mills attachment; and
- (v) $\eta_{H,\tau}^{\text{fix}}$ witnesses a point of the independently formed readiness domain with these charge and weak restrictions and the fixed Higgs identity; its image is the readiness entry of the same common ledger. Metric and vacuum-executed fields remain open.

The quantifiers following τ assert that the source and attachment obligations are satisfied; the elements of this set are gauge assignments, not full witness records. Thus two distinct common ledgers or readiness points supporting the same τ give one point of this set without being identified as ledgers or readiness data. Their actual values remain distinct members of the common-ledger family below. In particular, singletonhood of this gauge-assignment fibre implies neither a unique Higgs response, a unique mixed-incidence point, nor a unique codescent object. Source and attachment existence are not recovered merely by reading the three role names in Z_* .

Definition 12.16 (Lawful action and extension-orbit fibre). *Let $\Gamma_{B,z_*}^{\text{ext}}(\Gamma)$ be the subgroup of $\Gamma_g^{\text{lawful}}(\Gamma)$ whose induced SR7 redescription fixes z_* and which preserves the fixed Yang–Mills and Higgs attachment interfaces. If $\alpha \in \Gamma_{B,z_*}^{\text{ext}}$, its action on a raw fibre point is*

$$\alpha \cdot \tau := \alpha\tau.$$

Here $\alpha\tau$ means postcomposition on the candidate coordinate; the rigid role tags are fixed. The property witnesses are transported by α_* , so $\alpha\tau$ remains in the raw fibre. The BCRS extension-orbit fibre is

$$\text{Ext}_B(z_*) := \widehat{\text{Ext}}_B(Z_*) / \Gamma_{B,z_*}^{\text{ext}}(\Gamma).$$

Changing Z_* within its SR7 class gives a canonically isomorphic orbit fibre. Proof witnesses and presentation choices therefore do not multiply its standing content.

Definition 12.17 (Gauge-assignment fixed-extension closure predicate). *Define*

$$\text{FibreClosed}_B(z_*) = 1 \iff \widehat{\text{Ext}}_B(Z_*) \neq \emptyset \text{ and } \Gamma_{B,z_*}^{\text{ext}} \text{ acts transitively on } \widehat{\text{Ext}}_B(Z_*). \quad (12.9)$$

This is a gauge-assignment existence-and-fidelity condition. It is not included among the clauses of J_B , and it is not supplied by (12.6).

12.5 What the complete Standard Model chain fixes

The comparison uses the complete carrier-exhaustion chain, not only the gauge skeleton. The source dependence is acyclic. The AASC kernel, the Realized Physics and Gauge Realization sequences, PSM1–PSM6, and SR1–SR6 supply an independently instantiated SR7 comparison regime. SR7 Theorem 19.1

then supplies its unique minimal class, and the full SM-2 certificate may receive that result as a premise. Neither source theorem requires a BCRS quotient, assignment, or response count. Consequently the independent SR7–SM-2 chain may be received in the BCRS source registry before the finer response quotient is constructed. A later BCRS-to-SR7 comparison, such as Lemma 12.11, verifies that its reduct belongs to the same source regime; it need not create that regime for the first time. Circularity would arise only if the source antecedents were proved from the very BCRS compatibility or uniqueness conclusion being inferred from them. The full SM-2 certificate’s other source premises include the full Yang–Mills package, the Higgs and mass-fixation package, and the L1 electroweak equivalence package. The latter contains charged-current collapse, electroweak/electromagnetic bookkeeping, longitudinal/Goldstone equivalence, hypercharge/charge normalization, anomaly/Yukawa compatibility, neutral-current/FCNC discipline, and equivalence-space closure. A source interface must instantiate those premises on the same (D, Γ) ; merely naming SM-2 does not instantiate them. These are source-exact dependencies: the charged-current and electromagnetic equivalences concern declared low-energy or broken-phase descriptions; the longitudinal equivalence is restricted to its high-energy on-shell regime; and the Yukawa, FCNC, and charge-normalization results fix the declared chiral content and one electroweak Higgs doublet [53, 54, 55, 56, 57, 58, 59]. Mass fixation is used only through its structural role within the stated gauge-invariant bookkeeping scope [60]. None of these dependencies is erased by taking a premetric reduct.

Definition 12.18 (Complete post-SR7 carrier-source interface). *Let $\mathcal{C}_{\text{SM}}^3(\Gamma)$ be the dependent record of exact instantiated source interfaces which receives:*

- (i) *the SM-1 fixed-target normal-form and no-escape results;*
- (ii) *the full SM-2 gauge carrier certificate, after the independently established SR7 antecedent and base-class result;*
- (iii) *the closed SM-3 matter, SM-4 charge/coupling/transport, and SM-5 flavor packages, including their registered-status and boundary data;*
- (iv) *the generation-governance receipt: the original primitive duties, independently derived occurrence architecture, complete compatible occupation, and its source-defined root transport; and*
- (v) *SM-6 constructor adequacy, role-necessity exhaustion, and global carrier-exhaustion closure.*

Every interface retains its full source antecedent and its target map, as in Definition 1.6. This record contains no BCRS assignment, mixed-attachment witness, response-signature equality, or assignment-orbit conclusion.

The generation field receives the count-independent primitive-standing predicate and its exact classification

$$\text{Prim}_{\mathcal{P}_0}(I) \iff |I| = 3, \quad \text{ClosedGenCert}_{\mathcal{P}_0}(n) \iff n = 3$$

from [18, Def. 6.1, Thms. 6.2 and 5.4]. Its interpretation retains all original generation duties and their complete direct-comparison meaning. The source record carries the independently derived role architecture $R_{\mathcal{P}_0}$, a compatible injective assignment onto the entire claimed live occupant set, and one joint receipt for that assignment, as required by [18, Def. 5.3]. The SM-6 receiving construction identifies its physical occurrences by actual complete occupation [17, Sec. 3.8, Prop. 3.1].

The inherited FCGS source identifies the position object $I_3 = \text{Occ}_N$ with the received primitive occurrence architecture, retaining its external S_3 -action and positioned matter fibres [61, Thms. 3.6 and 4.5]. As a bare graph it is $V(K_3)$; $V(K_3)/S_3$ counts vertex orbits and is a different object. Its circuit rank one follows after this architecture and occupation have been established [18, Cor. 5.5]. For unchanged original generation work, a larger architecture gains no additional simultaneous primitive role merely from further consistent responses [18, Thm. 6.4].

A source may retain a nonempty relation of licensed primitive receipts without selecting a preferred embedded triangle. Primitive count is constant on that relation; any other original reader must also agree on each receipt fibre and with the receiving source's prescribed answer [18, Thm. 6.6 and Prop. 6.7]. These are retained source-use conditions. The generation field governs the matter/flavor target; it neither derives the three bosonic response roles nor adds a primitive bosonic carrier.

Lemma 12.19 (Exact carrier-exhaustion payload of the SM chain). *An inhabitant of $\mathcal{C}_{\text{SM}}^3(\Gamma)$ supplies a coherent closed carrier architecture $C_{\text{SM}}^{\text{prim}}$ and proves*

$$\text{RoleNecessity}_{\text{SM}}(x) \wedge x \in \text{Attr}(D_{\text{SM}}^3) \wedge \neg \text{TargetUpdate}_{\text{SM}}(x) \implies x \in C_{\text{SM}}^{\text{prim}} \vee x \downarrow C_{\text{SM}}^{\text{prim}},$$

and

$$\text{Surv}_{D_{\text{SM}}^3}^{\text{prim}}(\text{Attr}(D_{\text{SM}}^3) \setminus C_{\text{SM}}^{\text{prim}}) = \emptyset.$$

It does not identify two registered response, route, readiness, or continuation values merely because neither is a new primitive carrier.

Proof. SM-1 Theorems 13.1 and 13.6 give the normal-form and sector-relative no-escape discipline. After SR7, SM-2 Theorems 22.4–22.5 close the gauge carrier package and delete residual primitive gauge status. SM-3 Theorems 19.9 and 19.11 supply matter-package closure and its scoped primitive deletion. SM-4 Theorems 19.3 and 19.5 do the same for charge and transport, and SM-5 Theorems 18.5–18.6 for flavor. The generation theorem fixes the previously governed family field at three without adding a carrier. These are precisely the local packages received by SM-6. Its Theorems 19.10, 19.16, and 20.1 prove, respectively, role-necessity exhaustion, absence of a valid same-scope hostile primitive certificate, and the displayed global primitive deletion [12, 13, 14, 15, 16, 17, 18].

The final sentence preserves the codomain of those results. For example, SM-4 retains transport, calibration, running, and projection ledgers as registered statuses; SM-6 Corollary 20.2 expressly leaves the corresponding nonprimitive surfaces open. Their nonpromotion into a new carrier is not an equality theorem for their values. If a registered value is an admitted BCRS standing coordinate, its equality must therefore be proved at that coordinate rather than inferred from primitive deletion. $\square$

Definition 12.20 (Carrier-root projection of the response quotient). *Let*

$$\mathcal{Q}_{B,g}^+ := \mathcal{Q}_{B,Q}^+ \sqcup \mathcal{Q}_{B,C}^+ \sqcup \mathcal{Q}_{B,\text{EW}^{\text{pre}}}^+, \quad C_g^{\text{resp}} := \{C_Q, C_{\text{col}}, C_{\text{wk}}^{\text{pre}}\}.$$

The carrier-root projections are

$$\text{rt}_g^{\text{car}} : \mathcal{Q}_{B,g}^+ \longrightarrow C_g^{\text{resp}}, \quad [s]_{\text{pres}} \longmapsto \text{root}(s),$$

The Higgs projection $\text{rt}_H^{\text{car}} : \mathcal{Q}_{B,H}^+ \rightarrow \text{Hld}_B$ is the already defined map of Lemma 10.3, with value $\eta_H = [C_H]_{\text{id}}$. These maps forget response-level detail while retaining the already certified carrier identity. They are not further quotients proposed to replace the complete BCRS response identity relation.

Theorem 12.21 (Positive bosonic carrier-root classification). *At every source-certified common fibre supporting $\mathcal{C}_{\text{SM}}^3(\Gamma)$, the two root projections are well defined and satisfy*

$$\text{im}(\text{rt}_g^{\text{car}}) = \{C_Q, C_{\text{col}}, C_{\text{wk}}^{\text{pre}}\}, \quad \text{im}(\text{rt}_H^{\text{car}}) = \{\eta_H\}.$$

The gauge part consists of charge-response, color-transport, and pre-vacuum weak-chiral roots. The Higgs is the separate bridge/fixation identity. Common order, transport, and curvature support, anomaly and vacuum guards, and product compatibility are not additional peer response roots. Thus the carrier-root architecture has the separately typed cardinalities

$$(|\text{im}(\text{rt}_g^{\text{car}})|; |\text{Hld}_B|) = (3; 1).$$

This is a carrier-root classification, not the cardinality of the full response quotient, an assertion that either projection is injective, or a compatible-live response count.

Proof. Root and kind are coordinates of the complete signature, so both maps descend through $\sim_{\text{pres}}$. Positive charge, color, and weak source packets are supplied by Corollary 10.9; each witnesses a point over its root. The fixed Higgs package supplies a positive point over the unique identity of Theorem 10.2. Hence the displayed images are populated, rather than inferred from the presence of root names alone.

For exhaustiveness, the full SM-2 certificate separates its common gauge-order/transport/curvature spine, its charge/color/weak response roots, and its anomaly/vacuum guards. This is the typed decomposition of Proposition 2.5, received through the independent source chain rather than inferred from a BCRS count. SM-2's gauge deletion theorem and SM-6's role-necessity and global exhaustion theorems exclude an additional same-scope primitive gauge-response root outside that package. A support or guard cannot become a peer root without changing its type; a registered response or source variation over an existing root is not a new primitive root. The Higgs identity and its three distinct mechanism uses remain separated by Theorems 10.2–10.5. The sort distinction makes the three gauge roots pairwise distinct and keeps the Higgs identity separate. None of these arguments identifies distinct response signatures lying over the same root. $\square$

12.6 Source reception and value-level extension fidelity

The comparison distinguishes fixed target data, complete source laws, and marked evaluations of those laws. The anchors $D, K_D, \text{RPI}_D, \Gamma$ and boundary/continuation types are fixed before candidate formation. The SR7 class preserves the joint survivor records under full witness-preserving redescription; their complete gauge-law reduct is reconstructed in Section 13. The Yang–Mills attachment fixes the local-root and support-sector family. The Higgs attachment fixes its identity, order–deformation–fixation interfaces, and the full readiness family $\beta_{H,EW}^{\text{rd}}$, not a chosen point of that family's domain. A compatible realization supplies an actual point with its exact input restrictions. Its response, boundary trace, continuation, readiness, and failure values remain marked evaluations; their fixed types do not turn those values into target anchors.

12.7 Simultaneous source-relative occupation

The same-source requirement also controls the color attachment. A Yang–Mills theorem object chosen independently of the electroweak source would not suffice. In a full Standard Model source realization, however, the color closure is evaluated in the fixed gauge target, with its actual Yang–Mills antecedent and source target map. The following result uses that existing restriction.

Lemma 12.22 (Same-source color restriction). *For a full source realization Ξ , let $\mathcal{B}_{\text{YM},\Xi}^{\text{full}}(G_{\Gamma_\Xi})$ be the Yang–Mills source restriction used by its SM-2 color certificate. Then the color restriction of Ξ has a certified presentation s_{col} satisfying*

$$\text{Ante}_{\text{col}}(s_{\text{col}}, \text{ev}_B(s_{\text{col}}); D_\Xi, \Gamma_\Xi, G_{\Gamma_\Xi}).$$

Its color packet retains the restricted local-root/extended-support incidence of this very Yang–Mills object.

Proof. The full SM-2 color certificate supplies the non-Abelian transition/transport/response use and its dependence on the Yang–Mills package [13, Thm. 17.2 and its color certificate]. The A1–A5 and PSM1–PSM3 restrictions are evaluated on the same source gauge and response ledger as in Lemma 10.19. They supply the actual non-Abelian distinguishing witness and the complete algebraic-refinement record. The fixed Yang–Mills restriction supplies the local object, its theorem-scope global form, and the extended-support sector construction over that unchanged local object [47, Thm. II.1(A)–(E)].

Only the structural reduct $\mathcal{B}_{\text{YM}}^{\text{pre}}$ is carried into the packet. Its local-root identity, support/deformation equivalence, and sector-to-local incidence remain those of the source restriction. The vacuum vector,

Hamiltonian, spectral conclusions, and particle interpretation are not premetric coordinates. The all-instance classifiers therefore supply a certified source term with the exact color mechanism, boundary, continuation, role-incidence, and current failure values. These are all the clauses of Ante_{col} . Theorem 8.5 forms its positive packet. $\square$

Theorem 12.23 (A full source supplies a compatible BCRS pre-assignment). *Every full source realization Ξ determines*

$$\tau_{\Xi} \in \Sigma_g^{\text{pre}}(\Gamma_{\Xi}).$$

In particular, the compatible pre-assignment set and the compatible live residue are nonempty on the source-realized fibre. If the fixed Higgs and Yang–Mills interfaces are the restrictions of this same source, the associated fixed-extension fibre is also nonempty. This statement is existence; it does not identify the fine response signatures of different source-compatible pre-assignments. Its common ledger also contains the actual residual charge packet $\mathcal{R}_Q(h_{\text{res},\Xi})$ and its structural transport comparisons, obtained by restriction of this same source.

Proof. Let $x_Q, x_{\text{col}}, x_{\text{wk}}$ be the quotient classes of the three source restrictions just constructed, and put $\tau_{\Xi}(r_b) = x_b$. The packet theorems give $E_B(x_b, r_b)$. Since root sort is a rigid complete-signature coordinate, the three classes are distinct and τ_{Ξ} is injective.

Form ω_{Ξ} by restricting the one source tuple Ξ , retaining its three packet uses, the Higgs mixed witness, the common gauge, charge, anomaly, residual/vacuum, product, boundary, continuation, and diagnostic data. Check the compatibility clauses in their stated order. The source target map gives J_{Γ} , and restriction of its one gauge ledger gives J_{gauge} . Theorem 7.13 supplies $h_{\text{res},\Xi}$, its actual residual charge packet, and the inherited or lawfully reduced transport comparisons. These are retained fields of the source residual ledger, not a new independently chosen compatibility hypothesis. The charge and residual fields agree through the same SR6 shared pullback, giving J_{ch} . Its SR5 anomaly-neutrality and SR6 residual/vacuum survivor data give J_{anom} and J_{vac} . The inherited sector status and actual mixed ledger give J_{prod} ; lawful coupled status is retained, not replaced by an independence claim. The point d_{Ξ} of Theorem 10.20 gives J_H .

Boundary and continuation restrictions come from one source and therefore satisfy J_{∂} and J_{cont} . Every failure condition evaluated in this common admitted source ledger has its certified source verdict, and the positive current ledger has no failed preservation test, giving J_{fail} . Unexecuted conditional ports are not assigned passing values; their antecedents and implications are retained. Nor are the diagnostic outcomes of counterfactual continuation tests replaced by passing values.

For minimality, let Z_{Ξ} be the independent SR7 source survivor used in the SM-2 certificate. Its class is the load-minimal class of SR7 Theorem 19.1. The structural reduct $L_B(\omega_{\Xi})$ is that same survivor ledger up to its source-preserving realization equivalence: the additional Higgs and Yang–Mills attachments are forgotten by this reduct and do no work in establishing the SR7 class. Thus $\text{Min}_{\text{SR7}}(L_B(\omega_{\Xi}))$ holds. Selector freedom and no-repair preservation are inherited from the full source certificate, giving J_{min} exactly as in Definition 12.12. Hence all clauses hold on ω_{Ξ} , and $J_B(x_Q, x_{\text{col}}, x_{\text{wk}}; \eta_H) = 1$. This proves the pre-assignment claim. When the fixed attachments are the same source restrictions, the SR7 reduction and attachment identifications also supply every field of the fixed-extension definition. $\square$

Corollary 12.24 (Carrier-root role occupancy). *On a source-realized fibre, the carrier-root image of the live residue is*

$$\text{rt}_g^{\text{car}}(\text{Residue}_g^{\text{live}}(\Gamma_{\Xi})) = \{C_Q, C_{\text{col}}, C_{\text{wk}}^{\text{pre}}\}.$$

Every compatible pre-assignment induces the same role-preserving bijection from $\mathbf{R}_g^0$ to this image. Thus the carrier-root occupation has a single assignment and exactly three gauge roots, with the Higgs carrier identity counted separately. The full live response residue may contain distinct classes over one root; this corollary makes no assertion that the carrier-root projection is injective.

Proof. The source pre-assignment supplies one live class over each root. Exact eligibility sends the charge, color, and weak roles only to their respective roots, while source carrier exhaustion excludes an additional same-scope peer response root. Hence the displayed image is exact. Every compatible pre-assignment has these same three typed root values, so its projected assignment is the unique displayed bijection. This is an equality of carrier-root assignments and does not identify any two fine response classes in a projection fibre. $\square$

Definition 12.25 (Complete rolewise gauge-assignment signature). *For a raw fixed-extension point e , whose underlying pre-assignment is τ_e , set*

$$\text{ASig}_B(e) := (\text{Sig}_B(\tau_e(r)))_{r \in \mathbf{R}_g^0}.$$

Here Sig_B on a quotient class means its well-defined complete signature, and the role tags are retained. Witness choices certifying J_B , the SR7 comparison, and the attachment identifications remain property fields, as in Definition 12.15. This signature classifies the gauge assignment. It does not classify a chosen common ledger or a chosen point of the Higgs readiness domain.

Lemma 12.26 (Signature equality is equality of assignments). *If $e, e' \in \widehat{\text{Ext}}_B(Z_*)$ satisfy $\text{ASig}_B(e) = \text{ASig}_B(e')$, then $\tau_e = \tau_{e'}$ as functions into $\mathcal{Q}_B^+(\Gamma)$. Therefore the identity element of $\Gamma_{B,z_*}^{\text{ext}}(\Gamma)$ carries one raw fibre point to the other. No separate orbit-transitivity assumption is required after this equality has been proved.*

Proof. For each rigid role r , equality of assignment signatures gives $\text{Sig}_B(\tau_e(r)) = \text{Sig}_B(\tau_{e'}(r))$. The BCRS quotient was formed by equality of complete signatures, not by a smaller list of named presentation moves. Hence $\tau_e(r) = \tau_{e'}(r)$ in $\mathcal{Q}_B^+(\Gamma)$. Function extensionality gives $\tau_e = \tau_{e'}$. The raw fibre definition does not distinguish different proofs of its property fields, so the two raw fibre points agree. The identity automorphism preserves the target, every role tag, and both fixed attachments, and therefore belongs to the stated stabilizer subgroup. $\square$

Theorem 12.27 (Exact gauge-assignment fixed-extension criterion). *For the already activated BCRS target, gauge-assignment fixed-extension closure is equivalent to the following two assertions:*

$$\widehat{\text{Ext}}_B(Z_*) \neq \emptyset, \quad \left| \text{ASig}_B(\widehat{\text{Ext}}_B(Z_*)) \right| = 1.$$

In particular, a proof that every admitted witness/continuation value in $\text{ASig}_B(e)$ is the same target-relative restriction of z_ , $\eta_{\text{YM}}^{\text{fix}}$, and η_H^{fix} proves unconditional fibre closure once exact source reception has supplied a point.*

Proof. Nonemptiness and singleton assignment-signature image imply equality of every pair of raw fibre assignments by Lemma 12.26. The identity automorphism then proves transitivity, so Definition 12.17 applies. Conversely, the lawful action preserves every rigid role tag and every complete-signature coordinate. A nonempty transitive raw fibre therefore has one assignment signature. Activation was established in Theorem 5.2; SCIC Theorem 4.4 consequently identifies this complete-signature relation as the unique admissible same-scope identity relation. SCIC Theorems 9.1 and 9.4 identify its quotient with the signature image, without supplying the singleton-image premise [7]. $\square$

12.8 Source-level criteria for contextual fidelity

The signature criterion admits a sufficient test at the level of primitive operations. This test compares actual source data before taking the response quotient. Its hypotheses concern operation domains and observations, rather than assuming equality of the complete signatures to be proved.

For every port sort $b \in \mathfrak{S}_B$, let X_b be the full standing-semantic comparison domain at that sort before activity reduction. It includes the admitted auxiliary, intermediate, diagnostic, and semantically null values, wherever these occur in the declared target. Comparison-only values are not thereby assigned source generators. For each primitive operation schema μ and admitted structural parameter instance θ , retain its actual partial map

$$o_{\mu,\theta} : X_{b_1} \times \cdots \times X_{b_n} \dashrightarrow X_c.$$

The parameter θ retains the fixed structural data of that operation, including its interface requirements, but not source-proof names. All assertions below range over every admitted instance, not only a finite sample of source witnesses. A finite list of schemas may have infinitely many instances. Every target observation is also retained; a shorter observation basis may be used only after its completeness has been proved.

Definition 12.28 (Primitive-operation comparison). *A primitive-operation comparison is a family of relations $R_b \subseteq X_b \times X_b$ satisfying the following conditions.*

- (i) *Every actual value a which can occur as a fixed non-hole input satisfies a $R_b a$. The two evaluations use the same structural parameter θ , the same target test, and the same fixed auxiliary inputs.*
- (ii) *For every primitive operation instance and componentwise related tuples $x_i R_{b_i} y_i$,*

$$\mathbf{x} \in \text{Dom}(o_{\mu,\theta}) \iff \mathbf{y} \in \text{Dom}(o_{\mu,\theta}).$$

This equivalence includes all common-interface, boundary, continuation, and preservation conditions of the operation.

- (iii) *Whenever the equivalent domain conditions hold,*

$$o_{\mu,\theta}(\mathbf{x}) R_c o_{\mu,\theta}(\mathbf{y}).$$

Outputs may be auxiliary, diagnostic, inactive, or semantically null; they need not lie in a positive active quotient. The condition includes zero-arity operations.

- (iv) *For every target observation t and related values $u R_c v$, probe definedness and values satisfy*

$$\begin{aligned} u \in \text{Dom}(t) &\iff v \in \text{Dom}(t), \\ t(u) = t(v) &\quad \text{when defined.} \end{aligned}$$

The separate operation-domain and probe-domain observations of the fixed target are included. Current passing status does not replace agreement of counterfactual diagnostic observations.

Parameters and tests here are their full intrinsic structural values. Section 11 proves fixed-input covariance for source-proved skin changes by transporting the whole incident record; the same intrinsic parameter and fixed siblings are preserved. A change to a genuinely different argument, attachment, or probe is not such a comparison. Moving that context does not establish equality in the original fixed context.

Theorem 12.29 (Primitive comparison implies complete contextual fidelity). *Let $(R_b)_b$ be a primitive-operation comparison. If $u R_b v$, then for every admitted one-hole context $C \in \text{Ctx}_B(b; c)$, the values $C[u]$ and $C[v]$ are defined on exactly the same domain and are related by R_c when defined. Probe definedness also agrees separately, and*

$$\hat{t}(C[u]) = \hat{t}(C[v]) \quad (t \in \mathcal{T}_{B,c}).$$

Consequently, certified source presentations whose evaluated values are related have equal complete BCRS signatures.

Proof. Induct on the finite one-hole context. The identity hole returns the related inputs. At a composite node, precisely one input subcontext contains the hole. If that subcontext is undefined, the induction hypothesis makes it undefined on both sides. Every other input subdiagram is fixed: it has the same value on both sides if defined and is undefined on both sides otherwise. Hence failure to evaluate a required input is simultaneous.

When all input subdiagrams are defined, the induction hypothesis relates the hole-bearing outputs, and condition (i) relates each fixed sibling value to itself. Condition (ii) therefore gives exact domain agreement for the primitive node; condition (iii) relates its outputs whenever defined. This proves the induction step for every admitted finite arity, including contexts whose intermediate values leave the active response domain. It does not replace the context family by the unary-update submonoid.

Condition (iv) now gives probe-domain agreement independently of context definedness, and equality of every defined observation. If either stage is undefined, both tagged observations have the same marker $\perp_{\text{undef}}$, distinct from zero, an obstruction, or conditional readiness. Thus every coordinate of the already fixed complete signature agrees. $\square$

Corollary 12.30 (A sufficient source-to-response closure criterion). *Suppose the raw fixed-extension fibre is nonempty. Suppose, for every pair e, e' in that fibre, there is a primitive-operation comparison and certified source presentations s_r, s'_r of their rolewise assignments such that*

$$\text{ev}_B(s_r) R_{b(r)} \text{ev}_B(s'_r) \quad (r \in R_g^0).$$

Then $\text{ASig}_B(e) = \text{ASig}_B(e')$ for every pair, $\text{FibreClosed}_B(z_) = 1$, and $\text{ROC}_B(\Gamma) = 1$. On a source-realized fibre the common-source existence theorem supplies nonemptiness.*

Proof. The theorem gives equality of the complete rolewise signatures. Lemma 12.26 gives equality of assignments. Nonemptiness and Theorem 12.27 give fixed-extension closure, and the exact reduction and occupancy criteria below give $\text{ROC}_B(\Gamma) = 1$. The comparison relations may depend on the pair; neither an enumeration nor a common chosen source representative is needed. $\square$

The comparison hypotheses have been discharged for source-proved skin changes of the full pointed input record in Section 11. This includes the exact mixed H -charge-weak domain with the other intrinsic input legs fixed, and the retained order, boundary, continuation, and diagnostic values. The stronger quantifier over every pair of fixed-extension assignments requires more: all their marked uses would have to admit such an observation-preserving comparison. Rigidity of the whole gauge-law reduct and equality of the fixed attachment families do not identify different admitted argument or attachment orbits. Section 13 retains those marks in the complete source-network representation. No distinct compatible signature pair is inferred merely because its universal absorption has not been proved.

Proposition 12.31 (Primitive exhaustion is not value-level fidelity). *Lemma 12.19 does not by itself establish the singleton-signature assertion in Theorem 12.27. A distinction in an already admitted registered response or continuation coordinate is not excluded merely by the absence of an additional primitive carrier.*

Proof. The primitive-exhaustion statement has as its conclusion the emptiness of the complement of a carrier package. It permits distinct registered statuses over that package. Thus its conclusion remains true in an interpretation with one fixed carrier package and two registered target-admitted values $a \neq b$ of an existing response or continuation test. If two compatible extension presentations realize those values while having the same SR7 class and fixed attachments, no new primitive carrier has been introduced, but their complete BCRS signatures differ. SCIC then requires that the presentations remain distinct in the BCRS quotient. This is a logical countermodel to deriving signature equality from primitive exhaustion alone, not a claim that the actual corpus contains such a physical pair. To exclude

the pair in the actual target one must prove that the displayed coordinate is fixed by the retained source data, or prove that one of its values is incompatible with the already frozen target. Reclassifying the coordinate as an anchor after seeing the pair, or deleting it from the complete signature, would change the target. $\square$

The resulting distinction is exact. The SM1–SM6 chain and inherited generation handoff close the primitive carrier architecture, including its cross-sector, matter, flavor, transport, and governance branches. The bosonic root image is therefore the three gauge-response roots and the separate Higgs identity. The finer response quotient retains every admitted effect, mixed-readiness, boundary, continuation, and failure value. Its compatible gauge-assignment extension fibre is a singleton precisely under Theorem 12.27. When that criterion holds, the identity automorphism already supplies the required lawful orbit identification. When a retained witness separates two compatible extensions, SCIC preserves that distinction; primitive carrier closure remains valid, but it is not promoted into a stronger gauge-assignment rigidity claim.

Theorem 12.32 (Exact fixed-extension reduction criterion). *The one-way reduction separates extension existence from extension uniqueness as follows:*

$$\Sigma_g^{\text{pre}}(\Gamma) \neq \emptyset \iff \widehat{\text{Ext}}_B(Z_*) \neq \emptyset, \quad (12.10)$$

$$\tilde{\mathfrak{D}}_B(\Gamma) \cong \{*\} \iff \text{FibreClosed}_B(z_*) = 1. \quad (12.11)$$

SR7 Theorem 19.1 proves the singleton base $\mathbf{M}_{\text{SR7}}^{\min} = \{z_*\}$; it does not prove either right-hand condition above.

Proof. Given a compatible pre-assignment, choose its witnessing J_B -ledger. Theorem 12.14 supplies its SR7 equivalence to Z_* , while the color-packet and Higgs-readiness clauses of that same ledger supply the two fixed attachment identifications. This gives a point of the raw extension fibre. Conversely, every point of the raw fibre is, by the set-builder definition, a compatible pre-assignment. This proves (12.10) without constructing a section of Θ_B on arbitrary SR7 data. In fact, it proves equality of the underlying sets $\widehat{\text{Ext}}_B(Z_*) = \Sigma_g^{\text{pre}}(\Gamma)$.

By Lemma 12.8, the certificate-lawful group acts trivially on this set, and so does its fixed-attachment subgroup. Their orbits are therefore the same singleton point-orbits; no claim that a general subgroup has the same orbits as its parent group is needed. The common orbit set is a singleton exactly when the raw fibre is nonempty and the action is transitive, proving (12.11). $\square$

Lemma 12.33 (Conditional simultaneous packet). *A simultaneous rolewise eligible BCRS packet exists if and only if the raw fixed-extension fibre is inhabited. Thus pointwise packet inhabitation, the two independently fixed attachments, and the SR7 base class do not separately imply a simultaneous packet.*

Proof. This is Equation (12.10). The negative inference is exactly the distinction made by SR7 Proposition 3.1: marginal survival and separately available attachments do not supply one common witness ledger. $\square$

Corollary 12.34 (Totality under fixed-extension closure). *If $\text{FibreClosed}_B(z_*) = 1$, then every $\tau \in \Sigma_g^{\text{pre}}(\Gamma)$ is total on the live residue:*

$$\text{im}(\tau) = \text{Residue}_g^{\text{live}}(\Gamma).$$

Consequently $\Sigma_g^{\text{pre}} = \Sigma_g \neq \emptyset$, and the canonical map $\tilde{\mathfrak{D}}_B \rightarrow \mathfrak{D}_B$ is an equivalence.

Proof. Fibre closure gives a compatible pre-assignment and places every pair of compatible pre-assignments in one certificate-lawful orbit. If x is live, then $x = \tau'(r)$ for some τ' and some rigid

role r . Transitivity gives a lawful automorphism relating τ' to τ . That automorphism fixes the role tag and every complete-signature coordinate. Since candidates are already complete-signature quotient classes, its restriction at r gives $\tau'(r) = \tau(r)$. Hence every live element is in $\text{im}(\tau)$; the reverse inclusion is definitional. $\square$

12.9 Role-Occupancy Closure before cardinality

Theorem 12.35 (exact fixed-extension criterion for simultaneous compatible-live closure). *For every admitted Γ ,*

$$\text{ROC}_B(\Gamma) = 1 \iff \text{FibreClosed}_B(z_*(\Gamma)) = 1. \quad (12.12)$$

Thus BCRS-I Role-Occupancy Closure is equivalent to existence and lawful transitivity of the fixed-extension fibre over the unique SR7 base class. No gauge-response cardinality is evaluated in this theorem.

Proof. Target, source, use, activity, quotient, continuation, and failure completeness were established before this section. Exact eligibility is Theorem 12.2; simultaneous compatibility, the live residue, and the lawful automorphism action were then formed without using a count. These are the non-orbit fields required by the abstract Role-Occupancy Closure theorem [8, Thm. 7.4].

If $\text{FibreClosed}_B(z_*) = 1$, Corollary 12.34 supplies a nonempty compatible assignment set and totality on the live residue. Theorem 12.32 gives one lawful assignment orbit. ROC necessity and sufficiency therefore give $\text{ROC}_B = 1$.

Conversely, $\text{ROC}_B = 1$ supplies a nonempty compatible assignment and one certificate-lawful assignment orbit. Choose such a bijection $\sigma : \mathbf{R}_g^0 \rightarrow \text{Residue}_g^{\text{live}}$, and let τ be any compatible pre-assignment. For each role r , the value $\tau(r)$ is live, so surjectivity of σ gives a role s with $\tau(r) = \sigma(s)$. Both assignments preserve exact eligibility, and distinct role tags have distinct rigid root sorts. Consequently $s = r$, whence $\tau(r) = \sigma(r)$. Thus every compatible pre-assignment equals σ , not merely every bijective assignment. Theorem 12.32 now makes the raw fixed-extension fibre nonempty with one point, and the identity automorphism gives transitivity. Hence $\text{FibreClosed}_B(z_*) = 1$. Neither implication assumes the desired orbit conclusion inside J_B . $\square$

Scope 12.36 (Gauge-assignment alternatives). *Equation (12.12) classifies the gauge-assignment rigidity question at the fixed target. Inhabitation and a singleton complete assignment-signature image give the positive branch. An empty extension fibre gives a joint-realization obstruction; two distinct assignment signatures give a gauge-assignment rigidity obstruction while preserving the unique coarse carrier architecture. If neither a closure witness nor either obstruction is established, the numerical response-count status is NotReady. No one of these alternatives is inferred from the SR7 base singleton alone.*

Definition 12.37 (Occupied live-role fibres). *When $\text{ROC}_B(\Gamma) = 1$, define, for $r \in \mathbf{R}_g^0$,*

$$\text{Live}_r(\Gamma) := \{[(r, \sigma(r))] : [\sigma] \in \Sigma_g / \Gamma_g^{\text{lawful}}\},$$

and set

$$\mathcal{R}_G^{\text{live}}(\Gamma) := \text{Live}_Q \sqcup \text{Live}_C \sqcup \text{Live}_{\text{wk}}^{\text{pre}}.$$

The rigid role tag is part of each element. If ROC is not established, these are retained as unevaluated dependent fibres rather than assigned a cardinality.

Corollary 12.38 (Graded gauge-response cardinality). *If the equivalent conditions of Theorem 12.35 hold, their cardinality shadow is*

$$(|\text{Live}_Q|, |\text{Live}_C|, |\text{Live}_{\text{wk}}^{\text{pre}}|; |\text{Hld}_B|) = (1, 1, 1; 1), \quad |\text{Mech}_H(C_H)| = 3. \quad (12.13)$$

Consequently $|\mathcal{R}_G^{\text{live}}| = 3$. Without fixed-extension closure, the three gauge-response entries and $|\mathcal{R}_G^{\text{live}}|$ remain **NotFormed** as numerical conclusions; the independently proved one-Higgs identity and three-leg mechanism-fibre theorems remain valid. No cardinality is assigned to the full semantic response object, $\mathcal{Q}_B$, either object's connected components, the internal color or weak groupoids, or any heterogeneous union of gauge roles, Higgs identity, and mechanism fibres.

Proof. Under the closure condition of Theorem 12.35, the compatible assignment object has one lawful orbit and every assignment is a bijection from the three rigid role tags onto the live target-relative residue. Lawful automorphisms preserve the tags, so the occupied-pair fibre over each role has one orbit. This is the finite cardinality consequence of ROC, not a premise of it [8, Cor. 7.5]. The Higgs identity locus is the singleton of Theorem 10.2, and its mechanism fibre is the three-element typed set of Theorem 10.5. The semicolon and the separate mechanism equation record the sort boundary; no heterogeneous scalar sum is formed. $\square$

12.10 Network disposition on a simultaneous ledger

Let $\mathcal{A}_B^{\text{pre}}(\Gamma)$ be the family of pairs $A = (\sigma, \omega_A)$, where $\sigma \in \Sigma_g^{\text{pre}}(\Gamma)$ and ω_A witnesses its simultaneous compatibility. The following constructions are defined for every member of this family. Fix

$$H_A : \quad A = (\sigma, \omega_A) \in \mathcal{A}_B^{\text{pre}}(\Gamma). \quad (12.14)$$

Its common ledger is

$$\omega_A := \omega_B(\sigma(r_Q), \sigma(r_{\text{col}}), \sigma(r_{\text{wk}}); \eta_H)$$

witnessing $J_B = 1$. No uniqueness of the possible assignments is required. Under the closure condition of Theorem 12.35, the gauge assignment has one lawful orbit. Independently of that condition, each actual simultaneous ledger has its own network and codescent object. If $\mathcal{A}_B^{\text{pre}}$ is empty, no such object is asserted to exist. This dependence separates common-ledger existence from response rigidity.

Let

$$\text{Net}_g^{\text{live}}(A) := \text{NetSch}_g(\sigma(r_Q), \sigma(r_{\text{col}}), \sigma(r_{\text{wk}}))$$

with each declared interface interpreted by restriction of the one ledger ω_A . This is a three-vertex network presentation. It is not an additional response object.

Definition 12.39 (Witnessed live-network presentation). *For a submitted higher-arity candidate Y , let $\text{NetFact}_A(Y)$ mean that the following data have been exhibited:*

- (i) *every response-bearing input of Y is assigned, with its rigid role tag preserved, to an injective typed subrecord of one of $\sigma(r_Q), \sigma(r_{\text{col}}), \sigma(r_{\text{wk}})$;*
- (ii) *every comparison used by Y is the restriction of a declared interface in ω_A , so all of its equality, boundary, continuation, guard, and failure fields are read from that same ledger; and*
- (iii) *a finite expression T_Y , built from the licensed many-sorted constructors already present in $\mathcal{E}_B(\Gamma)$, has those subrecords as its response leaves and represents the same element of $\mathcal{Q}_B^{\text{port}}(\Gamma)$ as the active output of Y .*

Thus $\text{NetFact}_A(Y)$ is a factorization witness, not a consequence of the semantic-use factorization theorem. The latter says that each registered evaluator descends through presentation equivalence; it does not assert that an arbitrary source term or network has a normal form through $\text{Net}_g^{\text{live}}(A)$.

For diagnostic purposes, let $\text{LiveCompat}_A(Y)$ state that all role-bearing inputs of Y occur in $\text{im}(\sigma)$ and that all of the comparisons required by Y hold on ω_A . Failure of this condition concerns the chosen assembly. It does not by itself say that Y cannot occur in some other compatible assembly. Global

non-liveness requires a separate proof of absence from every simultaneous assignment. A failed equality on a licensed common coordinate is a definite incompatibility witness on that interface.

Definition 12.40 (Assembly-relative higher-arity classification). *For a submitted higher-arity candidate Y , let $\mathcal{D}$ be its supplied collection of verified certificates. Define $\text{Disp}_{>1,A}(Y; \mathcal{D})$ by the first line whose stated certificate is supplied in $\mathcal{D}$:*

- (a) *CoupledPresentation*, if Y is same-target, all of its registered failure tests on formed domains are neutral, and $\text{NetFact}_A(Y)$ is witnessed;
- (b) *TargetUpdate*, if Y carries a source-certified independent standing coordinate not contained in the fixed target signature;
- (c) *CertifiedObstruction*, if a registered failure use of Y is active;
- (d) *OutsideAssembly*, if Y is otherwise typed and source-registered but $\text{LiveCompat}_A(Y)$ fails, including the absence of its inputs from this assembly or disagreement on a licensed common coordinate;
- (e) *Inadmissible*, if a source-law violation or unauthorized load has been certified, including an independently standing repair, hidden parameter selection, or downstream projection import; and
- (f) *NotReady*, otherwise: the supplied collection contains no adequate certificate for any preceding line.

The ordering makes the reported values disjoint and total relative to the supplied verified collection. The last line asserts neither mathematical independence nor nonexistence of a certificate not supplied there; it is not silently identified with a coupled presentation. This is not a decision procedure for arbitrary raw submissions.

Theorem 12.41 (common-ledger network disposition and nonpromotion). *Under H_A ,*

$$\text{ProductCompatible}(\text{Net}_g^{\text{live}}(A))$$

is true and remains a proposition on the three live response vertices. It is not a fourth response occupant. Every submitted higher-arity candidate with a supplied verified collection has exactly one diagnostic value in Definition 12.40. In particular, no candidate classified at the fixed BCRS-I target acquires a fourth gauge role: a witnessed coupled presentation uses the existing three vertices, a new independent standing coordinate is a target update, and every other status licenses no new occupant.

Proof. The chosen witness of $J_B = 1$ contains the conjunct J_{prod} , so the displayed product-compatibility predicate is true on ω_A . Its codomain is proposition-valued, and the complete source grammar contains no source-certified standing-bearing coercion from that truth value to a new response root. Lemma 6.11 therefore proves nonpromotion.

For a candidate Y and supplied collection $\mathcal{D}$, inspect which of the five certificate classes in Definition 12.40 are supplied and verified. The finite priority order gives the first such class, or the residual **NotReady** value when none is supplied. The first-applicable convention gives uniqueness without asserting an ability to find a missing certificate. This is certificate-relative diagnostic totality, not a network-normal-form theorem. In case (a), the exhibited T_Y and its one-ledger interface data prove the coupled presentation. Case (b) changes a fixed standing coordinate and hence belongs to a re-instantiated or successor target. Case (c) records the value of an already registered failure use. Case (d) withholds realization on this assembly; marginal eligibility or pairwise compatibility does not replace its simultaneous ledger. It does not rule out a different compatible assembly. Case (e) violates the fixed admissibility and no-repair conditions. The residual case records absence of an adequate certificate from the supplied collection, not a theorem that no such certificate exists. None of cases (b)–(f) forms a response vertex, and case (a) contains only the three role-tagged vertices already present in $\text{Net}_g^{\text{live}}(A)$. $\square$

12.11 Set-codescent on intersections of one common ledger

Codescent is formed only from the simultaneous witness $A = (\sigma, \omega_A)$, not from independently chosen pairwise witnesses. Let K_A be the typed coordinate-key set of the dependent record ω_A , with value type V_k at key k , and write its coordinate graph as

$$\Omega_A := \coprod_{k \in K_A} \{(k, \omega_A(k))\}, \quad \omega_A(k) \in V_k.$$

For $a \in \{Q, C, \text{wk}, H\}$, let $K_a \subseteq K_A$ be precisely the keys belonging to the corresponding gauge-incidence packet, or to the fixed Higgs-readiness packet when $a = H$, and define the vertex packet used for codescent by the typed subrecord

$$X_a(A) := \coprod_{k \in K_a} \{(k, \omega_A(k))\} \hookrightarrow \Omega_A. \quad (12.15)$$

The map is injective because the coordinate key and its value type are retained. Equation (12.15) is the coordinate record of the already formed packet on the chosen simultaneous witness; it does not reconstruct a new packet from an evaluator. For the Higgs vertex, the retained subrecord includes the actual d_{ω_A} , its fine Higgs input and mixed-incidence class, its readiness value, and all retained domain and unexecuted-obligation coordinates. Replacing these data by the carrier identity alone would not be this vertex. For the source-realized assembly A_{Ξ} , the charge vertex also includes the keys of $\mathcal{R}_Q(h_{\text{res}, \Xi})$, its transport-comparison assignment, and their retained support and reduction dependencies. These are subrecord fields of the same ω_{Ξ} , so they belong to the declared union reconstructed below. This specified enrichment does not posit an executed residual packet at every arbitrary point of the base charge locus.

Definition 12.42 (Maximal common-ledger interfaces). *For two vertices of the live tuple set*

$$K_{ab} := K_a \cap K_b, \quad \mathcal{I}_{ab}^{\max}(A) := \coprod_{k \in K_{ab}} \{(k, \omega_A(k))\}.$$

Its two interface maps are the subset inclusions into $X_a(A)$ and $X_b(A)$. This interface is maximal only in the precise, target-relative sense needed here: every typed coordinate subrecord of ω_A that embeds in both vertices by coordinate-key-preserving inclusions has key set contained in $K_a \cap K_b$. Arbitrary injections between equal value types, or renamings of distinct keys, are not subrecord comparisons. For three vertices put $K_{abc} := K_a \cap K_b \cap K_c$. All restrictions to K_{abc} are literally the same inclusion of the same keyed ledger values, so the pairwise interfaces satisfy the cocycle identities strictly.

*For an arbitrary eligible tuple not yet known live, use the following failure discipline. If a licensed common key exists but the two proposed values disagree, its status is **Incompatible** and that interface is not glued. If the vertices have no typed common key, their status is **NonComparable** and no comparison arrow is formed. If the marginal vertices do not occur together on any ledger satisfying $J_B = 1$, the tuple has status **NonLive**, and no global codescent object is formed, even if every displayed pair happens to agree on a separately witnessed overlap.*

Let $\approx_A$ be the equivalence relation on the disjoint union $\coprod_a X_a(A)$ generated by

$$(a, k, \omega_A(k)) \approx_A (b, k, \omega_A(k)) \quad \text{whenever } k \in K_a \cap K_b. \quad (12.16)$$

Only copies of one and the same typed ledger coordinate are identified.

Definition 12.43 (Common-ledger set-codescent object). *The BCRS common-ledger set-codescent object of A is*

$$\text{Codesc}_B^{\text{Set}}(A) := \left(\coprod_{a \in \{Q, C, \text{wk}, H\}} X_a(A) \right) / \approx_A. \quad (12.17)$$

Lemma 12.44 (Overlap relation and vertex injectivity). *For elements of the disjoint union,*

$$(a, k, \omega_A(k)) \approx_A (b, \ell, \omega_A(\ell)) \iff k = \ell.$$

Consequently every canonical vertex map $q_A \circ \iota_a : X_a(A) \rightarrow \mathbf{Codesc}_B^{\mathbf{Set}}(A)$ is injective. Two vertex images meet exactly in the image of $\mathcal{I}_{ab}^{\max}(A)$.

Proof. Every generator in (12.16) preserves the typed coordinate key. Hence every finite zigzag in its symmetric and transitive closure preserves that key, proving the forward implication. Conversely, if an element of $X_a(A)$ and an element of $X_b(A)$ have the same key k , then $k \in K_a \cap K_b$ and they are related by one generator. Within a fixed vertex, equality of keys also fixes the unique ledger value, so the canonical vertex map is injective. The same equivalence proves the claim about the intersection of two images. $\square$

Theorem 12.45 (common-ledger codescent in **Set**). *For every chosen simultaneous witness A satisfying H_A , the object $\mathbf{Codesc}_B^{\mathbf{Set}}(A)$ is the colimit in **Set** of the diagram consisting of the vertex sets $X_a(A)$, the nonempty interface sets $\mathcal{I}_{ab}^{\max}(A)$, and their two inclusion maps. Explicitly, for every set Y and family of functions $f_a : X_a(A) \rightarrow Y$ satisfying*

$$f_a|_{\mathcal{I}_{ab}^{\max}(A)} = f_b|_{\mathcal{I}_{ab}^{\max}(A)}$$

on each formed interface, there is a unique function

$$\bar{f} : \mathbf{Codesc}_B^{\mathbf{Set}}(A) \longrightarrow Y \quad \text{such that} \quad \bar{f} \circ q_A \circ \iota_a = f_a \quad \text{for every } a.$$

The vertex maps are injective by Lemma 12.44. A lawful isomorphism of chosen common ledgers that preserves typed coordinate keys carries this overlap diagram isomorphically to the corresponding diagram and therefore induces a unique isomorphism of their set-codescent objects.

Proof. The strict cocycle identities follow from intersection of subsets of the one key set K_A ; no independently chosen comparison isomorphisms occur. The family $(f_a)_a$ defines a function on the disjoint union. Its interface equalities make it constant on every generator of $\approx_A$, hence on the generated equivalence relation. The quotient universal property gives $\bar{f}$, and surjectivity of the quotient map gives uniqueness. This is precisely the colimit universal property in **Set**. The final assertion follows by transporting keys, typed values, vertices, and intersections along the stipulated ledger isomorphism and applying the same universal property in both directions. $\square$

Corollary 12.46 (Exact reconstructed subrecord). *There is a canonical injection*

$$j_A : \mathbf{Codesc}_B^{\mathbf{Set}}(A) \hookrightarrow \Omega_A$$

whose image consists exactly of the keys in $\bigcup_{a \in \{Q, C, \text{wk}, H\}} K_a$. Hence the codescent object reconstructs the union of the declared packet subrecords. A global ledger field not lying in that union remains in ω_A ; it is not recovered from pairwise agreement.

Proof. Send the class of $(a, k, \omega_A(k))$ to $(k, \omega_A(k))$. Lemma 12.44 proves that this map is well defined and injective, and its stated image follows from the definition of the disjoint union. This proves the exact reconstruction claim. $\square$

The theorem asserts neither effective descent in an undeclared category of structured BCRS packets nor recovery of internal packet operations by a pullback from the quotient set. Such stronger claims require a category of packet morphisms and a separate preservation theorem. What is proved here is the exact set-colimit statement, strict overlap coherence, and injective recovery of every vertex as a typed coordinate subrecord. Incompatible interfaces are not glued, noncomparable pairs contribute no edge, and a non-live tuple does not generate a global codescent diagram.

13 Rigidity of the gauge laws and their response networks

A gauge law and a response obtained by evaluating that law are different objects. The connection assigns comparisons to admitted continuations; it is not one selected comparison. A residual interface is a dependent family over admissible support and reduction data; it is not one selected reduction. The distinction permits a stronger conclusion than carrier-root uniqueness without identifying response values that the sources still distinguish.

13.1 The complete source-law reduct

For an SR7 joint survivor Y , write $W(Y)$ for its full shared SR1 witness ledger. Its entries are boundary, quotient, transport, loop, field, particle, representation, charge, anomaly, vacuum, residual, and deformation data. Each standing assertion of a component is witnessed through the corresponding entry and is invariant under lawful redescription preserving this ledger [32, Defs. 4.3–4.4]. An SR7 comparison concerns the joint survivor records, whose witness maps agree on one common ledger [38, Sec. 5.1 and Def. 5.3]. Its product comparison retains the underlying SR1 realization class [33, Def. 5.2(xi)]. Consequently it supplies a full witness-preserving comparison, not merely agreement of seven role labels.

Definition 13.1 (Structured gauge-law reduct). *The structured gauge-law reduct $\mathfrak{L}_g(Y)$ is the joint source record consisting of:*

- (i) *the continuation site, admitted presentation and transition objects, invariant-role witnesses, restriction, inverse, and partial composition;*
- (ii) *the full transport-comparison assignment and closed-loop residue assignment, with their exact operand domains and endpoint data;*
- (iii) *the representation-response and charge-witness structures, including witnessed algebraic, Lie, Abelian, tensor, conjugate, chiral, dimension, multiplicity, ratio, and discrete specializations;*
- (iv) *the product and coupled-sector records, including the common sector restrictions and their mixed transition, transport, and loop residues;*
- (v) *the complete anomaly-probe and aggregation interfaces, their obstruction values, global and index comparisons, and repair-domain classification;*
- (vi) *the structural vacuum-compatibility and residual-response interfaces, retaining their complete support and reduction domains, induced residual records, restoration, and deformation data; and*
- (vii) *the boundary extraction, admitted continuation, conditional-domain, and failed-law rules inherited by these source structures.*

Every operation is retained as its typed partial graph, not as a sampled output or an assertion that some output exists. All rows retain their actual shared incidence maps. The record is therefore not a product of independently quotiented fields. Write $\mathfrak{L}_r(Y)$, for $r \in \{r_Q, r_{\text{col}}, r_{\text{wk}}\}$, for its indicated role restriction with those shared maps attached.

An intrinsic law record is the class of this joint record under the source-proved full-witness-preserving comparisons. A selected continuation, loop, probe, reduction argument, or mixed attachment is not identified with the whole rule that evaluates it.

These data are already explicit in the source sequence. A1 supplies the continuation and transition structure, with its partial operations [20, Defs. 3.1–3.14 and Props. 5.1–5.7]. A2 gives the map

$$\nabla_{\partial D} : K_D \longrightarrow \text{Arr}(\text{Gpd}_{\partial}(D))/\sim_{\partial G}, \quad \kappa \longmapsto [\Theta_{\kappa}],$$

where K_D is the admitted continuation family [21, Defs. 3.7–3.9 and Rem. 3.10]. Its intrinsic codomain is essential: a displayed representative is not a further law coordinate. Representation equivalence preserves transition action, transport action, loop response, invariant subcarriers, sums, tensors, conjugates, handed witnesses, dimensions, multiplicities, and the attached charge, anomaly, residual, and deformation ledgers [34, Def. 7.1]. Charge equivalence retains the full character, periodic, ratio, and discrete-witness structures [35, Def. 7.1]. Anomaly equivalence retains the admitted probes and aggregates, not only their neutrality verdict [36, Def. 7.1].

The residual family requires particular care. A6 retains actual non-null support, gauge-vacuum compatibility, reduction, and induced residual response [25, Defs. 5.2–5.10]. Its finite probes include continuation, transition, transport, loop, representation, charge, anomaly, deformation, and restoration data. Its joint normal form is a ledger class rather than an independently assembled product [25, Defs. 6.1–6.2 and Props. 6.3 and 6.6]. PSM6 closes the residual family under restriction, quotienting, transport, loop testing, representation and charge response, anomaly testing, support, and deformation [31, Props. 4.5–4.6 and 5.2]. SR6 preserves the actual support, reduction, residual, order, and deformation records [37, Def. 8.1]. No vacuum expectation value or selected physical residual particle enters this law reduct.

Lemma 13.2 (Source comparison of the complete law record). *An actual SR7 realization comparison $Y \sim_{\text{SM}} Y'$ induces*

$$[\mathcal{L}_g(Y)]_{\text{src}} = [\mathcal{L}_g(Y')]_{\text{src}}.$$

The induced role comparisons preserve and reflect every retained operation and probe domain and transport the corresponding output graphs. They preserve the joint dependencies between the roles.

Proof. The product-profile comparison preserves the underlying SR1 realization class, while the other profiles are compared over the same shared witness ledger. SR1 Definition 4.7 therefore gives a finite zigzag preserving all its standing-bearing entries, including transition, transport, loop, boundary, field, particle, representation, charge, anomaly, vacuum, residual, and deformation data. Restrict each step to the source structures in Definition 13.1, retaining all incident maps.

For transition and transport, a complete operand includes its continuation and endpoint witnesses. Transporting those witnesses preserves the A1 preservation predicate and the A2 comparison graph; reversing the source comparison reflects their domains. The same argument applies to the closed-loop datum and the full representation, charge, product, and anomaly predicates preserved by SR2–SR5. In particular an anomaly comparison transports the probe and its obstruction together, including the repair-domain record.

For the residual family, transport an arbitrary complete candidate (V, red, v) , not merely one previously selected residual output. The gauge-vacuum conditions of A6 Definition 5.4 and the admissible-reduction conditions of its Definition 5.8 concern exactly the retained transition, transport, loop, representation, charge, anomaly, particle, quotient, boundary, and deformation witnesses. Each condition is preserved. The reverse comparison proves the converse, and the induced residual fields are transported with the reduction that defines them. PSM6’s closure then retains the entire dependent family. This proves domain reflection without assuming that different reduction arguments coincide.

Boundary, continuation, conditional, and failure rules are restrictions of these same structures. Their values cannot be independently reassigned. Concatenating the restricted zigzag gives equality of the joint intrinsic law records. Quotient-compression steps use their admitted comparison spans; no inverse of a noninvertible raw map is introduced. Structured quotient descent preserves the corresponding restriction and partial-composition equations [52, Def. 4.4]. None of these steps uses the BCRS contextual signature relation. $\square$

Theorem 13.3 (Rigidity of the complete gauge-law reduct). *For every $e \in \widehat{\text{Ext}}_B(Z_*)$, the common source witness determines a well-defined intrinsic law value*

$$\mathcal{L}_g(e) = [\mathfrak{L}_g(F_B(e, \omega_e))]_{\text{src}} = [\mathfrak{L}_g(Z_*)]_{\text{src}}.$$

The value is independent of the admissible choices of ω_e and of its source comparison. Hence, when the fixed-extension fibre is nonempty,

$$\left| \mathcal{L}_g(\widehat{\text{Ext}}_B(Z_*)) \right| = 1. \quad (13.1)$$

Its three role restrictions are the unique complete charge, color, and pre-electroweak gauge-law records induced by that joint source class.

Proof. The definition of the fixed-extension fibre supplies an actual $\kappa_e : F_B(e, \omega_e) \sim_{\text{SM}} Z_*$. Lemma 13.2 gives the displayed equality. A second admissible common witness has its own comparison to the same Z_* , and therefore gives the same intrinsic law value. Different comparison paths may act differently on marked inputs, but cannot change the class of the whole law object. Nonemptiness gives equality, rather than just inclusion, of the image with the singleton. The role restrictions are restrictions of this joint equality, not separately selected marginal identifications. $\square$

13.2 The fixed color and Higgs law families

The independently fixed Yang–Mills attachment supplies more than the color root. Its structural family retains support and label types, support/deformation comparison, the actual sector assignment

$$\Xi_{G_\Gamma} : \mathcal{E}(G_\Gamma) / \sim_{\text{def}} \longrightarrow \mathcal{S}_{\text{ext}}(G_\Gamma),$$

and its incidence to the unchanged local theorem object [47, Prop. IV.17, Thm. VI.12, Cor. VI.14, and Thm. VIII.2]. Its deformation transports compose over that fixed local object [47, Def. VI.7 and Prop. VI.8]. The fixed Higgs family retains the full tuple

$$\mathcal{B}_H = (W_{\text{br}} / \sim_{\text{br}}, \Sigma_{\text{br}}, \beta_{\text{red}}, \rho_{\text{br}}),$$

including inherited gauge, order, deformation, residual, and guard interfaces [19, Defs. 6.2 and 7.1] [39, Thms. 7.4 and 7.6]. It is the complete canonical bridge law, not a carrier name or a chosen point of the readiness domain.

Corollary 13.4 (Whole-law family inventory). *At a nonempty fixed source-realized extension fibre, the joint gauge-law class, its independently fixed complete Yang–Mills family, and its independently fixed canonical Higgs family give one whole-law family class in each of the three gauge roles and one in the separate Higgs role. The corresponding law-family inventory is $(1, 1, 1; 1)$.*

Proof. Theorem 13.3 identifies the complete joint gauge-law record and its role restrictions. The two additional whole families are fixed source objects in the definition of the extension fibre. The common-source existence theorem supplies their actual occupation and a compatible use. No claim that all possible uses or relative attachments coincide is needed. $\square$

This inventory is not the earlier carrier-root count: its objects retain actual operation graphs, probe families, support-sector assignments, and dependent domains. It is also not a cardinality statement about the marked response quotient. In particular, the Yang–Mills source’s transport theorem does not lift every abstract automorphism of an independently retained gauge reduct. Relative comparisons are retained as actual legs of the source diagram. Different gluing orbits are not identified by comparing their marginal law families.

13.3 Marked source-use networks

The full law record determines how a supplied argument is evaluated. The argument and its actual incidence remain necessary data. We record them without introducing a representative selector.

Definition 13.5 (Decorated source-law network). *A decorated source-law network is a finite typed diagram whose vertices are actual source operations of the following kinds:*

- (i) *A1 transition, inverse, restriction, and admitted composition;*
- (ii) *A2 transport and A3 closed-loop response;*
- (iii) *A4 and PSM1, PSM3, and PSM4 witnessed representation, Lie, algebraic, and Abelian operations;*
- (iv) *PSM2 independent-product or coupled-sector composition;*
- (v) *A5 and PSM5 admitted obstruction probes, aggregates, and global or repair-domain comparisons;*
- (vi) *A6 and PSM6 structural support, compatibility, actual reduction, and residual-response operations;*
- (vii) *the fixed Yang–Mills support/deformation and sector-assignment operations; and*
- (viii) *the fixed Higgs order, deformation, fixation, and jointly witnessed mixed-incidence operations.*

Boundary extraction, admitted continuation, restoration/deformation, conditional readiness, and failed-law diagnostics are retained at their source operation. An edge carries its actual typed operand and common-interface map. A vertex carries the complete source structural parameter needed by its operation. It does not carry an independently assigned output value. Evaluation applies the indicated source map to those operands and computes its output and diagnostics.

Within each source instance, shared entries are restrictions of its actual composable hypothesis record. A general network retains all such source-instance records and the given common-interface maps joining them. For an eligible assignment, its common packet-law ledger is additionally distinguished. It is not thereby assumed that every internal constituent is a restriction of that one ledger. Additional color and Higgs data retain their actual relative maps. A mixed Higgs operation has the auxiliary output $\text{Inc}_{H,EW}^{\text{pre}}$; a product or guard output keeps its own declared sort. Neither is promoted to a new gauge role.

The decorations are actual source-use data: for example, an admitted continuation κ , a closed loop with its closing datum, a representation-sensitive probe, a complete structural reduction argument, or an actual mixed attachment. They are not a BCRS quotient class, a contextual signature, a paper label, or an output-atom placeholder. Source hypothesis witnesses certify that a decoration is admitted; their names do not supply additional physical coordinates.

Definition 13.6 (Relative source charts). *A source chart retains an actual source-instance record and its declared interface maps. An assignment chart additionally distinguishes its common packet-law ledger, comparison to the fixed SR7 class, and the actual relative maps by which its color and Higgs uses meet the fixed source families. A marked network retains the finite family of its source charts joined by their actual interface witnesses. We retain all such actual charts and their source-proved comparisons; no universal amalgamation into a single pointed record is assumed.*

A comparison acts on the whole decorated diagram, including its relative maps, not independently on marginal orbit sets. It exists only when the participating sources license that complete comparison. The chart system is not required to be connected. Automorphisms and their possible action on marks are retained.

Proposition 13.7 (Functorial evaluation on relative charts). *For every actual source chart comparison u , transport of decorated networks preserves and reflects the domain of each source operation and*

commutes with intrinsic source evaluation. For composable licensed comparisons,

$$T(1) = 1, \quad T(v \circ u) = T(v) \circ T(u).$$

Coherently composed chart transitions satisfy the corresponding cocycle equation. Arbitrarily chosen comparison paths are not asserted to act identically on marks.

Proof. Transport the full operand and structural-parameter record of a vertex. The source comparison preserves its exact admission predicate and its partial output graph; the reverse comparison reflects admission. For vacuum and residual vertices the candidate includes the actual support, reduction, and full probe record, as in Lemma 13.2. For color vertices use the source-proved support/deformation comparison over the retained fixed local object. A comparison of mixed vertices transports their complete joint attachment, as in Section 11. It does not follow merely from three marginal identifications.

Induction over the finite diagram gives preservation of every evaluation and domain record. Source identities and composition give the displayed equations; an admissible compression is represented by its comparison span and common refinement. Thus reversal does not require an inverse raw quotient map. A cocycle equation follows for transitions chosen by those coherent compositions. A different path may differ by an automorphism of the marked source object, whose action has not been discarded. $\square$

Definition 13.8 (Source-network atlas and evaluation). *Let $\mathcal{N}_B^{\text{src}}(\Gamma)$ be the class of decorated source-law networks with their actual relative charts and at least one declared response, obstruction, readiness, continuation, or failure output of Definition 2.6, modulo licensed full-diagram source-presentation comparisons. This output-profile condition fixes the codomain; it selects no output value or signature. Its constructors are those of Definition 13.5; neither a BCRS quotient class nor a contextual signature is a constructor or a decoration. The intrinsic evaluation is*

$$\text{ev}_B^{\text{law}} : \mathcal{N}_B^{\text{src}}(\Gamma) \longrightarrow \mathcal{R}_B^{\text{sem}}(\Gamma).$$

It applies the source operation at each vertex on its exact domain, retaining the output sort and all conditional and failure records. Proposition 13.7 and the intrinsic presentation semantics of Section 11 make this evaluation well defined. On the certified active domain, the resulting semantic response may then be sent to the contextual signature quotient. Equivalently, receive the evaluated network as an active source term of its retained sort and stratum and apply $q_{B,b}^{\epsilon}$; source-use factorization makes the two descriptions agree. Evaluation on the broader semantic domain does not itself confer activity.

Theorem 13.9 (Finite-use reconstruction over the law families). *Every finite source-admitted response diagram has a decorated source-law network with the same intrinsic evaluation. The network retains all actual operands, common attachments, auxiliary output sorts, partial-operation domains, boundary, continuation, readiness, and failed-law values. For a fixed-extension assignment, the distinguished common packet-law reduct is the unique class of Theorem 13.3.*

Proof. Induct on the source diagram. At a source mechanism instance, use its complete instantiated hypothesis tuple to identify the corresponding operation in Definition 13.5. Retain its actual structural operands and parameter. The vertex is the source map evaluated on those operands, not a new atom whose label is the desired result. A1–A6 and PSM1–PSM6 supply the gauge operation families, the Yang–Mills family supplies its support-sector map, and the Higgs family supplies its full mixed and unary mechanisms. The source-restriction rule keeps every later antecedent as an exact dependent interface unless it is actually supplied.

At a composition step, graft the networks already constructed using the given common-interface witness. The structural parameter and the input restrictions of a many-input operation are unchanged. Its actual source evaluation therefore supplies the same output. Boundary, continuation, readiness, and

failure values are evaluated by those same source rules. Associativity and the source partial composition laws give the evaluation of the full finite diagram. The comparison of a distinguished common packet-law reduct follows independently from Theorem 13.3. The constituent source records and their actual joining maps remain in the network. They are not replaced by restrictions of one universally amalgamated source record, and their decorations are not normalized to a single chosen argument. $\square$

Corollary 13.10 (Coverage of compatible assignments). *Every $e \in \widehat{\text{Ext}}_B(Z_*)$ is represented by three jointly compatible decorated source-law networks retaining their actual relative source charts. Applying their source evaluations and the contextual quotient maps gives exactly $e(r_Q)$, $e(r_{\text{col}})$, and $e(r_{\text{wk}})$. All eligibility and common-ledger conditions remain satisfied.*

Proof. Use the witnesses whose existence is required by the definition of this particular extension. Each complete packet supplies its actual active source term, composable hypothesis witness, role incidence, boundary and continuation records, and full passing failure value. The common ω_e supplies their shared gauge ledger, the actual color identification supplies its fixed local-root and sector leg, and the actual Higgs witness retains these very charge and weak restrictions. These distinguish the common packet-law ledger in the relative network. Every internal source instance also retains its given hypothesis record and joining witness. No comparison to a preferred marked chart or additional common-full-source lift for all constituents is asserted.

Theorem 13.9 replaces each finite source diagram by its operation network with identical evaluation. Theorem 3.5 ensures that this reception covers every standing-semantic instance, not only a selected source presentation. No argument, current boundary trace, continuation record, actual attachment, conditional antecedent, or failure value is changed. Consequently the same packet, activity, incidence, source-minimality, fixed-attachment, and common-compatibility witnesses still apply to the same evaluated assignment. Every extension is covered; no narrower eligible locus has been substituted. $\square$

Thus ev_B^{law} , followed by the contextual signature quotient rolewise, covers the full eligible-assignment image of $\widehat{\text{Ext}}_B(Z_*)$. No injectivity of this map, singletonhood of its marked fibres, or transport of every network into one common pointed chart is asserted. The map $\mathcal{L}_g$ instead records their uniquely reconstructed common gauge-law reduct.

13.4 Law rigidity and response cardinality

The reconstruction gives a genuine higher-arity description. Coupled products and mixed Higgs incidences are evaluations of their source operations over the reconstructed law families, with their full common input records retained. They require no new primitive response root. This statement is not NetFact_A through three selected response points: different decorations may give different responses over the same complete law family.

In particular, $\mathcal{P}_r(X) = (X; \dots)$ retains X itself. Theorem 13.3 identifies the structured source-law reduct of every role value occurring in the fixed-extension fibre, while Corollary 13.10 preserves X as an actual marked evaluation. Neither theorem deletes a witnessed distinction between two such evaluations. Equality of a transport map does not make it constant on its continuation domain; equality of a bridge law does not identify all of its actual attachments.

The remaining stronger question is therefore precise. It is whether every admissible marked use in a given role is absorbed, with all standing observations preserved, by one whole-role response class. That would require an additional source theorem about the marked arguments or an exhaustive normal-form theorem for all role-eligible output diagrams. It does not follow merely from ledger factorization. The complete gauge laws are rigid at the fixed source scope, and every compatible response has the decorated

network representation proved above. Singletonhood of the finer marked contextual-signature fibres is a separate assertion and has not been used in either result.

14 Bosonic response architecture and its realization boundary

A premetric Standard Model atlas must distinguish the identity of a carrier from the responses it supports. The source theorems determine the carrier-root architecture and the complete structured gauge-law reduct. The response construction retains every standing-relevant distinction over those laws, and the common-source construction supplies a simultaneous assembly. Uniqueness of a law is not constancy of its values on different admitted inputs.

14.1 The structural classification theorem

Write $\eta_H = [C_H]_{\text{id}}$ for the unique Higgs identity class and $\mathbf{I}_Q, \mathbf{I}_{\text{col}}, \mathbf{I}_{\text{wk}}^{\text{pre}}$ for the three packet maps. Their domains are the exact qualified packet loci, not chosen representatives. The Higgs readiness map retains its fine input packets and the mixed-incidence witness; its identity projection alone takes values in $\{\eta_H\}$.

Theorem 14.1 (BCRS-I structural classification). *Fix an admitted target $\tau_B(\Gamma)$ with its exact source interfaces. The following conclusions hold.*

- (i) *Every intrinsic, source-law-admissible response presentation has a certified source presentation. Equality of complete contextual signatures gives the canonical response quotient. Every admitted semantic use factors uniquely through that quotient, and every licensed constructor descends with its exact domain of definition. Null, positive, and failure outcomes are not conflated. Intrinsic source descent preserves all standing observations and fixed-input domains. In a mixed context the actual attachment and spectator inputs remain fixed; presentation changes do not become spurious standing discriminators.*
- (ii) *The charge, color, and pre-vacuum weak packet loci and their incidence maps are defined without particle labels or generator counts. The Higgs carrier-identity locus and mechanism fibre are*

$$\text{Hld}_B = \{\eta_H\}, \quad \pi_\mu^{-1}(\eta_H) = \{\text{Ord}_H, \text{Def}_H, \text{Fix}_H\}.$$

The full Higgs response quotient is not identified with this carrier-identity locus. On the complete A6–PSM6–SR6 reduction domain, the charge packet has a structural residual extension and boundary-fixed transport comparisons, with restriction and weak composition inherited from the source.

- (iii) *The simultaneous compatibility predicate, live response residue, assignment set, and lawful action are well defined on the response quotient. Their reduction to the independent SR7 minimal survivor has image contained in its singleton class $\{z_*\}$. Exact compatible gauge-assignment rigidity is characterized by*

$$\begin{aligned} \text{ROC}_B(\Gamma) = 1 \quad &\Longleftrightarrow \quad \widehat{\text{Ext}}_B(Z_*) \neq \emptyset, \\ &\text{and } \left| \text{ASig}_B(\widehat{\text{Ext}}_B(Z_*)) \right| = 1. \end{aligned} \tag{14.1}$$

- (iv) *For every full source realization Ξ , all restrictions being taken from that one source, there are a mixed-readiness point d_Ξ , a compatible pre-assignment τ_Ξ , and an actual residual charge packet $\mathcal{R}_Q(h_{\text{res}, \Xi})$ on its fibre (D_Ξ, Γ_Ξ) . The carrier-root image of the live residue is exactly*

$$\{C_Q, C_{\text{col}}, C_{\text{wk}}^{\text{pre}}\},$$

and every compatible pre-assignment induces the same role-preserving bijection to that image. Thus the jointly occupied carrier-root architecture has the separately typed invariant

$$(|\text{rt}_g^{\text{car}}(\text{Residue}_g^{\text{live}})|; |\text{Hld}_B|) = (3; 1), \quad |\text{Mech}_H(C_H)| = 3. \quad (14.2)$$

- (v) Every actual common ledger $A \in \mathcal{A}_B^{\text{pre}}$ supports its network disposition and an exact set-codescent object $\text{Codesc}_B^{\text{Set}}(A)$. Vertex maps are injective; the reconstructed image is precisely the union of the declared packet subrecords. This conclusion requires one common ledger, not uniqueness of all compatible assignments.
- (vi) Every member of $\widehat{\text{Ext}}_B(Z_*)$ has the same complete intrinsic gauge-law reduct

$$\mathcal{L}_g(e) = [\mathfrak{L}_g(Z_*)]_{\text{src}}.$$

This equality preserves the joint operation graphs, source probe families, anomaly and repair domains, and complete residual interfaces. On the nonempty source-realized fibre the law-family inventory, with the independently fixed color-support and Higgs families retained, is $(1, 1, 1; 1)$. Relative uses and gluing orbits are not identified by comparing their marginal law families.

- (vii) Every original compatible assignment is represented by finite decorated source-operation networks with its original evaluation. The representation retains all source operands, constituent charts, actual shared interfaces, partial domains, and failure records. Source-proved whole-diagram comparisons commute with evaluation. No unique marked chart or equality of different admitted response values is asserted.

The carrier-root invariant in (iv) is not a count of the fine response classes. If the equivalent conditions in (iii) also hold, the stronger compatible-live gauge-response cardinality follows:

$$(|\text{Live}_Q|, |\text{Live}_{\text{col}}|, |\text{Live}_{\text{wk}}^{\text{pre}}|; |\text{Hld}_B|) = (1, 1, 1; 1). \quad (14.3)$$

No scalar cardinality is assigned to a union of gauge responses, Higgs identity, mechanism types, support fields, and compatibility predicates.

Proof. The intrinsic response laws and all-instance source coverage give (i). Theorem 11.5 supplies source descent, including the fixed-sibling domain theorem for mixed instances. The complete contextual-signature relation is a congruence for defined operations, and its universal property gives unique semantic factorization. Activity reduction preserves all admitted tests; cancellation or a newly formed failure remains an explicit outcome rather than an implicit closure assumption. These constructions precede role eligibility and compatibility.

The three kindwise packet theorems prove (ii), together with the imported Higgs identity theorem and the mechanism fibration. Their domains and codomains remain separate. In particular, the identity projection forgets fine response information only when forming the carrier-identity object; it does not alter the response equivalence relation. The residual extension and its descent are Theorem 7.11 and Lemma 7.12.

The exact eligibility, compatibility, and assignment constructions give the objects in (iii). Sound SR7 reduction places every compatible ledger's minimal reduct in z_* . Equality of complete assignment signatures is equality of the rolewise quotient classes, so the identity automorphism already identifies assignments with equal signatures. Conversely every lawful automorphism preserves those signatures. The fixed-extension and Role-Occupancy Closure theorems therefore give (14.1) [38, 8, 7].

For (iv), the gauge restrictions and the Higgs incidence are certified from one source tuple Ξ , with its actual mixed-sector data retained. Theorems 10.20 and 12.23 construct d_Ξ and τ_Ξ . Theorem 7.13 constructs the residual packet from the same source's SR6 projection, without selecting a physical

vacuum or a particle realization. The source carrier-exhaustion chain excludes an additional same-scope peer root, and exact eligibility fixes the root of each role. Corollary 12.24 gives the unique projected assignment and (14.2). This argument does not collapse the fibres of the root projection.

Finally, one common ledger supplies every overlap in (v). Theorem 12.45 proves the set-colimit universal property, and Corollary 12.46 identifies its exact reconstructed subrecord. Theorems 13.3 and 13.9, together with Corollaries 13.4 and 13.10, prove (vi)–(vii). The conditional fine response count is Theorem 12.38; no count is used to form an earlier object. $\square$

14.2 Carrier, law, and marked-response uniqueness

Equations (14.2) and (14.3) answer different questions. The former identifies which carrier roots are occupied together in a source-realized Standard Model assembly. The latter would additionally say that no standing-relevant response distinction remains between compatible assignments over those roots. The first is proved by source construction and carrier exhaustion. The second has the exact criterion (14.1); it is not a consequence of the first. There is an intermediate, stronger positive conclusion: Theorem 13.3 fixes the entire structured gauge-law reduct, not just its carrier names. Its operation graphs and domains are preserved. Theorem 11.4 also settles the mixed-input presentation issue: a lawful re-presentation of one full intrinsic port preserves every fixed-attachment domain and output with the other legs fixed.

What remains distinct is a response evaluated at a different genuine source argument. The network representation retains such marks rather than replacing them by arbitrary output labels. To infer the fine singleton count would require a further theorem showing that all role-eligible marked evaluations have the same complete signature. Neither law rigidity nor source covariance asserts that a law is constant on its argument domain. No actual source-valid pair with different complete signatures is asserted here either.

This distinction is substantive even when every quantity is structural. A continuation test, a mixed-incidence class, or an extended-support datum can distinguish responses without introducing a new primitive carrier. SM-6 preserves precisely this distinction between carrier dependence and identity [17, Def. 2.9 and Thm. 20.1]. SCIC makes the complete-signature quotient canonical, while leaving its cardinality equal to the signature image [7, Thms. 4.4, 9.1 and 9.4]. The exact primitive-generation result enters the SM carrier closure through the source interpretation and complete occupation retained in Definition 12.18. Its three homogeneous positions govern matter incidence; they do not determine a bosonic response count.

Scope 14.2 (Scope of the structural endpoint). *Source formation, contextual quotienting, representative nonpromotion, one-Higgs identity, and mechanism typing are established at the declared target. The full source-realized branch also has positive mixed readiness, a simultaneous compatible assembly, actual structural residual charge transport, and unique carrier-root occupation. The complete gauge-law reduct is rigid, every compatible assignment has a source-operation network representation, and intrinsic fixed-input covariance is established. Each common assembly has exact set-codescent. A universal singleton claim for the finer compatible gauge-response assignments is not asserted: its source-relative condition is (14.1). Failure to establish that condition is not evidence that an extra carrier exists.*

The color conclusion retains the positive Yang–Mills extended-support sector layer over its fixed local theorem object. Its sector-to-completion filter remains attached; neither confinement nor an asymptotic free-gluon state is inferred [47]. The gravity role G_{AASC} is likewise retained as an ambient necessity of the nondegenerate comparison arena, not as an internal bosonic occupant [50].

14.3 The structural object passed to realization

The source-formed response architecture is the dependent record

$$\begin{aligned} \mathbb{B}_I^{2,0}(\Gamma) = & (\tau_B, \mathbb{I}_B, \mathcal{G}_B, \mathcal{S}_B, \mathcal{U}_B, \mathbf{A}_B, \mathcal{Q}_B^{\text{port}}, \mathcal{Q}_B^+, \mathcal{Q}_B^\times; \\ & \mathbf{I}_Q, \mathbf{I}_{\text{col}}, \mathbf{I}_{\text{wk}}^{\text{pre}}, \text{Hld}_B, \text{rt}_H^{\text{car}}, \text{rt}_g^{\text{car}}, \pi_\mu; \\ & (\text{RAnt}_Q(h_Q), \text{ResPort}_Q(h_Q))_{h_Q}; D_{H,EW}^{\text{rd}}, \beta_{H,EW}^{\text{rd}}; \\ & E_B, J_B, \text{Residue}_g^{\text{live}}, \Sigma_g^{\text{pre}}, \Gamma_g^{\text{lawful}}, \Theta_B, \widehat{\text{Ext}}_B, \text{ASig}_B; \\ & \mathcal{A}_B^{\text{pre}}, (\text{Net}_g^{\text{live}}(A), \text{Codesc}_B^{\text{Set}}(A))_A, \text{YM}^{\text{loc}}, \text{YM}^{\text{ext}}, G_{\text{AASC}}; \\ & \Phi_{\text{mix}}, \mathcal{L}_g, \mathcal{N}_B^{\text{src}}, \text{ev}_B^{\text{law}}). \end{aligned} \quad (14.4)$$

All entries use the same fixed target and global form. A source realization Ξ adds its constructed points $d_\Xi, \tau_\Xi, \omega_\Xi$, including $h_{\text{res},\Xi}$ and its residual transport packet, and the carrier-root occupation theorem. A proof of gauge-assignment rigidity adds (14.3); that equation is not inserted as a field of the general object.

The successor handoff records this architecture, its source provenance, and its exact realization boundary:

$$\mathbb{H}_{I \rightarrow II}^{2,0} = (\mathbb{B}_I^{2,0}, \text{SourceProvenance}_I, \text{Scope}_I, \text{SuccessorRule}_I). \quad (14.5)$$

The appendices record the source attribution, hypothesis dependencies, and correspondence with the arc's structural objectives.

Definition 14.3 (BCRS-I realization boundary). *The scope Scope_I consists of the source-law response classification and quotient, the identity and mechanism results, the source-relative common assembly, residual charge transport, and carrier-root occupation, intrinsic fixed-input source descent, complete gauge-law reduct rigidity, finite source-operation network coverage, the exact gauge-assignment rigidity criterion, and common-ledger set-codescent. It retains all typed failure, noncomparability, continuation, and source-scope data. It asserts no particle, polarization, generator, mass, width, pole, vacuum-value, coupling-value, confinement, asymptotic-state, graviton, or unrestricted physical-common-apex conclusion.*

Theorem 14.4 (Conservative passage to BCRS-II). *A BCRS-II realization of a common assembly A preserves $\mathbb{B}_I^{2,0}$, the identity projection to η_H , and the separate mechanism fibre. A new metric, vacuum, polarization, pole, confinement, or physical-projection assertion requires its own licensed realization map and antecedents. No later display creates, deletes, identifies, or recounts an earlier source or response class. In particular, a successor cannot replace the gauge-assignment rigidity criterion by an unproved singleton assertion.*

Proof. Every predecessor invariant is a fixed target field or an admitted semantic use descending through the complete-signature quotient. Preserving the source object therefore preserves these values and their typed failure outcomes. A changed invariant defines a different target or source instance; a new realization operation lies outside the predecessor's constructor domain. Fixed-authority no backflow excludes using either operation as an earlier-rank identity proof. This is the conservative successor discipline inherited from the fermionic arc [1]. $\square$

14.4 Conclusion

The premetric bosonic architecture consists of jointly supported charge, color, and weak carrier roots, together with one Higgs bridge/fixation identity. Order, deformation, and fixation are distinct mechanisms of that identity. A common Standard Model source supplies their compatible response assembly, residual charge transport, and mixed electroweak readiness without treating a physical boson display as a premise.

The source comparison fixes not only this carrier inventory but the complete compatible gauge-law reduct. Actual responses are represented by finite evaluations of source operations, with their arguments and relative attachments retained. Intrinsic covariance makes this description independent of presentation even in a fixed-input mixed context. Complete contextual signatures then determine which marked evaluations are indistinguishable. The resulting atlas combines a rigid law structure with the full response information needed for subsequent metric, vacuum, resonance, and confinement analysis.

A Scientific sources and retained dependencies

Each imported result is used with its own hypotheses, target, quotient, and conclusion. Restriction to the premetric target preserves those dependencies; it does not turn a conditional realization theorem into an unconditional structural claim. The principal source families and their mathematical roles are summarized below.

A.1 Principal source families

Source family	Retained mathematical role
Fermionic structural endpoint [1]	The shared Higgs identity, predecessor boundary and continuation data, and conservative passage to subsequent realization.
Admissibility and identity [2, 3, 4, 5, 6, 7, 8, 9, 10, 11]	Fixed-domain standing, complete semantic use, lawful quotient, continuation, role occupancy, and same-scope exhaustion.
Gauge realization [20, 21, 22, 23, 24, 25]	Boundary-fixed transition, transport, holonomy obstruction, representation and charge response, anomaly obstruction, and gauge-vacuum compatibility.
Gauge-sector specialization [26, 27, 28, 29, 30, 31]	Lie-type refinement, lawful product or coupled sector status, representation algebra, Abelian normalization, anomaly algebraization, and residual-sector normal form.
Realization rigidity [32, 33, 34, 35, 36, 37, 38]	Common-ledger survivor tests and the unique complete gauge-law reduct inherited through the SR1–SR7 comparison. Full operation graphs and domains are retained; law rigidity does not identify distinct marked response evaluations.
Standard Model carrier structure [12, 13, 14, 15, 16, 17, 18, 48]	Carrier classification and exhaustion, including the inherited generation-cardinality handoff. The full common source supplies the actual gauge and Higgs restrictions used in the assembly theorem; carrier exhaustion alone does not imply response-signature equality.
Higgs structural endpoint [39, 40, 41, 42, 43, 44]	One bridge/fixation carrier, its identity theorem, and the scale, vacuum, extension, and domain classifiers, with their typed mechanism and readiness restrictions.
Structural symmetry comparison [45, 46]	Comparison of quotient fidelity, target-signature coverage, and nonpromotion. These results are not constructors of the BCRS response quotient or assignment orbit.
Yang–Mills endpoint [47]	The fixed local theorem object, its distinct extended-support sector layer, and the exact sector-to-completion boundary.
Premetric gravity and transport [50, 51, 52]	The ambient nondegenerate physical interior, refoliation closure, faithful source descent with exact partial domains, and later smooth transport specialization. None is an extra bosonic occupant.

A.2 Source-use families and supporting theorems

The source family in Section 3 is generated by theorem conclusions and their full witness instances, not by mnemonic labels. Its semantic domain is the independently stated structural-law class of Section 2, not an arbitrary typed record. The following table records the proof support of every generator family.

A row licenses the indicated typed use; it does not turn every theorem named in that row into an additional response occupant.

Family and generators	Theorem pointers	Binding boundary
$G_{\text{spine}} = \{g_{\text{ord}}, g_{\text{tr}}, g_{\text{curv}}, g_{\partial}, g_{\text{cont}}\}$	Gauge order and transition: A1, Thms. 6.5 and 7.1; transport: A2, Thms. 7.1, 8.2, and 10.1; loop obstruction: A3, Thm. 11.1; boundary fixation and continuation: FE4 and FE3 [20, 21, 22, 10, 9].	Common structure is not a separately counted response occupant.
$G_Q = \{g_{\text{rep}}, g_Q, g_{Q\text{res}}, g_{\text{pol}}\}$	Representation and charge response: A4, Thms. 11.1 and 13.1; Abelian refinement: PSM4, Thms. 14.1 and 15.1, with SR4's complete survivor test. Residual response uses A6 and PSM6 only on their full domains; polarization remains a conditional port [23, 29, 35, 25, 31].	Q is a premetric Abelian response root. Physical vacuum realization, photon standing, and polarization execution remain outside BCRS-I.
$G_C = \{g_{\text{col}}, g_{\text{adj}}, g_{\text{self}}, g_{\text{YM}}^{\text{loc}}, g_{\text{YM}}^{\text{ext}}, g_{\text{conf}}\}$	Transport and loop response: A1–A3; witnessed actions and composites: PSM1, Thms. 10.4–10.5 and 11.1, PSM2, Thms. 11.2 and 12.1, and PSM3, Thms. 13.1 and 14.1; complete SR2–SR3 tests. Extended support: Yang–Mills Main Thm. II.1(A)–(E), Thms. V.12, VI.12, VII.8, and VIII.2 [20, 21, 22, 26, 27, 28, 33, 34, 47].	$g_{\text{YM}}^{\text{ext}}$ is positive structural sector information, not confinement or a free-gluon state; g_{conf} is readiness only.
$G_{\text{wk}} = \{g_{\text{wk}}, g_{\text{chiral}}, g_{\text{cc}}, g_{EW\ H}^{\text{pre}}\}$	Weak and chiral response: A4; PSM2, Thms. 10.2, 11.2, and 12.1; PSM3, Thms. 10.3, 13.1, and 14.1; complete SR2–SR3 tests. H17, Lem. 8.3 and Thm. 11.4, classifies a given joint bridge witness; A6/PSM6 do not supply missing electroweak attachments [23, 27, 28, 33, 34, 39, 25, 31].	$g_{EW\ H}^{\text{pre}}$ is an incidence-readiness generator, not an executed vacuum/mass map, a second Higgs carrier, or a W/Z particle.
$G_H = \{g_H, g_{H\text{ord}}, g_{H\text{def}}, g_{H\text{fix}}, g_{\text{exc}}\}$	Bridge-carrier admissibility and necessity: H17, Thms. 7.6 and 8.4; protocol classification: H17, Lem. 8.3 and Thm. 11.4; identity normal form: H18, Thms. 5.6 and 8.1; layered classifier and endpoint: H22, Thms. 5.2, 8.3, and 9.1 [39, 40, 44].	Order, deformation, and fixation are mechanism fields of one identity; g_{exc} is a readiness/failure port.
$G_{\text{guard}} = \{g_{\text{anom}}, g_{\text{prod}}, g_{\text{vac}}, g_{\text{fail}}\}$	Product status: PSM2, Thms. 10.2, 11.2, and 12.1, and SR1; anomaly: A5, Thm. 10.1, PSM5, Thms. 14.1 and 15.1, and SR5; vacuum/residual: A6, Thms. 10.1 and 12.1, PSM6, Thms. 15.1 and 16.1, and SR6; fixed-domain diagnosis: Bivalence, SCIC, and same-scope closure [27, 32, 24, 30, 36, 25, 31, 37, 6, 7, 11].	A source-level interface or obstruction is not the later BCRS compatibility truth value and is not a response occupant. Missing evidence is not a proved obstruction.

A.3 Limits of imported conclusions

Interface	Imported theorem content	Promotion excluded
Fermionic structural endpoint	The same $C_H = \mathcal{H}_{\text{br/fix}}$, predecessor boundary and continuation data, retained theorem hypotheses, and conservative successor discipline	No bosonic response cardinality, product-network disposition, confinement, pole, or physical-apex result
Fixed-domain kernel	Standing, admissibility, lawful redescription, continuation, and same-scope exhaustion	No empirical selector, numerical parameter, gauge response, or particle identity follows from the kernel alone

Interface	Imported theorem content	Promotion excluded
Gauge A1–A6	Transition groupoid, standing-preserving transport, loop obstruction, response witnesses, anomaly obstruction, and vacuum compatibility	Potentials, generators, curvature components, representation dimensions, charge labels, and vacuum coordinates do not self-promote
PSM1–PSM6	Smooth specialization and simultaneous algebraic compatibility of an already admissible premetric gauge package	Lie dimension, factor notation, basis multiplicity, trace notation, or a residual chart is not a primitive carrier count
SR1–SR6; later SR7	Direct common-ledger survivor tests, followed by the exact SR7 final-assembly comparison only on its full domain	Separate marginal survivors cannot be glued after the fact; SR7 is not a premise of early source coverage and supplies no mass parameter
SM-2; SYM comparison	Gauge-root terminology and later closed-certificate, quotient, and status comparisons	SR7-dependent SM2 capstones do not establish pre-SR7 semantic coverage; shared support and joint guards are not extra response occupants
Higgs H17–H22	One bridge/fixation identity with order, deformation, and fixation mechanism fields	The three fields are not three Higgs carriers; no primitive scale, vacuum selector, or physical Higgs pole is imported
Yang–Mills endpoint	Positive extended-support sector structure over the fixed local theorem object and its admissible-completion filter	The positive nonlocal sector layer does not itself entail confinement or a free-gluon asymptotic state
Gravity, WDW, and covariant differentiation	The ambient nondegenerate comparison arena and downstream realization context	Gravity is not a BCRS carrier, and smooth metric or differential structure is not inserted into the premetric source grammar

A conclusion beyond the stated scope of an imported result requires a separate theorem at the present target or an explicitly enlarged target. Source attribution does not enlarge a theorem’s conclusion.

B Structural scope and dependency summary

The semantic class is fixed by intrinsic structural laws before source certification. Theorems 3.5 and 3.11 establish coverage and complete contextual evaluation. Activity reduction then yields the canonical quotient and its exact partial constructor domains (Theorems 4.6 and 4.8). Licensed display changes preserve these data; a familiar particle or field notation does not supply a new standing identity (Theorem 6.3).

The packet constructions retain the exact charge, color, and pre-vacuum weak source hypotheses. Higgs identity and Higgs response remain different comparison levels: Theorems 10.2 and 10.5 establish one identity with three mechanism types, without identifying all fine Higgs responses. A full common source supplies its actual mixed-readiness point, compatible gauge pre-assignment, and structural residual charge transport (Theorems 10.20, 12.23, and 7.13). Corollary 12.24 then fixes the jointly

occupied carrier-root invariant $(3; 1)$ on that source's fibre.

The SM-1–SM-6 carrier-exhaustion comparison retains the primitive-generation receipt of Definition 12.18: the complete original duties, their derived architecture, and actual complete occupation. The received position object is $V(K_3)$ with its external S_3 -action; its vertex-orbit quotient counts a different object. The generation receipt does not determine a bosonic response count [12, 13, 14, 15, 16, 17, 18, 61]. The SR7 comparison also determines the complete structured gauge-law reduct (Theorem 13.3). The source chart construction represents every compatible assignment by its actual finite operation network (Corollary 13.10). Intrinsic presentation comparisons preserve fixed-input mixed domains and all standing observations (Theorem 11.5). These conclusions preserve, rather than suppress, distinct admitted arguments and relative attachments. The finer gauge-assignment singleton is equivalent to nonemptiness of the fixed-extension fibre and a singleton complete assignment-signature image (Theorems 12.27 and 12.35). This is the role of the complete-signature and occupancy principles [7, 8]; it is not an additional consequence of primitive carrier exhaustion.

Finally, every actual common ledger supports network nonpromotion and set-codescent independently of that singleton condition (Theorems 12.41 and 12.45). The reconstructed set is exactly the union of its declared packet subrecords, with injective vertex maps (Corollary 12.46). Actual Higgs inputs, mixed attachments, and common-ledger values remain in the assembly family throughout.

C Notation and status glossary

Symbol or status	Meaning
$\mathfrak{B}_B = \{Q, C, \text{EW}^{\text{pre}}, H\}$	Frozen kind index for Abelian charge response, color, pre-vacuum weak, and Higgs response questions; it is not a four-carrier assertion and Q is not yet a vacuum-descended photon sector.
$\mathcal{S}_B, \mathcal{U}_B$	Certified source terms and admitted semantic uses. Complete signatures test all typed one-hole contexts, including each input of a many-input constructor and its domain of definition.
$\mathfrak{S}_B, \mathcal{Q}_B^{\text{port}}$	Full root and auxiliary port sorts, and their active contextual quotient. Constructor composition is partial when an operation is undefined or its output is inactive.
$\text{Act}(\mathcal{S}_B)$	Activity reduction after semantic uses have been proved complete.
$\sim_{\text{pres}}, q_B^{\text{port}}, \mathcal{Q}_B^+, \mathcal{Q}_B^\times$	Complete contextual-signature equivalence, its active full-port projection, and the root-fibre positive and failure exports.
$\text{Residue}_g^{\text{live}}, \mathcal{R}_G^{\text{live}}$	The union of responses occurring in compatible pre-assignments, and the occupied-role orbit fibres used for counting after ROC. Neither is the carrier-root image.
J_B, ROC	Simultaneous compatibility on the common witness ledger and the exact Role-Occupancy Closure criterion for gauge assignments. Compatibility existence alone does not establish ROC.
Θ_B, z_*	One-way reduction to the unique minimal SR7 class. It has no asserted inverse on arbitrary SR7 data and does not itself establish extension existence or uniqueness.
$\widehat{\text{Ext}}_B(Z_*), \text{ASig}_B$	Gauge assignments admitting all fixed source and attachment obligations, and their complete rolewise signatures. Actual readiness and ledger witnesses are retained in the assembly family, not identified by this gauge-assignment quotient.
$\Xi, \mathcal{A}_B^{\text{pre}}$	A full common Standard Model source realization, and the family of actual compatible assemblies $A = (\sigma, \omega_A)$. The source supplies $d_\Xi, \tau_\Xi, \omega_\Xi$ and the actual residual witness $h_{\text{res}, \Xi}$ on its own realized fibre.
$\mathcal{L}_g, \mathfrak{L}_g(Z_*)$	The intrinsic complete gauge-law reduct and its unique source class, including jointly assembled operation graphs, probes, and domains.
$\mathcal{N}_B^{\text{src}}, \text{ev}_B^{\text{law}}$	Decorated finite source-operation networks in their actual relative charts, and their intrinsic evaluation. Marks retain actual operands and relative attachments; a source-law class is not a marked value.
$\text{RAnt}_Q(h_Q), \mathcal{R}_Q(h)$	The full structural support/reduction domain over a charge witness, and its evaluated residual charge-transport packet for $h \in \text{RAnt}_Q(h_Q)$. Neither is a physical photon.
$C_H = \mathcal{H}_{\text{br}/\text{fix}}$	The single same-scope Higgs bridge/fixation carrier inherited without duplication from the Higgs and FCGS theorem chains.
$\text{Ord}_H, \text{Def}_H, \text{Fix}_H$	Order, deformation, and fixation fields of one Higgs mechanism fibre; these are neither three identities nor three particles.

Symbol or status	Meaning
$\text{Disp}_{>1,A}$	Common-ledger higher-arity disposition relative to supplied verified certificates. A factorization through the three vertices is witnessed, not inferred from semantic factorization alone.
$D_{H,EW}^{\text{rd}}, \beta_{H,EW}^{\text{rd}}$	The exact mixed-readiness domain and its premetric incidence map. A domain point retains the fine Higgs input, both gauge inputs, and the actual mixed attachment $[\lambda]$.
$\text{Codesc}_B^{\text{Set}}(A)$	The set-colimit of packet subrecords on one actual common ledger, canonically isomorphic to their union with every vertex injected.
$\beta_{H,EW}$	The already reserved partial bosonic descent map. BCRS-I supplies nonmetric domain readiness but neither redefines nor evaluates this map at a vacuum-fixed metric or pole target.
$\Gamma \in \mathfrak{G}_{\text{src}}$	A fixed source-admitted global form. The stronger common-source conclusions apply to the image (D_{Ξ}, Γ_{Ξ}) , not to every formal parameter value or a splice of different fibres.
Structural	A theorem about source, quotient, response, compatibility, incidence, or realization architecture rather than a numerical or terminal projection.
Positive	The asserted structural object exists and satisfies its stated theorem predicate at the frozen target.
Ready, NotReady	The required named inputs are established, or are not established by the supplied evidence. Readiness is not execution; a NotReady diagnostic does not prove that the missing certificate cannot exist.
NotFormed	The named object is deliberately outside the target or has not been constructed; no cardinality or truth value is assigned to it.
Incompatible	Candidates share a comparison target but fail a required lawful compatibility condition.
NonComparable	No common licensed comparison interface exists; no verdict may be obtained by coercing the objects into one fibre.
TargetUpdate	An identity-bearing change to target, source authority, quotient, granularity, continuation semantics, or global-form fibre.
source-formed	Every intrinsic-law standing witness has a source certificate with the same structural values. Arbitrary typed submissions need not satisfy those laws.
quotient-complete	All licensed presentations with the same complete signature are identified, and every licensed constructor descends with its exact partial domain.
compatible-live	A quotient class survives the simultaneous compatibility predicate and has a lawful role assignment.
assignment orbit	An assignment modulo the lawful automorphism group; raw labelled assignments are not counted separately.

D Scope of the result and passage to realization

Theorem D.1 (Scope of the BCRS-I classification). *At the frozen premetric target, the structural classification theorem gives:*

- (i) *a source-law-complete response construction and its canonical contextual quotient, with exact partial constructor domains;*
- (ii) *one same-scope Higgs identity C_H , carrying the distinct mechanism fields Ord_H , Def_H , and Fix_H ;*
- (iii) *for every full source realization Ξ , positive mixed Higgs–electroweak readiness, actual structural residual charge transport, and a compatible simultaneous assembly whose carrier-root occupation has the separately typed invariant $(3; 1)$;*
- (iv) *an exact criterion for finer gauge-assignment rigidity: the fixed-extension fibre is nonempty and its complete assignment-signature image is a singleton;*
- (v) *for every actual common ledger, network nonpromotion and an exact set-codescent construction, with injective vertices and reconstructed image precisely the union of its declared packet subrecords;*
- (vi) *rigidity of the complete structured gauge-law reduct, retaining actual operation and probe graphs and their exact domains; and*
- (vii) *intrinsic fixed-input covariance and source-operation network coverage of every original compatible assignment, without identifying different actual response arguments or attachments.*

The criterion in (iv) concerns gauge assignments, not uniqueness of the fine Higgs input, mixed attachment, common ledger, or codescent object. Only when it holds does the fine live gauge-response count become $(1, 1, 1)$. No cardinality is assigned to the heterogeneous union of responses, carrier identities, mechanism types, and compatibility guards.

Proof. These are the conclusions of Theorem 14.1, with its source-realized and fixed-extension scopes retained. The actual assembly family records the data existentially required, but not identified, by the gauge-assignment fibre. The exact reconstruction assertion is Corollary 12.46. $\square$

The unconditional finer response count requires a further theorem at the same structural target; it is not supplied by whole-law rigidity. The classification also does not entail a physical-particle count, a physical photon, $W^\pm$ or Z^0 projection, a Higgs resonance, a polarization or generator multiplicity, a preferred global form, a numerical coupling or vacuum value, an unstable-pole theorem, confinement, an asymptotic free-gluon state, a graviton, or a physical common apex. Those are successor-target questions and require their own licensed premises and maps. A successor may extend the handoff conservatively, but it may neither alter the meaning of $\beta_{H,EW}$ nor use a later display to regenerate an earlier source, quotient, role, or cardinality.

E Reconstruction and robustness of the source presentation

Several source uses may be restrictions of one structural object, and a stored value may be recovered from its retained inputs and operation. Source reconstruction therefore distinguishes a redundant presentation, an incomplete or inconsistent structural record, and a change of target. The source grammar need not be an irredundant primitive basis. Only a separately specified deletion with a complete-signature separation witness establishes loss of a standing use in the sense of Definition 3.13.

E.1 Source schemas and their uses

The source family of Definition 3.2 has 28 schema families. The 31 named uses below consist of 29 mechanism uses from those families, including two uses of the same gauge-order schema, and two target anchors. The numbers index source uses; they are not response cardinalities. In particular, transition coherence and gauge order are restrictions of one A1 continuation-site/groupoid package. Structural residual charge is a use of $g_{Q_{\text{res}}}$, not a second name for the pre-residual Abelian refinement supplied by g_Q .

Use	Schema and retained object	Exact source interface
01	g_{ord} : transition coherence	A1 continuation site and groupoid, Definitions 3.1 and 3.8–3.10, Propositions 5.1–5.3 and 5.5, Theorem 7.1 [20].
02	g_{tr} : transport	A2 Theorems 7.1 and 10.1; actual comparison assignments on admitted continuation domains [21].
03	g_{curv} : loop residue	A3 Theorems 11.1 and 13.1; retained transport composite and endpoint-closing datum [22].
04	g_{rep} : representation	A4 Theorems 11.1 and 13.1 and PSM3 Theorem 13.1; the actual witness-preserving action, not its label [23, 28].
05	g_Q : charge	PSM4 Theorem 14.1 and Corollary 14.2; the complete Abelian refinement of the charge-witness ledger [29].
06	g_{anom} : anomaly	A5 Theorem 10.1 and PSM5 Theorems 14.1–15.1; formed common-ledger obstruction probes [24, 30].
07	g_{prod} : joint interface	PSM2 Theorems 10.2, 11.2, and 12.1; one co-scoped sector tuple with its independent or coupled status [27].
08	Fixed Γ	Global-form target anchor; not a source production. Extended comparison retains the same fibre [27, 47].
09	g_{vac} : vacuum interface	A6 Theorems 10.1 and 12.1; non-null support, preservation, and admissible-reduction interfaces [25].
10	$g_{Q_{\text{res}}}$: residual charge	A6 Theorems 10.1 and 12.1 and PSM6 Theorems 15.1–16.1; the induced charge restriction of the actual residual ledger [25, 31, 37].
11	g_{col} : color transport	A1–A3 and the complete PSM3 refinement; actual color-response restrictions [20, 21, 22, 26, 28].
12	g_{adj} : adjoint action	PSM3 Definition 4.2 and Theorem 13.1; its witnessed action map [28].
13	g_{self} : self-response	A3 loop composition or a PSM1/PSM3 witnessed bracket specialization, with the source comparison map retained [22, 26, 28].
14	$g_{\text{YM}}^{\text{ext}}$: extended support	Yang–Mills Main Theorem II.1(A)–(E), Theorems V.12, VI.12, VII.8, and VIII.2; fixed extended-to-local incidence [47].
15	g_{wk} : weak response	The actual weak action, lawful sector status, and handed splitter under PSM2 and PSM3 [27, 28].
16	g_{cc} : charged current	A4/PSM3 witnessed current action; the L1 charged-current restriction when that common source is instantiated [23, 28, 53].
17	$g_{H_{\text{ord}}}$: Higgs order	Higgs-bridge Theorems 7.4 and 7.6, Lemma 8.3, and Theorem 11.4; actual order field over C_H [39].
18	$g_{H_{\text{def}}}$: deformation	The same bridge tuple and its actual order-to-deformation map; the Higgs-domain classification [39, 44].

Use	Schema and retained object	Exact source interface
19	$g_{H\text{fix}}$: fixation	Bridge/fixation output with Higgs-identity preservation and domain restrictions [39, 40, 44].
20	g_{pol} : mode readiness	Exact unexecuted projection-authority domain; the A4/PSM3 specialization boundary, not a physical polarization [23, 28].
21	g_{conf} : completion interface	The Yang–Mills sector-to-completion filter; a sector is not automatically a filtered completion [47].
22	g_{exc} : excitation readiness	Higgs-domain Theorems 5.2, 8.3, and 9.1; the exact downstream excitation authority type [44].
23	g_{fail} : formed failure	Actual source obstruction tests and the derived aggregate of Definition 3.7 [24, 30].
24	Ambient gravity role	The nondegenerate physical target and its localization condition; not a bosonic production [50, 49].
25	g_{∂} : boundary restriction	FE4 fixation and A1 Definition 3.1(iv),(vi); the actual upstream boundary tuple [10, 20].
26	g_{cont} : continuation	FE3 and A1 Definitions 3.1–3.2; admitted upstream continuation objects and typed morphisms [9, 20].
27	g_{ord} : gauge order	The same A1 package as use 01, especially its admitted composition domain and Proposition 5.5’s unit law [20].
28	$g_{\text{YM}}^{\text{loc}}$: local object	Yang–Mills Proposition II.2 and Main Theorem II.1(E); the fixed local base of the source sector tuple [47].
29	$g_{EW\ H}^{\text{pre}}$: mixed incidence	The actual common DefMed graph and its three input restrictions, as in Theorem 10.16 [39, 13].
30	g_H : Higgs identity	Higgs-identity Theorems 5.6 and 8.1 and the inherited identity projection; not a fine-response singleton [40, 1].
31	g_{chiral} : handed splitter	PSM3 Definition 10.1, Proposition 10.2, and Theorem 10.3; an actual handed-response distinction [28].

E.2 Reconstruction from retained structural data

Proposition E.1 (Recovery of a stored source restriction). *Let w be a retained structural source tuple and let p be a retained licensed operation on its original domain. If a stored use is $a = p(w)$, deleting that stored copy while retaining w and p does not lose its complete contextual signature. This statement uses the actual structural components of w , not its citation, theorem name, or proof identifier.*

Proof. Re-evaluation gives the same $p(w)$, with the same domain and interfaces. Every enclosing one-hole context therefore has the same definedness and value. Substitution at every occurrence is the same-witness case of Theorem 3.14. In particular, reclosure by arbitrary retained many-input constructors does not invalidate the reconstruction: their other inputs and exact common-interface conditions are unchanged. $\square$

The proposition has the following concrete applications.

- (i) The transition and order uses 01 and 27 are projections of one A1 tuple. Retaining the full tuple under either use recovers the other projection. A2 identity transport is recovered as the identity comparison class. A3 loop residue is recomputed from its actual comparison classes and endpoint-closing datum, regardless of whether that residue is trivial.
- (ii) A retained full representation, Abelian, color, weak, or current tuple recovers its corresponding restricted response. This applies to uses 04–05, 11–13, 15–16, and 31 only on that same source instance. A represented bracket can be recovered as $\text{ad}_X(Y)$ when that exact action map is retained. A carrier name, dimension, or two-part decomposition supplies none of these maps by itself.

- (iii) Uses 06 and 23 are evaluated from the actual obstruction maps, their formed domains, and their input ledgers. Aggregate pass is derived from all the individual neutral-test equalities; it is not an independent flag.
- (iv) A full joint tuple recovers use 07. A full gauge–vacuum tuple recovers use 09. An actual admissible reduction and full residual refinement recover use 10 by restriction of the induced residual ledger. Neither separate sector marginals nor pre-residual Abelian data reconstruct an unprovided joint or reduction witness.
- (v) A retained Yang–Mills sector tuple recovers its extended-support fields and fixed local base, uses 14 and 28. The strictly local object alone is not asserted to reconstruct its global-sensitive extended layer. The completion filter, use 21, remains a separate interface.
- (vi) An actual bridge tuple recovers the order, deformation, and fixation fields, uses 17–19. The actual joint graph supplied by one source Ξ recovers use 29 and its input-leg identifications. The canonical identity projection recovers use 30, but does not determine every fine Higgs field.
- (vii) The readiness declarations in uses 20–22 can be copied or restricted without executing their consequent. What is recovered is their exact domain and output type. Upstream boundary and continuation tuples similarly recover their actual restrictions, uses 25–26; no downstream history is used to supply them.

Each statement identifies the retained data needed for recovery. None asserts that every smaller source witness extends to a larger common source, or that deletion of the entire source family is harmless.

E.3 Incomplete submissions and inconsistent replacements

Definition E.2 (Specified relation erasure). *Starting with an actual admitted tuple, a specified relation erasure removes a named structural field or incidence while retaining the original target declaration. Its output is a raw submission. A law-inconsistent replacement instead assigns a domain or relation value incompatible with the retained actual input and its source law. Deleting a proof token of a relation that the structural data still satisfy is neither operation.*

A raw submission lacking a required field does not itself furnish the original certification. This is an incompleteness statement, not a proof that the underlying physical structure is inadmissible. A licensed reconstruction changes the outcome to Proposition E.1. By contrast, the following replacements contradict specified laws on actual inputs.

Proposition E.3 (Source-law negative controls). *The following substitutions cannot certify the unchanged source profile.*

- (i) *At an admitted A1 presentation, declare identity composition undefined while retaining the original identity input and continuation interface.*
- (ii) *At an admitted A2 identity continuation, declare its transport comparison undefined or deny its equality to the identity class.*
- (iii) *On an actually formed failure-test domain, declare that same test unformed while retaining its original domain witness. On a positive common source whose formed tests are all neutral, replace the derived aggregate pass value by a failure assertion without changing any test value.*
- (iv) *In the common graph of Theorem 10.16, deny one of its actual input-leg identifications while retaining the same graph, input restriction, and equality witness.*
- (v) *Count the three Higgs mechanism fields of an admitted tuple as three distinct carrier identities while retaining their common identity projection to η_H .*

Proof. A1 Proposition 5.1 gives the identity transition at every admitted local presentation, and its identity composition has the required admitted interface and unchanged role data. Declaring it undefined contradicts that domain and Proposition 5.5’s unit law [20]. A2 Theorem 7.1(i) gives the identity comparison value on the admitted identity continuation [21]. These are actual identity inputs, not postulated nontrivial transitions.

For (iii), domain formation is an independent coordinate of the complete signature. Its contradictory replacement cannot describe the same input. Definition 3.7 fixes aggregate pass from all the individual formed tests, so changing only that derived report cannot change the evaluated obstruction ledger. A nonneutral value reported as neutral would likewise be rejected whenever an actual nonneutral source test is supplied; no such inhabitant is inferred from this conditional statement.

For (iv), the common graph carries exactly the restrictions named by its input legs. Denial of an equality while retaining both its actual endpoints and its witness contradicts the unchanged graph. Replacing the graph by separately certified marginals instead leaves the joint domain uninstantiated; Proposition 12.5 does not manufacture that missing witness.

For (v), the order, deformation, and fixation fields all project to the same C_H . The Higgs-identity normal form and Lemma 10.3 identify their primitive identity projections. Treating the mechanism index as a second identity coordinate contradicts the preserved identity relation [40, Theorems 5.6 and 8.1]. It does not contradict the fact that the mechanism fields themselves are differently typed. $\square$

The remaining relation-erasure checks are equally local. Removing an action map from uses 04, 11–13, 15–16, or 31 does not supply the corresponding witness-preserving action. Removing the reduction incidence from use 10 does not supply a residual ledger induced by that reduction. Removing an extended-to-local incidence from use 14 does not furnish the declared complete color packet. Removing a bridge edge from uses 17–19 does not furnish its mediation or fixation. Removing an upstream restriction or morphism typing from uses 25–26 does not furnish the claimed boundary or continuation comparison. These are checks of the specified submissions, not independence proofs about all alternative source expressions.

Readiness has an additional distinction. Suppressing the authority domain in uses 20–22 and reporting an executed polarization, completion, confinement, or excitation output is not a source-preserving update. The missing antecedent has not become false or true: the execution is unlicensed. A declared typed implication remains meaningful even when its antecedent has no demonstrated inhabitant.

E.4 Anchors and the strength of the conclusion

Changing the actual global-form anchor in use 08, or removing the ambient nondegenerate physical target in use 24, leaves the original target fibre. Deleting their printed names while preserving the underlying anchors changes no standing use. Neither experiment deletes a bosonic production or produces an additional carrier.

These results distinguish recoverable source descriptions from incomplete records, law-inconsistent replacements, and target changes. They do not assert an irredundant primitive basis of all 28 schemas. Such a claim would require a separately specified deletion of full production families, complete retained reclosure, and an actual complete-signature separation witness. The source coverage, contextual factorization, and compatible-assembly theorems do not depend on that additional minimal-basis claim.

References

- [1] A. J. Maley, *Structural Universality and Exact Terminal Bifurcation: One-Higgs Naturality, Projection Firewalls, and the Branch-Complete Fermionic Endpoint*. FCGS-III, Primitive-standing reference edition, 21 September 2026.
- [2] A. J. Maley, *Non-Degenerate Construction and the Kernel of Admissibility: An Exhaustion Theorem for Standing, Reference, and Irreversibility*, hardened edition, 16 June 2026.
- [3] A. J. Maley, *The Structure of Admissibility: Ametricity, Bivalence, and the Unique Admissible Interior*, hardened WA edition, 19 June 2026.
- [4] A. J. Maley, *A Fixed-Domain Exhaustion Theorem: Standing, Admissibility, and the Determination of Consequence under Typed Boundary Discipline*, 22 April 2026.
- [5] A. J. Maley, *The Structure of Coherent Construction: A Bivalent Trajectory Theory*, 2026.
- [6] A. J. Maley, *The Bivalence Theorem for Non-Degenerate Construction: Standing Conservation, Governance Necessity, and Local Closure*, 2026.
- [7] A. J. Maley, *Standing Carrier Identity Closure: A Same-Scope Exhaustion Theorem for Source-Distinct Physical Realizations under AASC*, August 2026.
- [8] A. J. Maley, *Role-Occupancy Closure under Admissibility: Why Role-Cardinality Matching Is Only the Cardinality Shadow of Compatible Realization*, May 2026.
- [9] A. J. Maley, *Continuation-Locus Induction under Admissibility-Bearing Governance: A Fixed-Domain Persistence Theorem under AASC*, May 2026.
- [10] A. J. Maley, *Boundary-Trace Fixation of Continuation Loci under AASC: A Role-Anchor Theorem for Fixed-Domain Persistence*, May 2026.
- [11] A. J. Maley, *Closure by Exhaustion for Same-Scope Operators under Admissibility: A Self-Contained Derivation from Graph Rigidity, Family Classification, and Interaction Closure*, March 2026.
- [12] A. J. Maley, *Primitive Carriers, Derivative Skins, and Classifier-Level Prediction Closure in the Standard Model*, July 2026. [doi:10.5281/zenodo.21213269](https://doi.org/10.5281/zenodo.21213269).
- [13] A. J. Maley, *Gauge Carriers: Photon, Gluon, and Weak-Boson Standing*, 1 July 2026. [doi:10.5281/zenodo.21213644](https://doi.org/10.5281/zenodo.21213644).
- [14] A. J. Maley, *Matter Carriers: Quarks, Leptons, and Neutrinos*, July 2026. [doi:10.5281/zenodo.21213753](https://doi.org/10.5281/zenodo.21213753).
- [15] A. J. Maley, *Charges, Couplings, and Renormalization as Standing Transport*, July 2026. [doi:10.5281/zenodo.21213845](https://doi.org/10.5281/zenodo.21213845).
- [16] A. J. Maley, *Flavor Transition, Mixing, and Generation Cardinality*, July 2026. [doi:10.5281/zenodo.21214028](https://doi.org/10.5281/zenodo.21214028).
- [17] A. J. Maley, *The Standard Model Carrier Exhaustion Closure*. Primitive-standing reference edition, 21 September 2026. Earlier version: [doi:10.5281/zenodo.21214336](https://doi.org/10.5281/zenodo.21214336).

- [18] A. J. Maley, *Generation Cardinality as a Forced Role-Occupancy Endpoint: Primitive necessity, exact-three occupation, and nonpromotion under enrichment*. Primitive-standing reference edition, 21 September 2026. Earlier version: [doi:10.5281/zenodo.21215285](https://doi.org/10.5281/zenodo.21215285).
- [19] A. J. Maley, *Higgs-Like Bridge and Scale-Anchor Rigidity over the SR7 Skeleton*, May 2026.
- [20] A. J. Maley, *The Gauge Transition Groupoid Theorem: Boundary-Fixed Local Presentation Rigidity under Realized Physical Admissibility*, 12 May 2026.
- [21] A. J. Maley, *Connection as Standing-Preserving Transport Geometry: Minimal Transport-Comparison Structure for Physical Gauge Realization*, 12 May 2026.
- [22] A. J. Maley, *Curvature as Transport-Holonomy Obstruction: Boundary-Detectable Loop Residue in Physical Gauge Realization*, 12 May 2026.
- [23] A. J. Maley, *Representation Admissibility and Charge-Witness Rigidity: Standing-Preserving Representation Response in Physical Gauge Realization*, 12 May 2026.
- [24] A. J. Maley, *Anomaly as Standing-Failure Obstruction: Cancellation as an Admissibility Requirement for Representation-Bearing Gauge Realization*, 12 May 2026.
- [25] A. J. Maley, *Gauge-Vacuum Compatibility: Vacuum Standing, Symmetry-Breaking Admissibility, and the Bridge to Mass-Generation Rigidity*, 12 May 2026.
- [26] A. J. Maley, *Lie-Type Gauge Realization Admissibility: When Boundary-Fixed Gauge Transition, Transport, and Loop-Residue Structures Admit Smooth Infinitesimal Specialization*, 12 May 2026.
- [27] A. J. Maley, *Product Gauge-Sector Compatibility: When Multiple Lie-Type Gauge Ledgers Can Coexist in One Standing Domain*, 13 May 2026.
- [28] A. J. Maley, *Representation Algebra Specialization: When Representation-Response Carriers Admit Algebraic Representation Spaces*, 13 May 2026.
- [29] A. J. Maley, *Abelian Charge-Witness Normalization: Charge Lattice Readiness, Normalization Skin, and Abelian Factor Rigidity*, 13 May 2026.
- [30] A. J. Maley, *Anomaly Ledger Algebraization: When Standing-Obstruction Neutrality Becomes Trace, Sum, Probe-Induced Charge, and Index Cancellation*, 13 May 2026.
- [31] A. J. Maley, *Vacuum-Compatible Residual Sector Normal Form: Residual Gauge Structure after Admissible Vacuum-Mediated Reduction*, 13 May 2026.
- [32] A. J. Maley, *Simultaneous Constraint Collapse of the Admissible Gauge Arena: Arena-Rigidity Foundations for Standard-Model Realization under AASC (SR1)*, 13 May 2026.
- [33] A. J. Maley, *Product-Sector Rigidity in the Admissible Gauge Arena: Low Product-Role-Rank Collapse under Simultaneous AASC Constraints (SR2)*, 13 May 2026.
- [34] A. J. Maley, *Representation-Response Rigidity in the Admissible Gauge Arena: Asymmetric, Conjugate, Tensor, and Multiplicity Discipline over the SR2 Product Survivor Class (SR3)*, 13 May 2026.
- [35] A. J. Maley, *Abelian Charge-Ratio Rigidity in the Admissible Gauge Arena: Normalization, Ratio, and Abelian Role-Standing over the SR3 Representation Survivor Class (SR4)*, 13 May 2026.

- [36] A. J. Maley, *Anomaly-Ledger Rigidity in the Admissible Gauge Arena: Mixed, Nonlinear, and Global Neutrality Collapse over the SR4 Charge Survivor Class (SR5)*, 14 May 2026.
- [37] A. J. Maley, *Residual Vacuum-Sector Rigidity in the Admissible Gauge Arena: Residual Normal-Form Collapse over the SR5 Anomaly Survivor Class (SR6)*, 14 May 2026.
- [38] A. J. Maley, *Standard-Model Realization Rigidity in the Admissible Gauge Arena: Final Assembly of Product, Representation, Charge, Anomaly, and Residual Survivor Ledgers (SR7)*, 14 May 2026.
- [39] A. J. Maley, *The Higgs Bridge Carrier: Kernel Consequences for Electroweak Mass-Response*, 28 June 2026.
- [40] A. J. Maley, *Endpoint on Higgs Structural Identity: Bridge-Carrier Occupation and No Fifth Identity Ground*, 28 June 2026.
- [41] A. J. Maley, *No Primitive Higgs Scale: Carrier-Indexed Scale Standing and Hierarchy Orientation*, 28 June 2026.
- [42] A. J. Maley, *Higgs Vacuum Route Status: Metastability, Lower Branches, and Standing Closure*, 28 June 2026.
- [43] A. J. Maley, *The Higgs Extension Classifier: Extra Scalars and Same-Scope Carrier Preservation*, 28 June 2026.
- [44] A. J. Maley, *The Higgs Domain Endpoint: Carrier, Identity, Scale, Vacuum, Extension, and Projection*, 28 June 2026.
- [45] A. J. Maley, *Gauge Architecture and Gauge Skin in the Standard Model: Integrated Role-Occupancy Closure Release*, 13 July 2026.
- [46] A. J. Maley, *The Standard Model Symmetry Atlas and Exhaustion Closure: Target-Indexed Normal Forms, Physical Role, and Same-Scope Exhaustion*, 13 August 2026.
- [47] A. J. Maley, *Admissibility and the Well-Definedness of Yang–Mills Endpoint Constructions: Determinacy, Identity, and Uniqueness on Fixed Domains*, 4 April 2026.
- [48] A. J. Maley, *The Standalone Standard Model Structure from First Principles: An AASC Derivation of Gauge-Product, Representation, Higgs/Bridge, Yukawa, Charge, Anomaly, Vacuum, and Parameter-Status Structure*, 20 May 2026.
- [49] A. J. Maley, *The Realized Physical Interior: Locality, Metric Role, Fields, Particles, Gauge Structure, Mass, Vacuum, Time, Causality, and the Standard Physics Stack under AASC*, June 2026.
- [50] A. J. Maley, *Gravity as Structural Necessity: An AASC Account of the GR-Type Physical Interior*, 2026.
- [51] A. J. Maley, *WDW-3 Normal Deformation as Constrained Refoliation: Groupoid-Faithful Hypersurface-Deformation Closure under AASC*, July 2026.
- [52] A. J. Maley, *Covariant Differentiation from the AASC Kernel: Standing-Preserving Transport, Identity Principal Symbol, and Levi-Civita Closure*, 12 July 2026.
- [53] A. J. Maley, *Equivalence Completion and Forced Collapse of Charged–Current Degrees of Freedom*, L1 equivalence-completion series, Paper 1, undated manuscript.

- [54] A. J. Maley, *Equivalence Completion Between Electroweak and Electromagnetic Coupling Bookkeeping*, L1 equivalence-completion series, Paper 4, undated manuscript.
- [55] A. J. Maley, *Equivalence Completion for Longitudinal Gauge Modes and Goldstone Degrees*, L1 equivalence-completion series, Paper 6, undated manuscript.
- [56] A. J. Maley, *Anomaly Freedom $\iff$ Yukawa Realizability under Minimal Chiral Load*, L1—Equivalence Completion, Paper 7, undated manuscript.
- [57] A. J. Maley, *Absence of Tree-Level Flavor-Changing Neutral Currents under Admissible Redescription*, L1—Equivalence Completion, Paper 10, undated manuscript.
- [58] A. J. Maley, *Hypercharge Normalization and Electric Charge Quantization under Admissible Redescription*, L1—Equivalence Completion, Paper 11, undated manuscript.
- [59] A. J. Maley, *Exhaustion and Closure of the Equivalence Space*, L1—Equivalence Completion, Paper 12, undated manuscript.
- [60] A. J. Maley, *Mass Fixation and Inertial Response: Why the Higgs Was Structurally Unavoidable*, manuscript, 5 January 2026.
- [61] A. J. Maley, *Premetric Exact-Three Root Structure and the Generation-Governed Standard-Model Matter Bundle*. FCGS-I, Primitive-standing reference edition, 21 September 2026.